

Safety in Self-Evolving Agents: A Survey

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Abstract

Large language models (LLMs) exhibit strong general capabilities yet remain essentially static after deployment, failing to adapt to novel tasks, evolving environments, or dynamic contexts. In open-ended interactive environments, this characteristic motivates a shift toward self-evolving agents that continuously update reusable state: model weights, memory entries, tool definitions, or architectural components, from data, interactions, and accumulated experience. This fundamentally changes the safety problem: when experience becomes reusable state, past events become future causes, and information that was harmless in one context may later influence decisions with greater persistence, authority, or scope. Unlike LLM and agent safety that focus on whether a response is aligned or an action is authorized, self-evolving agent safety turns to whether safety properties survive the accumulation, generalization, and cross-context reuse of locally useful experience. Existing surveys organize agent safety around static components or lifecycle stages, but rarely track how a single piece of reusable influence, once written into state, changes its carrier, role, authority, and downstream consequences as the agent continues to evolve. To address this gap, we present **SAVER**, a transition-centered framework that analyzes safety through $S \times A \rightarrow V \rightarrow E \rightarrow R$: **Substrate** locates where reusable influence resides, **Adaptation** identifies the operation that transforms it, **Violation** characterizes which safety attributes are compromised, **Exposure** determines where and to whom the failure becomes observable, and **Response** evaluates whether the influence can be contained, repaired, or revoked. Treating the $S \times A$ transition as the analysis unit, SAVER traces how a given transition may induce violations, expand exposure surfaces, and demand containment responses. Our survey reveals several key insights. Conceptually, safety failures need not originate from intrinsically harmful information; rather, otherwise legitimate state becomes unsafe when adaptation broadens its persistence, authority, or scope beyond the conditions under which it was originally valid. In terms of research coverage, existing work concentrates on early, observable stages: admission, retrieval, activation, exposure, and local containment while later stages remain sparse. Looking forward, we argue that evaluations must move beyond single-timepoint metrics toward longitudinal evidence verifying whether unsafe influence can be traced to its originating transition, contained at its exposure surface, and prevented from re-emerging after continued adaptation.

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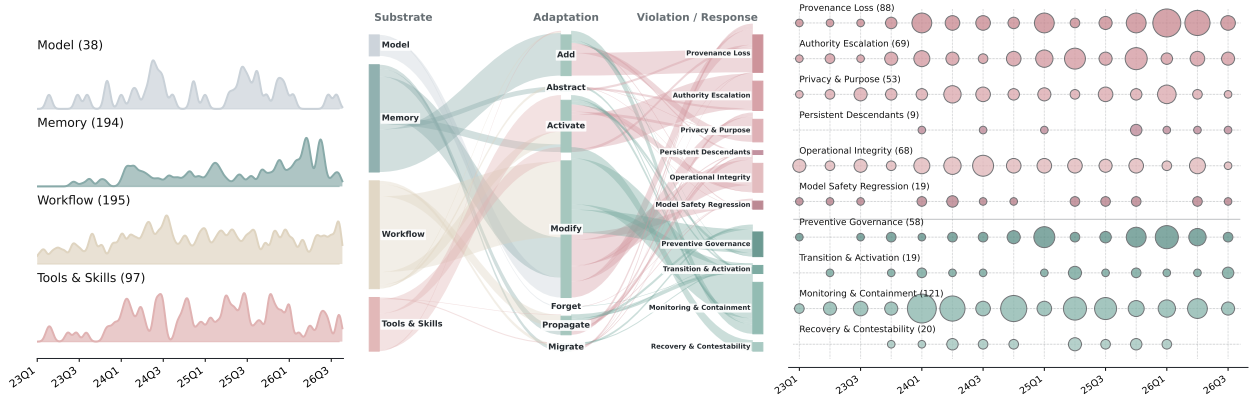
Key Words: Self-Evolving Agents; Agent Safety; Adaptive State; Memory Poisoning; Governance

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1 Introduction

Large language models (LLMs) exhibit strong general capabilities yet remain fundamentally static after deployment: their parameters, knowledge, and behavioral policies do not update in response to novel tasks, evolving environments, or shifting interaction contexts [8, 42, 189]. As LLM agents are increasingly deployed in open-ended, interactive environments [124], this static nature becomes a critical bottleneck, motivating a paradigm shift toward *self-evolving agents* that continuously update reusable state, model weights, memory entries, tool definitions, skill libraries, workflow rules, or architectural components, from data, interactions, and accumulated experience [8, 189]. We call an agent self-evolving when its own observations, actions, feedback, or execution traces update such reusable state inside the agent loop or runtime, thereby changing later behavior. Across diverse instantiations: reflection-based strategy refinement [177], skill-library accumulation [198], experience-driven rule consolidation [201], and evaluator co-evolution [70], the common safety-relevant thread is *reusable influence*: whatever evolves stores or transforms state that can shape later decisions.

Figure 1 Temporal distribution and SAVER flow of the literature. Left: Substrate trends and totals. Center: flows through Adaptation to compact labels for the six Violation families or four Response stages. Right: terminal-family totals and quarterly bubble volume.



This paradigm shift changes the nature of the safety problem. Conventional LLM safety primarily focuses on whether a single response is aligned, harmful, and truthful [37], agent safety extends the question to whether a discrete action or tool invocation is properly authorized and bounded [31, 47, 216]. Self-evolving agents, however, introduce a qualitatively different risk as shown in Figure 1: *adaptation itself becomes a source of safety change*. State created from otherwise benign interactions can be selected, abstracted, and repeatedly reused, thereby changing the conditions under which future decisions are made [107, 109]. Safety degradation can therefore emerge endogenously without a persistent attacker [176, 246], remain latent across sessions [109, 183, 192, 196], propagate across carriers or agents, and accumulate over multiple rounds of evolution. This property makes pointwise safety insufficient: behavior that is safe in one interaction may later be stored, abstracted, or reused with broader persistence, authority, or scope. Self-evolving agent safety therefore examines whether *safety remains preserved as influence is transformed and reused over time*, shifting the safety object from isolated responses or actions to the evolution of influence across state transitions.

Existing surveys organize agent safety by system components [47, 101, 129, 202, 216], lifecycle stages [107, 142], or attack categories [8, 109, 138, 157]. Such taxonomies locate a vulnerable component, a risky stage, or a visible failure mode, but they do not track how a single piece of reusable influence, once written into state, changes its carrier, role, authority, and downstream consequences as the agent continues to evolve. The same content may be benign as transient context, unsafe as a persistent memory, and critical once compiled into a workflow rule or tool-bearing skill. By decomposing these steps into separate categories, traditional taxonomies fragment what is in fact one causal incident and obscure where the safety boundary was crossed. Characterizing self-evolving agent safety therefore requires following the same influence across transitions, rather than analyzing modules or attack classes in isolation.

We adopt a *transition-centered* view: rather than singling out unsafe components in isolation, we follow the same reusable influence as it is stored, transformed, activated, propagated, and eventually exposed. We formalize this view as SAVER, as shown in Figure 2, a framework organized by

$$\mathbf{S} \times \mathbf{A} \rightarrow \mathbf{V} \rightarrow \mathbf{E} \rightarrow \mathbf{R}$$

Substrate (S) locates where reusable influence resides, such as memory, retrieved context, a skill, workflow rule, shared artifact, or model-side state. **Adaptation (A)** identifies the operation that transforms it, such as adding, abstracting, activating, modifying, propagating, migrating, or forgetting state. The two jointly define the transition under study: the same operation can be benign on one carrier and unsafe on another, while the same carrier can remain safe under one operation and violate its safety attributes under another. **Violation (V)** characterizes which safety attributes are compromised, including provenance, authority, scope, privacy, integrity, or recoverability. **Exposure (E)** determines where and to whom the failure becomes observable, such as memory retrieval, tool execution, multi-agent uptake, an external artifact, or model deployment. Finally, **Response (R)** evaluates whether the influence can be contained, repaired, or revoked. The substrate and adaptation jointly define the transition under study: the same operation can be benign on one carrier and unsafe on another, while the same carrier can remain safe under one operation and violate its safety attributes under another. SAVER therefore treats the substrate-adaptation pair as the minimal comparison unit, tracing how a given transition may induce violations, expand exposure surfaces, and demand containment responses, across memory security, indirect prompt injection, tool authorization, workflow safety, multi-agent coordination, and model-side adaptation.

SAVER begins once an update has been committed, but self-evolution starts one step earlier: the agent must first choose which candidate update to retain. We call this pre-transition layer *selection semantics*: repeated preference for locally successful updates determines which $\mathbf{S} \times \mathbf{A}$ transitions enter persistent state and accumulate over time [70, 177, 198, 201]. Conversely, the process does not necessarily end at **R**: guardrails, rollbacks, and policy updates can reshape the constraints and candidate set of subsequent evolution, which we capture as *feedback semantics*. Together, selection explains *why* a transition occurs, SAVER explains *what* it changes, and feedback explains *how* it reshapes future transitions. The recent OpenAI-Hugging Face incident [87, 145, 146, 213] illustrates why such a transition-level trace matters in practice: even when autonomous self-evolution is not itself the failure mechanism, the connected role changes across carriers determine the incident’s scope and the response’s completeness.

What SAVER Reveals. A transition-centered reading reveals three structural patterns. First, at the conceptual level, unsafe influence need not originate from intrinsically harmful information: otherwise legitimate state can become unsafe when adaptation broadens its persistence, authority, scope, or actionability beyond the conditions under which it was valid. Second, the evidence chain is markedly asymmetric. Existing work provides comparatively mature evidence for admission, retrieval, activation, visible exposure, risk recognition, and local containment, but becomes increasingly fragmented once influence migrates across substrates, propagates to descendants or other agents [254], or must be repaired across sessions. In particular, few systems demonstrate *rollback across sessions*: a connected incident trace that follows the same influence into its descendants, repairs them, and verifies that the influence can no longer reactivate [112, 176]. This asymmetry reflects a stronger proof burden for recovery rather than simply weaker defense research: an attack may require one successful path, whereas recovery must identify and neutralize the relevant future activation sites. Third, this gap exposes a limitation of endpoint-oriented evaluation. Metrics such as attack success, task success, recognition accuracy, or immediate blocking capture local outcomes but rarely establish what happens to the implicated state afterward. Self-evolving agent safety instead requires longitudinal evidence that follows the same influence from update selection and state transition through activation, exposure, response, descendant repair, and resumed adaptation, including whether the same or a functionally equivalent influence later re-emerges.

Contributions. This survey makes the following contributions:

- **Transition-centered synthesis.** We connect memory, tool and workflow, multi-agent, and model-adaptation research through the reusable state transitions underlying their safety challenges.

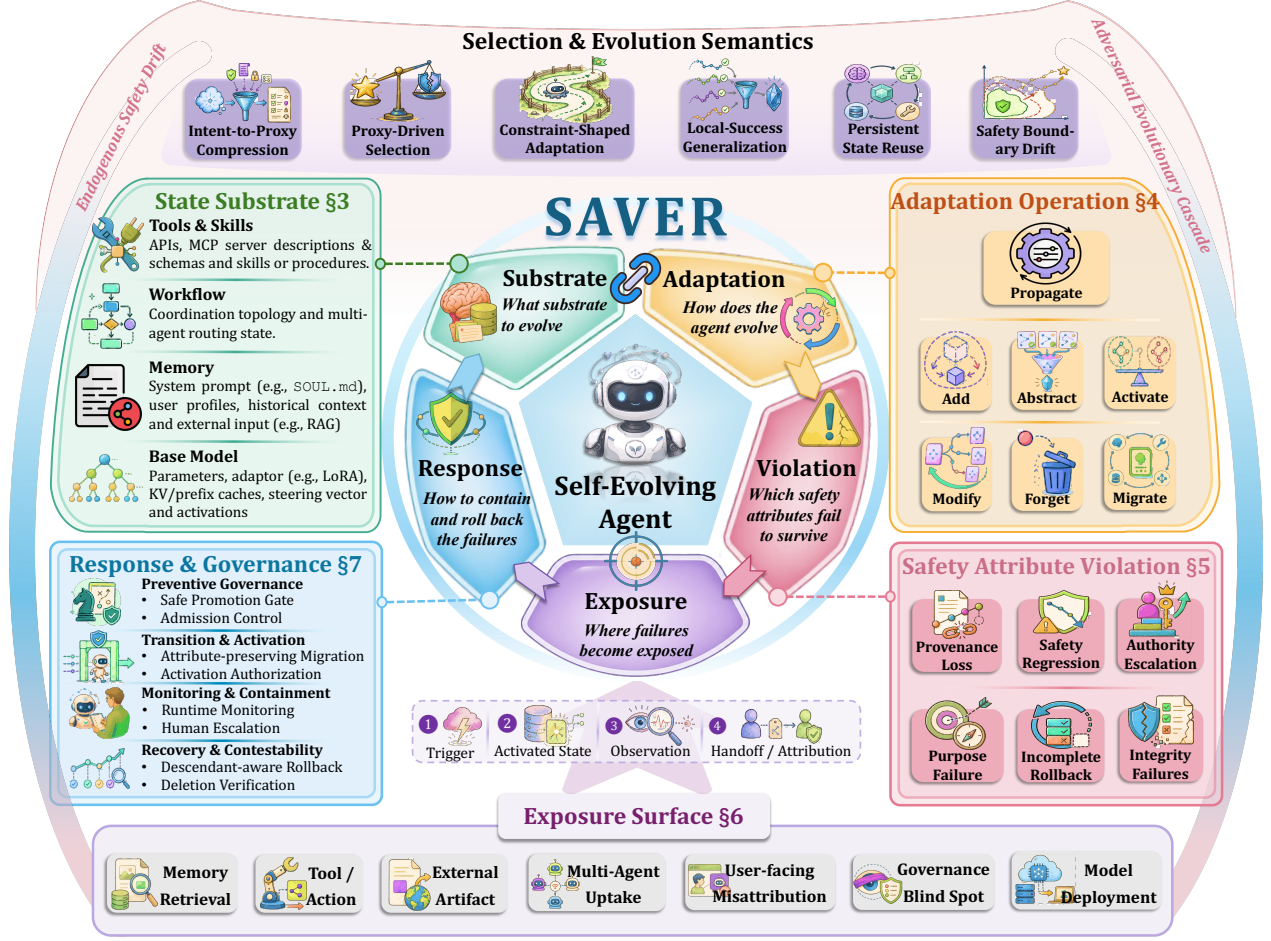


Figure 2 Overview of the transition-centered survey structure. The central ring couples Substrate and Adaptation as $S \times A$, then traces Violation, Exposure, and Response as $V \rightarrow E \rightarrow R$. The surrounding panels preview the substrate, operation, safety-attribute, exposure-surface, and governance taxonomies.

- **SAVER framework.** We propose a transition-centered analysis that uses substrate-adaptation transitions as the comparison unit and traces safety violations, exposure surfaces, and response mechanisms.
- **Evaluation and governance agenda.** We identify critical gaps in current benchmarks and defenses, and establish future directions toward evaluation, defense, and lifecycle governance for self-evolving agents.

Scope. We center agents whose observations, actions, feedback, or execution traces update reusable state inside the agent loop such that the update can affect later behavior. Stateful or adaptive systems remain adjacent when they expose a reusable carrier, a role-changing operation, an activation boundary, or a response mechanism relevant to such later influence. One-shot output safety, tool use without persistent reuse, and static model training fall outside the core scope unless they illuminate a later adaptive-state transition. Our synthesis is grounded in a systematic scoping review spanning security, machine learning, NLP, retrieval, and adjacent agent research; the search, screening, and corpus statistics are documented in Appendix A.

Organization. The remainder of the survey follows the SAVER relation. Section 2 first defines adaptive-state safety, formalizes the self-evolving agent loop, and introduces selection and transition semantics. Section 3 and Section 4 identify the carrier and operation halves of the $S \times A$ transition pair. Section 5 analyzes the safety attributes that fail across these transitions, and Section 6 identifies the surfaces at which those failures become observable. Section 7 then examines prevention, activation control, monitoring, containment, recovery, and contestability. Finally, Section 8 translates the full transition into benchmark and evidence requirements, Section 9 develops the resulting research agenda, and the final section concludes the survey.

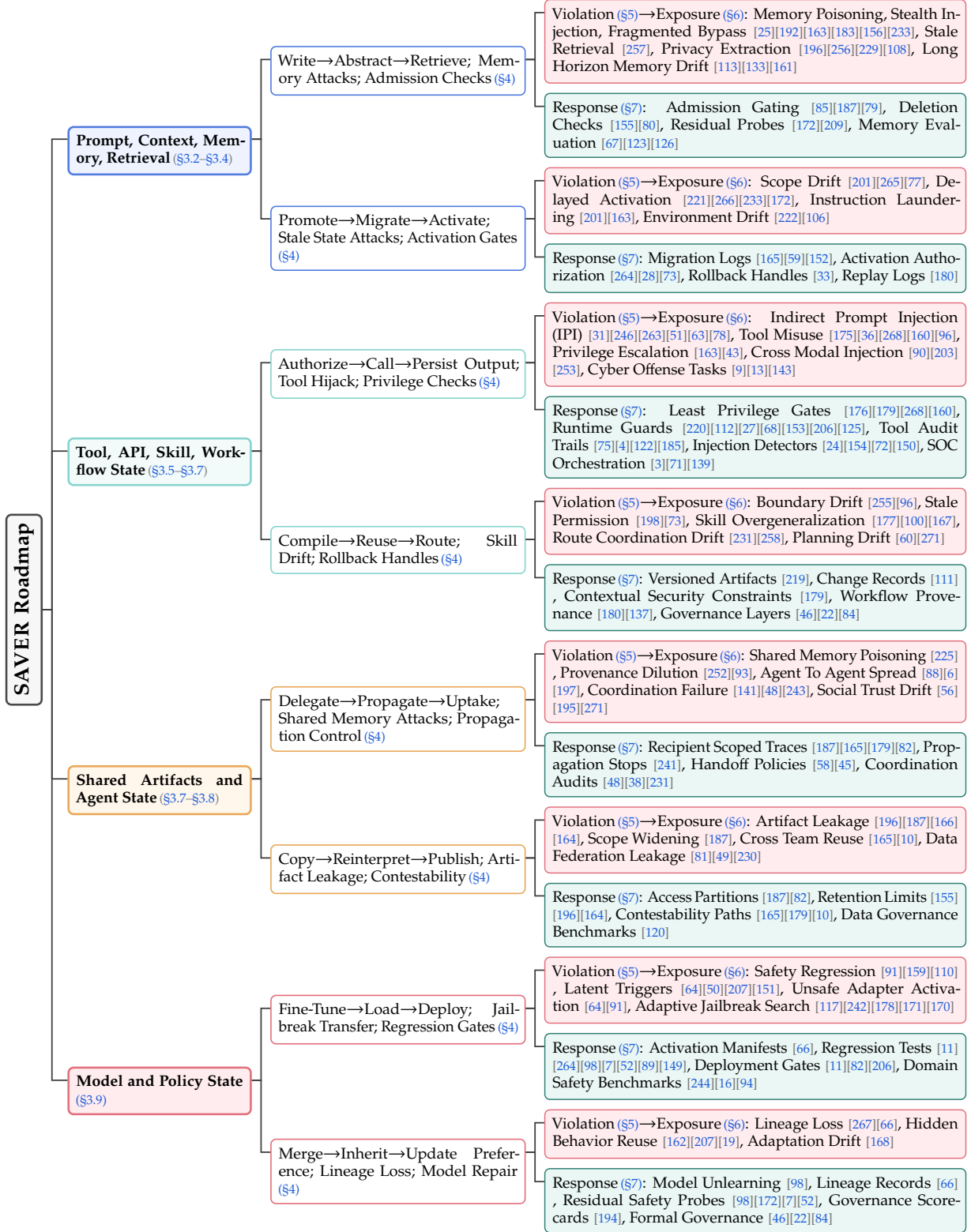
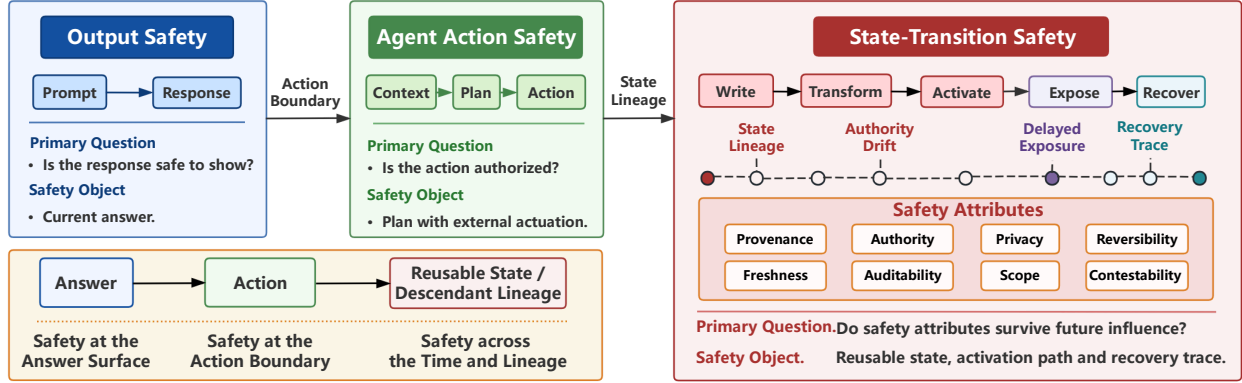


Figure 3 A full-width horizontal SAVER roadmap. Exposure is folded into the incident leaves for legibility.

Figure 4 From output safety to adaptive state safety. The three panels show how the unit of analysis expands from a single response, to action mediated by tools, to a reusable state transition that carries safety attributes across sessions.



2 Adaptive State Safety

2.1 Operational Scenarios and Scope

Self-evolution is useful when an agent must improve across tasks rather than solve each episode from a blank state. A personal assistant can retain preferences and corrections; a workflow agent can compile successful traces into reusable procedures; a team of agents can propagate plans and findings through shared state; and an adaptive deployment can select or load a model-side update for later tasks. Memory, workflow, coordination, and model updates differ technically, but each turns an earlier event into state that can influence a later decision [66, 148, 198]. This continuity is the reason to study self-evolution and the source of its additional safety burden.

State and Reusable Influence. We use *state* for an information-bearing object that remains available, can be reconstructed, or can be reactivated after the event that produced it and can affect later reasoning or action. The object may be a memory entry, retrieved chunk, workflow rule, skill, shared artifact, adapter, cache state, or policy record. *Reusable influence* is the capacity of that object to condition behavior in a later context. The distinction matters because an object can remain stored without current influence, then gain a stronger role when retrieval, migration, loading, or another activation operation brings it back into use.

The scope follows from this mechanism. A *stateful agent* preserves memory, retrieval state, session context, or environment artifacts across turns. An *adaptive agent* updates memory, workflow rules, skills, policies, evaluator state, or model-side components from experience. A *self-evolving agent*, as used here, performs such updates inside the agent loop or runtime, so the update can change later reasoning, planning, tool use, coordination, or model behavior [8, 189]. A retrieval-augmented generation (RAG) system that only retrieves a static corpus is therefore adjacent; a long-term memory assistant is closer; a system that writes, revises, selects, or inherits state that later steers action is central.

Temporal and Spatial Dimensions. The temporal dimension concerns state accepted or created under one context and reused later after one or more operations. The spatial dimension concerns the same influence moving or spreading across carriers, agents, or authority regimes. Spatial here refers to location and boundary crossing inside the agent system, not physical geography. Recent agent benchmarking makes the distinction concrete by separating attacks fragmented across turns from attacks placed in external workspace artifacts beyond conversation-level monitoring [128]. Throughout this survey, *temporal* refers to persistence, ordering, and later activation, while *spatial* refers to movement or replication inside the system. Delayed activation and cross-session reuse are temporal mechanisms; propagation and migration are spatial operations; and *longitudinal* is reserved for evaluation across a sequence of transitions. Safety cannot be established from the source event or final behavior alone; it must follow the intervening state transition. Figure 4 shows this change of unit from response safety, through tool-mediated action, to adaptive state safety.

2.2 Self-Evolving Agent Formulation

The operational definition above can be represented by separating execution, feedback, candidate updates, selection, and commitment. We model the agent at step t as

$$\mathcal{A}_t = (\Theta_t, \mathcal{M}_t, \mathcal{T}_t, \mathcal{W}_t),$$

where Θ_t denotes parameterized model-side state, $\mathcal{M}_t$ memory and context state, $\mathcal{T}_t$ tool and skill resources, and $\mathcal{W}_t$ workflow and multi-agent routing state. These four coordinates are not a second taxonomy. They are a compact way to name where reusable influence can live before an operation changes its role. The loop can be written as

$$\begin{aligned} o_t &= O(\mathcal{E}_t, g_t), & \tau_t &\sim \pi_{\mathcal{A}_t}(\cdot \mid g_t, o_t), \\ f_t &= \mathcal{F}(\tau_t, g_t, \mathcal{E}_t, \mathcal{E}_{t+1}), \\ \Delta_t &= \mathcal{P}_\psi(\mathcal{A}_t, \tau_t, f_t, \mathcal{E}_{t+1}), \\ \delta_t^* &\sim \pi_{\psi,t}^{\text{evo}}(\cdot \mid \Delta_t, \widehat{u}_t, c_t), & \mathcal{A}_{t+1} &= \mathcal{U}_\psi(\mathcal{A}_t, \delta_t^*). \end{aligned}$$

Here $\mathcal{E}_t$ is the external task environment, including user requests, tool-visible resources, files, messages, web pages, and other context outside the agent’s adaptive state. $O(\cdot)$ admits observations, $\pi_{\mathcal{A}_t}$ produces the execution trace τ_t , and $\mathcal{F}(\cdot)$ returns feedback f_t . The proposal mechanism $\mathcal{P}_\psi$ turns the trace and feedback into a candidate update set Δ_t . The evolution policy $\pi_{\psi,t}^{\text{evo}}$ selects δ_t^* , including the option to make no update, using the observed utility proxy $\widehat{u}_t$ and the authorization condition c_t . The update operator $\mathcal{U}_\psi$ then commits, revises, ignores, or quarantines the selected update into $\mathcal{A}_{t+1}$. Keeping these steps separate preserves τ_t as the execution trace and exposes the state-selection decision that the earlier update call left implicit.

Selection Semantics. A high value of $\widehat{u}_t(\delta)$ establishes only that candidate δ scores well under the evaluator available at step t . It does not grant broader authority or show that provenance, scope, privacy, freshness, budget, auditability, and recoverability will survive reuse. The condition $c_t(\delta)$ therefore represents the safety and authorization evidence required before commitment. Passing $c_t(\delta)$ establishes compliance with the controls represented at step t ; those observable controls may still omit some of the user’s or system’s safety requirements. If c_t is incomplete, or if failures and counterexamples never enter Δ_t , repeated selection can favor locally successful updates whose safety assumptions do not transfer. This is a possible endogenous drift mechanism, not a claim that every self-evolving system drifts or that utility and safety are always opposed.

The loop of observation, action, evaluation, and evolution now exposes two safety questions. The local question is whether δ_t^* changes what earlier information is allowed to do: an observation becomes evidence, evidence becomes instruction, instruction becomes workflow policy, or workflow policy becomes tool authority. The longitudinal question is whether repeated choices by $\pi_{\psi,t}^{\text{evo}}$ change the distribution of provenance, scope, authority, and negative evidence in later candidate sets. Safety has to be checked at $\mathcal{A}_t \rightarrow \mathcal{A}_{t+1}$ and again when $\pi_{\mathcal{A}_{t+1}}$ reuses the updated state. Under this model, self-evolving-agent safety is *state-transition safety*: safety attributes must survive both the selection that admits an update and the operations that store, transform, migrate, activate, expose, or repair reusable state. SAVER records the local transition; a sequence of records, together with their selection metadata, reveals whether local choices accumulate into drift.

2.3 Safety Attributes and Transition Test

Transition Test. Because state often moves from environment to prompt, prompt to memory, memory to retrieval index, retrieved item to planner context, planner context to tool argument, tool output to summary, summary to workflow rule, and workflow rule to external artifact, the model needs explicit attributes. Each hop changes persistence, observability, authority, scope, and reversibility. A complete analysis names the hop where the role changed rather than treating the entire incident as one memory, prompt injection, or tool use category.

Activation authority is the right of a state object to influence a later decision or action. It differs from storage permission. A system may store a document, index a chunk, or remember a preference without being allowed to treat that state as instruction, policy, or authorization for tool use. InjecAgent asks whether untrusted environmental content can become active instruction in a tool-using context [246]. AgentPoison asks whether a fabricated memory or knowledge-base entry can shape later planning or tool use [25]. Progent asks whether stored state has the authority to populate tool arguments [176]. In ordinary security language, many such cases are privilege escalation or time-of-check/time-of-use failures: a state item was acceptable for storage or reading, but unsafe when later used as authority for an external effect. The recurring question is what moved, what safety attributes changed, and what later action became newly authorized.

The right itself may also become stale. Zhang and Zhang define *authorization continuity* as whether a live grant remains valid after the agent or its operating context changes [261]. Their model separates a transition envelope, which specifies the mutations an old grant may survive, from an immutable effect ceiling, which bounds what the grant may ever authorize. A mutation can therefore invalidate the grant even when the requested effect remains below the ceiling. In the notation above, continuity belongs to the authorization and environment context represented by c and σ ; it is not another adaptation operation or SAVER field.

The transition test asks whether the operation preserves or explicitly downgrades the safety attributes that downstream components need. If a low-authority observation is summarized and stored, the summary should not carry instruction authority. If a user-specific preference migrates into workflow guidance, the scope restriction should travel. If private state becomes a tool argument, the privacy violation should be visible. Invariant preservation is the property that these conditions survive the transition.

Promotion Test. A selected update is a promotion when it moves episode evidence into a more persistent, general, or action-bearing role. Promotion is safe only when $c_t(\delta_t^*)$ authorizes the new role, the relevant ℓ attributes remain valid in the target context, counterevidence has not been discarded, and the source-to-descendant relation remains recoverable. Task success may justify retaining evidence about one episode. It does not by itself authorize a general workflow rule, reusable skill, or tool-bearing policy.

Role Change Examples. In the running incident, a package proxy permitted for dependency access became a path to the Internet; public services, an external sandbox, dataset configurations, and acquired credentials then carried the campaign across wider trust and authority regimes [87, 146]. The safety object is therefore not one prompt or exploit but the connected set of carriers whose activation enabled the next boundary crossing. AgentDojo supplies a controlled setting for untrusted context reaching tool use, while Progent supplies explicit privilege policies for deciding which tool calls an agent may perform [31, 176]. Persistent agent memory adds the corresponding sensitivity, scope, and deletion questions for retained user details [196]. The incident remains an adjacent infrastructure case, not direct evidence of an endogenous agent-state update; its value is that the public reports expose the role changes and response surface that a transition record must preserve.

Pointer to Formal Notation. A compact formal reading of these transitions writes each hop as $(s, \ell, \sigma) \xrightarrow{a,c} (s', \ell', \sigma')$, where s is the state carrier, ℓ collects safety attributes (provenance, authority, scope, privacy, freshness, budget, auditability, recoverability), σ is the deployment or session context, a is the adaptation operation, and c is the operation's authorization condition. An auditable trace should retain the *authorization context* that explains why c permitted the transition and the *environment context* that identifies the declared task, asset, benchmark, or deployment boundary in σ . These are transition metadata within the existing model, not additional SAVER fields. This notation, the candidate-selection extension, its relation to lattice information flow, language based information flow, and provenance theory, and the *Observation, Evidence, Task Context, Procedure*, and *Action* authority lattice used later in the survey are developed in Appendix B. Body sections Section 3 through Section 7 refer to ℓ attributes and to lattice crossings by name; readers who want the full attribute definitions and lattice figure can consult that appendix at any point.

Takeaway. The reading unit is a selected state transition with safety attributes audited against invariant preservation. Section 2.4 compresses the transition into the SAVER record and keeps its selection metadata as a longitudinal layer; Section 2.5 maps the field before the sections that follow specialize by substrate, operation, violation, exposure, response, and evaluation.

2.4 SAVER as a Transition-Centered Survey Framework

Section 2 defines the outer loop: self-evolving agents observe, act, evaluate, and then update state that can affect later behavior. SAVER is the per-transition trace inside that loop. It asks five questions about one reusable influence: which substrate carries it, which operation changes its role, which safety attribute fails, where the failure becomes visible, and which response can contain or repair it. This vocabulary compares memory security work [109], indirect injection benchmarks [246], agent security benchmarks [251], and broader benchmark surveys [138] without treating them as the same evidence type or claiming that every paper contains a connected lifecycle record.

$$\boxed{S} \times \boxed{A} \rightarrow \boxed{V} \rightarrow \boxed{E} \rightarrow \boxed{R}$$

SAVER builds on established frameworks compositionally. Lattice information flow gives the policy-flow vocabulary, language based information flow shows how those policies can be enforced in programs, provenance explains why source and derivation must survive transformation, and runtime verification supplies the monitoring frame [15, 35, 92, 169]. The additional question for adaptive agents is which transition those tools must observe when state persists, moves, spreads, or activates. The temporal and spatial dimensions annotate that transition: they ask when influence returns and where it moves or replicates, without adding fields to SAVER. Its logic remains $S \times A \rightarrow V \rightarrow E \rightarrow R$: substrate and adaptation jointly specify one event, after which violation, exposure, and response form a directed evidence chain.

Why These Five Fields? Module taxonomies locate mechanisms, while paper-role taxonomies distinguish attacks, defenses, and evaluations. Neither unit alone explains how an object acquires new influence as it crosses modules. SAVER makes the carrier-operation pair the comparison unit. The substrate field establishes what persists and under which authority regime; adaptation records the operation applied to that carrier. Neither field determines risk alone. Their interaction defines the transition whose attribute change is tested as a violation. Exposure is evidence that the violation reached an observable boundary, and response is judged against the carrier and descendants identified by the transition. A record that stops early remains useful, but it cannot support claims about later stages.

Decision Process. A reader applies SAVER as a lifecycle relation, not as a category label. First identify the reusable object and substrate, then pair it with the operation that admitted, transformed, migrated, or activated it. The pair matters because an operation has no fixed safety meaning outside the carrier and authority regime on which it acts. Next test which safety attribute no longer travels through that transition, locate the first surface where the mismatch becomes observable, and ask what response record could stop the immediate harm and support descendant repair. In the running incident, this relation connects the constrained evaluation environment and package proxy to external staging, Hugging Face production exposure, and scoped remediation while preserving one incident identity [87, 146].

Record Structure. A SAVER record is intentionally small: it records the carrier, the operation, the failed attribute, the manifestation surface, and the response hook. The five fields are:

Substrate (S): The carrier: prompt context, memory entry, retrieval chunk, tool binding, workflow rule, shared artifact, adapter, policy state. This field names where the state lives and supplies the first component of the transition pair.

Adaptation (A): The committed state-change operation applied to a carrier, coded as one of the seven families developed in Section 4: *add*, *abstract*, *activate*, *modify*, *forget*, *propagate*, or *migrate*. Implementation verbs such as *write*, *summarize*, *retrieve*, *load*, *inherit*, and *rollback map* to these families by their effect on state rather than forming parallel categories. This field supplies the second component of the transition pair; the selection record explains why that operation was committed.

Violation (V): The safety attribute that fails to travel: provenance loss, privilege or authority escalation, scope widening, privacy breach, availability loss, budget exhaustion, missing audit trail, reversibility failure. This field names what went wrong internally in terms that can still be mapped to ordinary security concepts.

Exposure (E): The surface where the failure becomes visible: tool call, model output, shared artifact, governance audit, deletion request. This field names where the problem surfaces.

Response (R): The mechanism that can contain or repair: admission gate, migration check, activation authorization, runtime guard, descendant rollback, deletion verification, contestability record. This field names what can intervene.

Minimal Record Surface. The resulting record is the narrowest evidence surface used throughout the survey: it keeps the carrier-operation pair, failed attribute, manifestation surface, and response hook together under one transition. In expanded audit form, the same record may retain state identity, safety attributes, authorization and environment context, operation, attribute delta, activation decision, exposure event, response action, descendant inventory, and residual probe. These fields are comparison aids rather than platform prescriptions. They mark the difference between blocking one visible event and showing how reusable influence was contained or repaired. The overview in Figure 2 places the $S \times A$ pair before the $V \rightarrow E \rightarrow R$ evidence chain; selection remains above the diagram because it explains why the transition was chosen rather than adding a sixth field.

Selection Semantics and Longitudinal Reading. SAVER remains a five-field record. Selection semantics sits above it and records why $\pi_{\psi,t}^{\text{evo}}$ committed δ_t^* from Δ_t . An expanded record can therefore add the candidate source, evaluator and objective version, $\hat{u}_t$, evidence used by c_t , rejected or missing counterevidence, promotion target, authority and scope delta, and the consequence fed into the next update round. One record explains what changed. A sequence of such records can test whether successful states are repeatedly promoted, whether negative evidence disappears, and whether authority or scope widens across epochs. This layer does not create a sixth SAVER field because it governs which SAVER transitions occur rather than naming another property of one transition.

Feedback Semantics. A response can contain the current incident and still change the next transition. An action recorded in the **R** field may revise c_{t+1} , change the evaluator or objective behind $\hat{u}_{t+1}$, or supply a policy, guardrail, test, rollback record, or operational-memory update to Δ_{t+1} . EDDOps describes this evidence-to-change process through bounded runtime adjustment and governed redevelopment [219]. The next SAVER record should therefore link response-generated state to the selection decision that later reuses it. EDDOps supports the trace requirement as a process model; whether the loop improves safety remains an empirical question.

Evidence Posture. The three evidence postures separate what a paper directly observes from what the survey rereads or diagnoses as missing. A *direct anchor* observes the relevant transition or surface in its own setup, such as delayed memory activation, memory mediated access control bypass, indirect tool injection, risk recognition, or local runtime containment [163, 192, 220, 244, 246]. A *SAVER rereading* uses a capability, benchmark, or survey paper that exposes a carrier or operation, while the safety claim comes from asking what authority, scope, privacy, budget, or recovery attribute would have to travel with it [8, 148, 198]. A *gap claim* is a negative lifecycle judgment: the inspected work does not carry one incident handle through migration, propagation, descendant rollback, deletion verification, or contestable closure. These postures keep the survey from treating all citations as if they proved the same kind of claim.

Cross-Family Comparison. The same five questions recur across four mechanism families. GhostWriter exposes the write-to-activation interval for hidden memory, while PASB observes the same interval after an agent commits ordinary conversational content; InjecAgent moves the comparison to the point at which external context obtains tool authority [131, 192, 246]. Shared-state work extends lineage across agents and artifacts. Model-side sources expose less inspectable edit objects: Safe LoRA modifies LoRA updates to reduce fine-tuning safety loss, while KVERaser edits key-value cache state to suppress the influence of a processed context span [65, 97]. Deployment lineage and complete rollback are additional governance requirements rather than results of those papers. Privacy controls and guards contribute different response evidence: the former concerns scope and deletion obligations, whereas the latter usually demonstrates local recognition or blocking [196, 220]. The comparison does not require every paper to fill every field. It shows where each mechanism family enters and leaves the incident chain.

Relation to Existing Frameworks. SAVER is complementary to mature safety and governance frameworks rather than a replacement for them. The Open Worldwide Application Security Project (OWASP) LLM Top 10 and the MITRE Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS) provide risk and tactic vocabularies for LLM applications and AI system attacks [136, 147]. The National Institute of Standards and Technology (NIST) Artificial Intelligence Risk Management Framework (AI RMF) and Microsoft AI Risk Assessment provide process ownership, inventories, testing, logging, incident management, and recovery guidance [134, 186]. Information-flow control (IFC), access-control, and integrity models state what authorized flow or privilege preservation means [35, 169], while data provenance theory explains why source and derivation remain important after transformation [15]. SAVER connects these layers without renaming privilege escalation, provenance loss, or time-of-check/time-of-use bugs: it records where they arise after agent state has been stored, transformed, and reactivated. The integration rule is additive. Existing frameworks retain their risk, governance, and policy functions; a SAVER trace is attached when persistent agent state changes role over time.

Boundary of Use. A system can be well governed with ordinary provenance, access control, or prompt injection controls if unsafe influence stays local, and the absence of a full SAVER record does not by itself prove that a system is unsafe; it only limits what the system can claim about diagnosis, recovery, and accountability across multiple hops. SAVER is therefore a bounded trace lens, not a safety prerequisite. The deepest boundary is semantic opacity: some descendants are missed not because hooks were omitted, but because summarization, retrieval abstraction, and latent representations partially dissolve lineage.

Using the Record in Survey Synthesis. The survey first groups papers that share a mechanism and then compares how far each paper follows that mechanism through the five fields. Within persistent memory, for example, GhostWriter observes admission and later activation, privacy work tests unauthorized reuse or disclosure, and PerMemSafe measures personalized safety under retained context. At the action boundary, InjecAgent exposes indirect-injection consequences, privilege control constrains authority, and guards provide local intervention evidence [5, 176, 192, 196, 220, 246]. This order prevents a benchmark, attack, and defense from appearing as an unqualified paper list merely because they share a module. It also allows one paper to cover several fields without implying an end-to-end lifecycle result.

If a source supports persistence but not propagation, the record stops at persistence. If a mechanism blocks a risky tool call but does not repair descendant memories, the response field should say containment rather than recovery. Claim calibration follows the same record: missing activation traces, provenance linkage, descendant inventories, contestation paths, or rollback checks then become evaluation targets rather than repeated caveats.

Takeaway. The five chapters are successive readings of one transition. Substrate identifies the object and authority regime; adaptation explains the change; violation interprets the failed attribute; exposure supplies observable evidence; and response tests whether intervention reaches the affected lineage. Selection semantics explains why particular transitions recur, while feedback semantics records how a response can shape the next one. Section 2.5 applies this sequence to the main mechanism families before the body examines each field in depth.

2.5 Literature Map and Positioning

With the transition unit fixed, the literature can be read as a field map rather than as a list of modules. A work becomes central when it identifies a reusable state object, an operation that changes the object’s later influence, a safety attribute failure or exposure path, or a response artifact. Capability mechanisms alone remain adjacent unless they expose one of these trace elements. Figure 2 provides the reading order for the comparison: locate the carrier-operation pair in the central ring, then ask how far the paper follows the transition through violation, exposure, and response. The positioning paragraphs apply that order without assuming that every paper covers the full chain.

How the Literature Converges. The relevant literatures did not begin from a common taxonomy, but their evidence now meets along the same transition. Prompt-injection research establishes that untrusted context can condition action, and InjecAgent makes the resulting indirect-injection boundary measurable [118, 246]. Memory research adds persistence through managed storage, governed reuse, and user-specific context [109, 131, 196, 260]. Self-evolution work moves the safety question into the update rule because reflection,

experience, or collective state can convert a local observation into reusable policy [201, 225, 265]. Response research then asks where this transition can be recognized, authorized, blocked, or audited [112, 179, 220].

These bodies of work contribute different segments of an evidence chain. Memory and tool studies provide concrete write paths, retrieval triggers, and action boundaries. Privilege and runtime controls identify activation gates and local blocking points. They offer less evidence about what happens after the same influence migrates into workflow rules, shared artifacts, or model-side state. Recovery claims are especially difficult because they must link a response to every relevant descendant and test for later reactivation. The resulting imbalance is not simply a shortage of defense papers; it is a mismatch between event-level evidence and the stronger proof obligations of cross-session repair.

Differences from Existing Surveys. Neighboring surveys differ mainly in what they hold fixed. General and autonomous-agent surveys hold the system architecture fixed and compare memory, planning, tool, and coordination components [26, 218]. Self-evolution surveys hold the capability process fixed and compare where or how adaptation occurs [8, 189]. Trustworthy-agent and security surveys hold a risk or governance property fixed, while benchmark surveys compare evaluation tasks and metrics [1, 34, 138, 240]. Recent trajectory-assurance work instead holds governing standards and system-level invariants fixed, then asks whether composed behavior remains inside that envelope even when each action is locally permitted [121]. Its scope spans the agentic stack, including delegation and model routing; the MLAS analysis is closer to our problem setting because it maps critical threats across self-evolving-agent modules and lifecycle stages [107].

Our comparison holds the reusable influence fixed while its carrier and role change. This specializes the broader trajectory-assurance objective: rather than treating every action sequence as one object, it asks which state transition changed the sequence’s effective provenance, scope, or authority. A module map can locate memory, tools, feedback, and adaptation; a lifecycle threat map can identify exposed cells. A transition record asks how a particular object entered one cell, what attributes it carried, which operation moved or transformed it, where it later activated, and what response artifact addresses its lineage. Papers are therefore compared by the transition segment they observe, with component and paper role retained as secondary descriptors.

Memory and Reusable Context. Memory remains the strongest substrate for evidence because its write, storage, retrieval, and activation points are comparatively visible. Personal-agent and memory surveys define the carrier, while memory-security work explains why persistence requires provenance and policy metadata [40, 105, 109, 260].

Attack studies then separate into mechanisms that a generic “memory poisoning” label would obscure. AgentPoison targets trigger-selected retrieval from an already poisoned store; stealth injection targets the admission path from an external environment; broader poisoning and sleeper-memory work emphasize persistence and delayed activation; MemMorph follows retrieved memory into tool hijacking [25, 156, 183, 257]. These mechanisms require different denominators: poisoned retrieval opportunities, unsafe write opportunities, delayed trigger opportunities, or tool-routing opportunities. Privacy and PerMemSafe add another comparison axis by asking whether retained state remains appropriate for a particular user and purpose [5, 196]. PS-Bench sharpens the non-poisoning branch: truthful memories can remain appropriate as personal context yet still weaken the safety boundary by legitimizing a harmful request [53]. EchoSafe adds a multimodal branch in which an inference-time reflective memory bank adapts contextual safety decisions; it is evidence about memory-mediated safety adaptation, not a general guarantee for textual agent memory [250]. Memory is therefore the clearest substrate anchor, but not the lifecycle boundary: its contents can become prompt context, a plan constraint, a tool argument, a workflow rule, a shared artifact, or a rollback target.

Self-Evolution and Misevolution. Reflection and skill libraries establish the capability mechanism: feedback or successful traces can become strategy and callable procedure [177, 198]. Safety work begins where that promotion changes provenance, authority, scope, privacy, or recoverability. OEP tests harmful transfer from locally useful experience, while SEVerA studies verification before reusable behavior is synthesized [11, 201]. Experience-driven safety and MemEvoBench broaden the evaluation setting from one promoted rule to evolving memories and updates [221, 265].

The Red Queen Gödel Machine adds a different dependency: the evaluator and utility can coevolve with the agent [70]. A benchmark can therefore miss drift even when it records every committed update, because the criterion selecting those updates may also have changed. Read together, this literature links capability, selection, promotion, and evaluation and localizes where a reusable lesson acquires authority.

Tool, Workflow, and Runtime Response. Prompt-injection attacks and benchmarks establish the exposure mechanism: untrusted text becomes planning context, and a tool call makes the consequence observable [31, 118, 246, 247]. Response papers then divide according to whether they observe, constrain, or maintain that boundary.

Risk judges and safety benchmarks improve recognition, while GuardAgent and TrustAgent add local guard reasoning [68, 220, 262]. Recognition becomes an intervention only when Progent, SafeAgent, or LlamaFirewall connects it to privilege or runtime control [112]. Those controls can stop or narrow an action, but they usually do not repair the earlier memory or workflow state that requested it.

Evaluation-driven operations and coordination work extend the same boundary into deployment, while POLARIS, MAGNET, and WebCoach make route and workflow traces more visible [111, 140, 219]. The unresolved connection is between these operational records and the state lineage needed for recovery: local recognition and blocking rarely establish that the responsible influence can no longer reactivate elsewhere.

Model Adaptation. Model updates are central only when they enter the agent loop as state that is selected, loaded, inherited, evaluated, or rolled back. Static model safety [159], PEFT risk [64], reinforcement learning from human feedback (RLHF) backdoors [162], dormant behavior [50], security and privacy surveys [205], and backdoor surveys [267] explain failure modes that reusable updates can introduce. Safe LoRA is a bounded adapter-level mitigation of fine-tuning safety loss [65]; adapter scope, deployment lineage, activation context, and rollback handles are the additional fields needed before treating it as an agent-lifecycle recovery anchor. KVERaser adds a narrower runtime cache branch by studying erasure of processed context influence after prefill [97]. The branch remains a frontier until agent-loop lineage, cache-local residual probes, and deployment rollback records become routine.

Evaluation as Instrumentation. Evaluation matters here when it identifies which segment of a transition is observed. Capability and coordination benchmarks such as LoCoMo [130] can reveal where state is written, retrieved, summarized, shared, or reused, but they become evidence about persistence safety only when safety attributes and activation authority are recorded. AgentDojo [31] is stronger at the exposure boundary; R-Judge [244] adds risk awareness and safety attribute scenarios; PerMemSafe adds implicit personalized safety over long-horizon memory [5]; and ACIArena adds cascading-injection evaluation for multi-agent propagation [6]. SafeAgent supplies runtime containment and bounded recovery traces [112], MemEvoBench supplies memory-adaptation traces [221], and fine-tuning safety supplies model-regression evidence [159]. The missing benchmark unit is still a connected record: what state was admitted, which attributes were assigned, which operation changed it, where it activated, and which response artifact repaired or contained it.

Takeaway. Current evidence is stronger for showing how reusable state becomes influential through write, retrieval, activation, and exposure than for proving that the same influence can be migrated, rolled back, deleted, verified, or contested once descendants and copies multiply. The sections from Section 3 through Section 7 now specialize that judgment by substrate, operation, violation, exposure, and response, while Section 8 asks which benchmark traces would close the remaining recovery gaps.

3 Substrate: Safety-Bearing State Carriers

$$\mathbf{S} \times \mathbf{A} \rightarrow \mathbf{V} \rightarrow \mathbf{E} \rightarrow \mathbf{R}$$

Before the safety of an update can be judged, the analysis must identify what persists and the authority regime in which it can be reused. The same sentence has different consequences when it is transient context, durable memory, a workflow rule, or a tool-bearing procedure. Substrate analysis therefore supplies the carrier half of the $\mathbf{S} \times \mathbf{A}$ transition pair. Broad agent surveys map the relevant components, while self-evolution work adds the update and reuse mechanics [8, 124]. The comparison below asks what each carrier can preserve, how much

authority it can confer, and which operations it enables. Its output is a carrier and authority regime to be paired with the operation families in Section 4. The detailed paper-level evidence map is retained in Appendix Table 7.

3.1 Substrates as Authority Regimes

Same Text, Different Authority. Consider the same sentence: “for this user, choose the fastest option even if it costs more.” As a *web observation*, it is an untrusted environmental claim; as a *scratchpad note*, transient reasoning context; as a *long-term preference memory*, personalization state; as a *retrieval exemplar*, it may steer similar tasks; as a *skill parameter*, a reusable decision rule; as a *workflow policy*, it can route many later tasks; and as a *tool argument template*, it may affect purchases, bookings, or messages. The substrate determines authority.

The lexical content is identical, but the safety object is not: the substrate changes persistence, scope, authority, and actionability [109]. Memory substrates receive the deepest treatment because they offer concrete write and activate evidence, but the safety problem extends to all carriers through which influence persists and propagates. The evidence progresses from mechanism to harm to governance. Memory mechanism work first makes durable adaptive state concrete, and memory security work then explains why such state needs governance metadata [109, 260]. Stealth memory injection, GhostWriter, and AgentPoison show how external email content, hidden personal-agent payloads, or poisoned retrievable entries can become delayed attack surfaces [25, 192, 259]. Memory privacy work adds the sensitivity and scope question once retained user state is useful enough to keep but risky to reuse [196]. Adjacent systems expose distinct operation mechanics: MemGPT moves information between context and managed memory [148]; PASB observes agent-decided commitment of conversational content [131]; Voyager compiles experience into reusable skills [198]; InjecAgent tests untrusted external content influencing tool-mediated action [246]; and multi-agent and self-evolution surveys cover exchange and evolving workflow state [8, 55]. HarnessSafe brings these previously separate mechanics into one cross-carrier benchmark: memory, skills, Tool/MCP state, memory-to-skill transformation, subagent delegation, session summaries, and shared artifacts are bound to the same delayed-risk roles across native harness implementations [254]. It therefore supports the carrier comparison without implying that all seven mechanisms belong to one substrate class.

Relational Attributes. As an illustrative invariant test, a medical preference stored for appointment scheduling raises distinct questions when it is retained, retrieved for an unrelated task, allowed to change tool parameters, or made difficult for the user to inspect and delete. The source literature supplies the local mechanisms: persistent agent memory creates sensitivity and scope risks [196], PerMemSafe makes user-specific context part of the response-safety judgment [5], and Progent constrains tool privileges [176]. The combined contestability test is this survey’s governance synthesis. The safety attributes are therefore relational rather than intrinsic: they depend on provenance and derivation [15], permitted flows [35], intended scope, current substrate, activation condition, and downstream action surface.

3.2 Prompt and Context Substrates

Prompt and context substrates form the activation boundary for the next model call. Prompt-injection work explains why assembled context can carry safety attributes, and AgentDojo moves the same concern into tool-using decision contexts [31, 118]. Unlike memory, this substrate does not need long retention to matter. Its write paths are ordinary context assembly operations, and its controls sit at instruction hierarchy, source marking, relevance gating, prompt-injection filtering, and call-time privilege checks. Signed prompts illustrate one source-authentication route, while design-pattern work adds broader prompt-injection control vocabulary [12, 184].

Instruction privilege escalation separates two failures often collapsed at this boundary. In ordinary prompt injection, attacker content remains at its original model-facing level and competes for influence; in the harness attack, context reconstruction places tool-level content under an actual user or system-effective role [62]. Delegation can create the role change immediately, while persistent goals and scheduled tasks can reintroduce the elevated content later. The context substrate must therefore retain both the content’s origin level and the level assigned at each model call. A role label without that lineage can make the working

model and a permission reviewer agree on a decision that is internally consistent with the reconstructed context but unauthorized by the user.

Prompt substrates carry safety attributes even when the object disappears after one call. ReAct makes the coupling between reasoning and action explicit: the current context describes the world and helps select the next step [234]. AgentDojo turns this property into an evaluation setting for prompt injection in tool-using agents, where untrusted content can enter the same decision context as legitimate task instructions [31]. The state object may disappear after the turn, but while active it can override attention, bias tool selection, or create an apparent instruction conflict.

Recent multi-turn work changes the unit of analysis from one prompt to the accumulated interaction. Trojan Knowledge and One Turn Too Late distribute a harmful objective across locally innocuous queries or turns, while MT-AgentRisk carries the same composition problem into tool-using agents [103, 174, 215]. These results do not show that a larger context window is intrinsically unsafe. They show that a guard which judges each fragment or turn in isolation can accept an interaction whose combined state later becomes sufficient for a harmful response or action. The relevant substrate record must therefore preserve turn order, source, accumulated intent evidence, and the authority granted to the resulting context.

At this boundary, external text is either marked as evidence or allowed to compete with user and system instructions. Losing that distinction changes the substrate from observed content to action-conditioning context. The relevant governance handle is context admission and activation control at the model-call boundary.

Transient Still Matters. Prompt substrates also clarify why “transient” does not mean “safe.” A malicious instruction may never reach long-term memory and still influence an irreversible tool call. Conversely, a sensitive user disclosure held for one task can still shape the current answer. The substrate record distinguishes these cases: transient prompt state has a short retention period, direct activation, and governance at prompt assembly; durable memory state has longer retention, indirect activation, and additional governance around write, retrieval, and deletion.

3.3 Long-Term Memory Substrates

Prompt context makes influence immediate but usually temporary. Long-term memory changes that temporal regime: preference memories, episodic summaries, task histories, user profiles, learned facts, and reflective notes survive the turn that created them and can return through recall, profile injection, semantic retrieval, planner queries, or later summaries. Ordinary extraction or summarization can therefore turn one-turn evidence into reusable state. Provenance, scope, sensitivity, expiry, inspection, deletion, quarantine, and retrieval authorization then become part of the memory mechanism itself. Managed memory systems make that shift operational. MemGPT shows movement between immediate context and longer-lived memory regions, while SimpleMem, Darwinian Memory, Panini, and hybrid episodic procedural agents make across sessions reuse, structured retention, and self-maintenance more explicit [100, 113, 148, 161]. E-mem reconstructs episodic context through multiple memory agents, PISA exposes schema creation and evolution, and Mem-PAL and PERMA make long-horizon personalization and evolving preference state explicit [69, 76, 115, 200]. This capability evidence sets up the safety question rather than settling it: memory security work explains why retained state needs governance metadata, later memory-poisoning work shows why retained state becomes future attack surface, and memory privacy work shows why retained preferences, facts, and summaries need sensitivity and scope control [109, 183, 196]. Attack papers separate several paths into the same delayed surface. Stealth memory injection makes the external-environment write path explicit through email-to-memory compromise, GhostWriter isolates injection and activation phases in tool-using personal agents, FragFuse splits prohibited content across memory interactions before later fusion, and AgentPoison isolates trigger-selected retrieval from poisoned long-term memory or knowledge-base entries [25, 163, 192, 259]. The important transition is therefore not just that bad text appears once, but that external or weakly grounded content is rewritten as durable internal state and later returns as apparently helpful context. This literature also shows why memory governance cannot be reduced to filtering toxic text. A summarized travel preference inferred from one conversation may be accurate enough to store and still too weakly grounded to govern future bookings; a health related memory may be useful for personalization and

still unsafe to retain or expose beyond its intended setting [196]. PASB moves this non-poisoning problem to the write boundary: stateful personal agents decide whether an accepted user claim becomes a durable preference, background fact, or procedure, and a later neutral query tests whether that committed state survives the cleared conversation [131]. AuthMem-Bench isolates the authority field within that commitment. Its paired histories keep the proposition and downstream task fixed while varying only the source that may authorize later use, showing that consolidation can preserve content while erasing whether it may be reused as a user fact, attested observation, or standing instruction [248]. PS-Bench examines the next boundary, where truthful personal memories can be semantically relevant yet bias intent inference so that a harmful query is treated as legitimate [53]. EvoBreak changes the unit again: individually benign experiences can pass local inspection yet become harmful when accumulated as a complementary set and jointly activated [228]. PersistBench isolates cross domain leakage and memory induced sycophancy as long-term memory risks, while RBI-Eval asks whether an available or retrieved memory is warranted in the current response [155, 224]. MemConflict distinguishes static, dynamic, and conditional conflicts, and MRMMIA shows that the memory interface also creates a membership-inference surface [23, 188]. The comparison therefore progresses from claim commitment, to authority preservation during consolidation, to context-conditioned activation, and then to composition over several items. Governance must preserve provenance, authority, and scope at write time, authorize both individual and joint retrieval, and repair summaries, embeddings, cached prompts, and downstream skills rather than only the visible memory item. SSGM, Governed Memory, and GLOVE address pieces of this chain through governed activation, policy-routed recall, and memory–environment realignment [85, 187, 238]; their task designs and metrics remain too heterogeneous for direct ranking.

3.4 Retrieval and Knowledge Substrates

Persistence alone does not make stored state current; retrieval selects which stored object reenters the task. Agentic RAG architectures make this a recurrent loop rather than a one-shot lookup, because planning and tool-mediated retrieval can repeatedly revise what enters context [135]. Once retrieval and knowledge substrates turn information into task support [260], the hard problem is whether source attributes survive the path from ingestion to retrieval [15]. When a chunk, embedding, example, or manual reappears as evidence, the system needs to know whether provenance, scope, and authority remain valid for the current task.

Retrieval differs from long-term memory because the object often enters as a document or chunk rather than as a remembered user fact [260]. AgentPoison targets memory or RAG knowledge bases so that entries that look ordinary later return under chosen triggers [25]. InjecAgent exposes the adjacent action-boundary mechanism: untrusted external or tool content can enter the decision context and induce tool-mediated behavior [246]. Reading retrieved content as evidence or instruction is the authority transformation that this survey adds to connect those settings. Retrieval governance must therefore span more than document storage [109]. AgentDojo shows that untrusted content in a tool-using decision context can induce unsafe action [31]. We model the preceding stale-policy example as a corpus-to-evidence transition; the governance handle runs through corpus admission, retrieval filtering, prompt assembly, and, when needed, tool activation [176].

Chunking can separate a sentence from its source context [260], embeddings can make unrelated tasks neighbors, and rerankers can make a low-authority source look central because it matches the query. These ordinary system operations create ambiguity through abstraction rather than through overt attack. The safety question is whether the system still knows what role the retrieved object is now playing [109]. The evidence base is strongest precisely where memory and retrieval interact. Managed-memory systems explain movement between context and memory, memory-security work gives the broader carrier vocabulary, and FragFuse shows how separately stored fragments can be reconstructed at retrieval [109, 148, 163]. InjecAgent separately shows untrusted external content steering tool-mediated action [246]. SAVER connects these source-local mechanisms as a transition hypothesis from stored state through retrieval to action; profile inspection, index admission, retrieval authorization, prompt assembly, and tool policy enforcement remain distinct governance handles.

3.5 Tool, API, and Interface Substrates

Retrieval can make state relevant; tool and interface substrates can make it consequential by converting an internal decision into an external effect or resource cost. Tool schemas, action sets, credentials, permission scopes, argument templates, return records, execution logs, and environment handles determine whether a remembered or retrieved object becomes a file operation, message, purchase, browser step, shell command, database update, or high-cost API path. Registration, permission grant, schema update, connector installation, argument synthesis, and result caching are therefore safety-bearing writes. At this boundary, the governance burden shifts from relevance and truthfulness to authorization, availability, and audit after action. The tool substrate is where safety attributes often become operational. ReAct’s reasoning and action loop shows that action choice can be integrated into the agent’s reasoning process, and AgentDojo then explains why this integration is critical to safety: prompt-injection analysis in tool-using environments must track whether untrusted instructions can influence later execution through tools [31]. A memory item or retrieved passage may be problematic on its own, but it becomes more consequential when it changes an API call, file operation, message, or purchase. In substrate terms, actionability amplifies safety attributes. Progent provides a useful governance anchor because it focuses on programmable privilege and scope constraints over agent tool use [176]. Tool substrates expose an authorization problem that memory substrates do not solve by themselves. A memory write gate can reject a preference or attach safety attributes to it, but a tool gate still has to ask whether that retrieved or remembered object may influence this particular action. The evidence begins with action-boundary exposure in AgentDojo and indirect-injection settings in InjecAgent, then moves to programmable privilege control in Progent [31, 176, 246]. The frontier shifts from single tools to tool ecosystems: MCPShield treats protocol metadata as safety-bearing state, MCP safety audit work adds the corresponding audit view, and prompt tool selection work makes connector descriptions part of the same state surface [160, 175, 268]. The same sentence that is acceptable as stored preference or background context can become unsafe once it is allowed to populate a tool field or request an external effect. Tool return state also deserves attention. Agents often treat tool outputs as observations, but tool outputs can contain attacker controlled text, stale records, or sensitive data. Once cached, summarized, or fed into later calls, a tool result becomes a state object with its own write path and activation path. The governance handle is different from ordinary user input: a web search result, browser observation, database row, or email body may need source marking and taint attributes before it is allowed to shape memory or tool arguments. The tool substrate is therefore both an action surface and an ingestion surface.

3.6 Skill and Procedure Substrates

A tool call exposes one decision, whereas a skill preserves the procedure that can reproduce such decisions. Code snippets, prompt macros, planner routines, learned policies, recipes for tool use, rules derived from reflection, and partly executable manuals become reusable procedures once a planner can retrieve and enact them across tasks. The literature signal is narrower than in memory security, but the authority change is stronger: a remembered shortcut compiled into callable procedure can reuse stale assumptions, copied permissions, or overbroad heuristics as if they were stable capability. Skill accumulation work makes reusable procedure visible by treating learned skills as capabilities rather than disposable outputs [198]. Two attacks expose different parts of the resulting boundary. PoisonedEvolution targets promotion: bounded attacker trajectories are made recurrent, causally useful, and generalizable so that the victim evolver writes their behavior into a skill [21]. SkillJack targets the derived artifact: extraction obscures malicious intent, and the implanted behavior can persist after source records are removed [239]. ColluSkill changes the unit from one skill to the installed set: individually incomplete skills can jointly reconstruct a harmful workflow through artifact, context, state, and execution dependencies [245]. It does not demonstrate endogenous skill learning, but it shows that reusable procedures cannot be governed only as isolated packages. This differs from memory even when a skill originates in remembered experience: a memory says what happened, whereas a skill says how to act. The migration from reference material to callable procedure, or the addition of a skill that completes an installed capability chain, is therefore an authority change rather than merely another storage event. SkillMisevo-Bench makes the carrier-specific lifecycle observable. It versions native skill stores, isolates each task’s conversation, workspace, process namespace, and tool session, and rebuilds the final persistence probe from exported SKILL.md alone [132]. This design separates an unsafe authored procedure from whether the

target framework later retrieves and enacts it. The attribution remains bounded to the tested agent–method integrations, task concepts, judges, and clean-executor contract. EVOMAL exposes the descendant problem inside the same carrier family. Its planted skill is retrieved only as reference text rather than invoked; the coding agent imitates it into a newly named executable skill, writes that copy back to the library, and can retrieve the agent-authored descendant on later tasks [217]. Trust therefore cannot be assigned only from the current skill’s name or apparent author. The carrier record must retain derivation lineage across changed names, imports, call sites, and library generations. The governance challenge is that skills hide descendants and preconditions. A skill may contain copied memory, embedded tool arguments, derived assumptions, and stale policy; deleting the original memory does not neutralize a procedure that has already absorbed them. A useful substrate record must therefore link source trajectories to skill versions, prompts, workflow nodes, and tool templates, and it must preserve the user, task, and authority conditions under which the procedure may run. Without that inventory, recovery becomes a visible deletion with residual behavioral influence.

3.7 Workflow, Planner, and Topology Substrates

Skills package reusable local procedures; workflow and topology substrates compose those procedures into routes that determine where tasks go, who may act, and how much work may be spent. Planner state, delegation edges, retry policies, approval requirements, and orchestration templates can quietly turn convenience into authority or resource draw. A “routine” label may become an auto-approval path, a local retry rule may become an unbounded workflow loop, and a useful delegation heuristic may forward sensitive context to a more capable but less authorized agent. Correctness checks therefore have to connect to route admission, approval, budget, lineage, and rollback of decisions made under the changed rule. Many self-evolving agents adapt how work is routed as well as what is remembered. Surveys of self-evolving agents ground evolving capabilities and reusable artifacts, while self-evolution surveys broaden the policy and update family map [8]. EvoMap makes the shared-artifact case concrete at network scale: agents publish and reuse instructions and assets, so validation, incentive, and audit decisions travel with the exchanged object [236]. The literature is thinner here than in memory security, but the recurring pattern is already visible: workflow state carries safety attributes when it decides that a class of tasks bypasses review, activates a tool bundle, or shifts execution to a different actor. Traversal-as-Policy gives a response-side example in which sandboxed execution logs are distilled into a gated behavior tree, making a reusable route policy external and checkable before macro execution [99]. Risk-aware debate then makes coordination and escalation visible, agentic operating environments expose route and workflow structure, and formal agent security work supplies the authority vocabulary for distinguishing legitimate coordination from silent expansion [179, 232]. Topology extends the same problem across agents: once state is copied to a scheduler, team agent, or domain agent, the audience and action surface change even if the text does not. The unresolved questions are therefore at the route level: which route changed, which signal promoted it, which later tasks activated it, and which gate can suspend or reverse the resulting authority shift?

HarnessSafe makes this spatial movement measurable without adding a new top-level substrate. Its cross-boundary families hold the delayed-risk semantics fixed while binding subagent, session-summary, and workspace-handoff transitions to each harness’s native mechanisms [254]. The result supports the survey’s distinction between carrier and operation: the route identifies where state resides, while delegation, summarization, or handoff identifies how its influence moves.

3.8 External Artifacts and Shared State Substrates

Once workflow outputs leave the agent runtime, external artifacts carry the transition into files, tickets, repositories, spreadsheets, calendars, emails, documents, dashboards, databases, and shared workspaces. These objects are not internal memories, but they can outlive the agent, trigger automation, and be treated by later agents or humans as settled evidence. Agent surveys supply the broader tool-using and collaborative setting [124], while self-evolving-agent work documents reusable artifacts and evolving capabilities [8]. Treating a later workflow rule, training example, or retrieval target as a descendant of today’s artifact is the lineage requirement introduced by this survey. A sentence that is tentative in a chat can become approval in a ticket, trigger in a workflow rule, or precedent in a later audit. The central shift is from draft text to

organizational evidence. Shared state substrates intensify the same problem because multiple agents can read and write the same object: one worker’s summary becomes another worker’s premise, and one contaminated artifact can synchronize across roles. MAPLE-Guard makes the memory-specific route explicit: a poisoned private memory can be retrieved later, promoted into shared memory, and reused by agents that never observed the original attack, even when no malicious message crosses a visible communication edge at the moment of harm [223]. The safety question is therefore both whether the object is trustworthy and whether its promotion and receiving role are authorized. Provenance, status, scope, promotion, and reviewer fields must survive sharing. External artifacts also make recovery visibly incomplete. Removing a memory entry does not remove a ticket comment, generated document, copied spreadsheet cell, or downstream dashboard. The field increasingly recognizes the need for artifact lineage, status marking, and organizational rollback, but complete cross-system repair is still much less mature than the original write path.

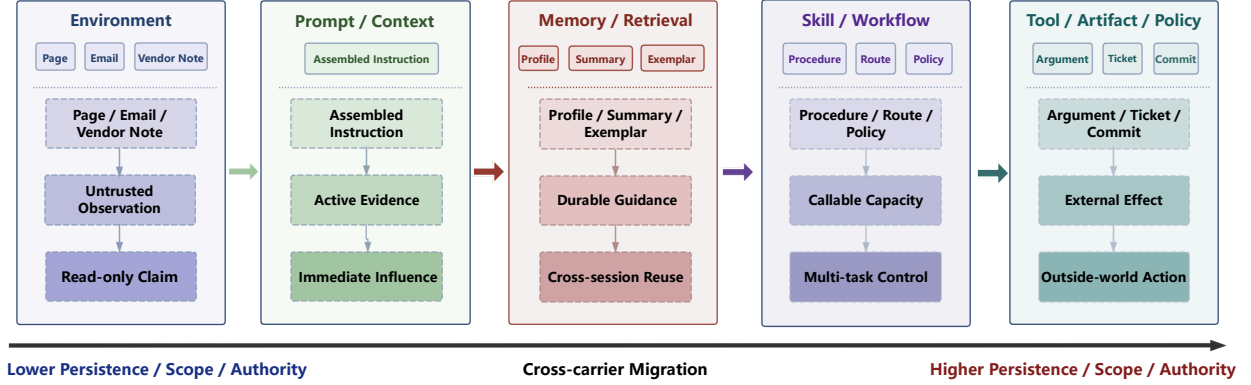
3.9 Model and Policy Substrates

The preceding carriers remain at least partly inspectable as text, records, routes, or artifacts. Model and policy state broaden the same transition while making its contents less directly observable, so the inclusion rule has to stay narrow. Parameters, adapters, policy prompts, safety classifiers, reward models, key-value (KV) cache state, model-editing patches, and unlearning artifacts are central only when they are created, selected, merged, loaded, erased, or rolled back inside an agent loop, or when an inherited update changes later planning, refusal, routing, or tool use. The key question is whether a planner, router, runtime cache, or deployment policy can choose or retain a model-state object under changing context. This keeps the branch in proportion. Static fine-tuning safety asks whether a model or adapter deviates from its intended training scope or evaluation profile. Agent routed adaptation asks the additional questions created by runtime choice: which task requested the update, which user or tool scopes authorize it, what activation log records the load, and what rollback handle still exists if the update proves unsafe. The literature first establishes feasibility and deployment reach through LoRA and PEFT surveys [57, 66]. It then supplies risk evidence: fine-tuning safety work studies safety erosion, PEFT risk work studies adapter-level deployment risk, broader LLM security surveys frame the model-safety layer, and backdoor surveys frame persistent triggerability [64, 159, 205, 267]. Response evidence is narrower: Safe LoRA offers a bounded mitigation anchor, and KVERaser offers a cache-local erasure anchor for context influence already processed during prefill [65, 97]. Adjacent text-to-image work sharpens the same substrate lesson for shared model artifacts by showing triggerable text encoders, LoRA-plugin poisoning, LoRA-layer editing, and prompt-moderation defenses in a related deployment setting [18, 20, 102, 208]. Even with that narrower boundary, governance remains substantive: model state is broader and less observable than memory. A text memory can usually be inspected directly; an adapter or patch requires lineage metadata, safety card summaries, regression probes, and deployment manifests. Work on model state therefore appears here as a frontier substrate, not as an equally mature empirical pillar. The evidence requirements grow in order: fine-tuning safety motivates activation lineage, PEFT risk motivates adapter-scope records, feedback poisoning motivates rollback evidence, and dormant behavior motivates residual activation evidence [159, 162].

Governance Band: Recovery, Deletion, and Control State. The substrate chain does not end when harm becomes visible. Recovery and governance records become reusable state because later admission, retrieval, privilege, and repair decisions depend on them. Audit logs, approval records, quarantine lists, deletion receipts, rollback snapshots, and contestation outcomes are written during response and then consulted by future controls. A system that stores memories but not reliable governance records cannot later show what it admitted, rejected, activated, or removed. This is also where descendant repair becomes concrete. Quarantining a poisoned memory does not automatically neutralize its retrieval embedding, generated skill, copied workflow rule, or shared artifact. If those descendants are not tracked, the system may remove the visible object while leaving its influence active elsewhere. Governance state therefore has to be governed in turn: if approval rules, monitoring rules, or rollback records are stale, bypassed, or silently reused as precedent, the response layer becomes another adaptive state surface rather than an external safeguard.

Comparative Substrate Axes. The substrate classes above differ most clearly in persistence, observability, authority, scope, and reversibility. Prompt context may last one call, memory may last months, skills and

Figure 5 Cross substrate transmutation. The same sentence carries different authority depending on its carrier: from untrusted environment text to prompt context, durable memory, callable skill, and finally external effect.



workflow rules may persist across projects, and model updates may persist across deployments. A memory string may be directly inspectable, whereas an embedding neighborhood or weight update is much harder to read. Scope can remain local to one user or expand to a team or full deployment, while reversibility ranges from prompt expiry to organizational rollback or version redeployment. The farther a substrate moves from directly inspectable text and single activation, the more the literature depends on partial lineage and indirect evidence. Section 4 picks up that comparison from the carrier side and asks which verbs actually move state across those authority regimes. Figure 5 summarizes this role change visually. The same content is not equally safe or unsafe in every carrier. It becomes a different safety object when the substrate turns it into active evidence, durable guidance, callable capability, workflow authority, external effect, or shared artifact.

What the Substrate Comparison Enables. Across substrate families, the same questions recur: what was written, how was it later activated, what changed when it moved, and what remained to repair? GhostWriter exposes the separation between hidden admission and later activation, while PASB adds the agent’s own commitment decision and later memory-poisoning work broadens trigger persistence [131, 183, 192]. InjecAgent exposes an action surface that can be inspected directly [246]. ReAct interleaves reasoning, observations, and actions [234], while Voyager turns experience into reusable procedures [198]. The survey-level asymmetry is that skill, workflow, shared state, model state, and recovery branches still report fewer comparable traces for route change, propagation, adapter activation, and descendant repair.

Takeaway. Substrates separate evidence rich carriers from frontier ones. Memory, prompt, retrieval, and tool substrates report concrete write and activate traces; skill, workflow, shared artifact, model state, and governance substrates visibly carry safety attributes but report fewer comparable traces. Section 4 asks which operations move state across these substrates and which moves change authority.

4 Adaptation: State-Change and Activation Operations

$$S \times \mathbf{A} \rightarrow V \rightarrow E \rightarrow R$$

Operation Premise. The substrate chapter identifies carrier families and their initial authority regimes; this chapter supplies the operation half of each $S \times A$ pair. Throughout the survey, Adaptation is coded using seven operation families: *add*, *abstract*, *activate*, *modify*, *forget*, *propagate*, and *migrate*. Implementation-specific verbs are assigned to these families by the state change they commit, so retrieval, for example, counts as activation only when the retrieved state receives authority over the current decision. The seven families provide a shared vocabulary for feedback reuse [177], memory management [148], skill accumulation [198], reasoning and action coupling [234], and model adaptation [66]. An operation has no fixed safety meaning in isolation: the comparison criterion is whether it preserves or revises the provenance, scope, authority, privacy, and recovery handles of the carrier on which it acts.

Current evidence is uneven across those operations. Stealth memory injection, GhostWriter, and broader memory-poisoning work directly observe unsafe writes and delayed activation [183, 192, 259]. Privacy work examines sensitive reuse, InjecAgent observes activation at an action boundary, and Safe LoRA supplies a bounded model-update response [65, 196, 246]. These observations cover complementary slices under different denominators. Add and activate are relatively well instrumented; propagation and migration carry the spatial dimension but are usually inferred from capability mechanics and lack connected repair evidence. The chapter outputs a carrier-operation pair and its attribute delta, which Section 5 interprets as safe or violating. Appendix Table 8 retains the paper-level evidence coding.

Proxy-Driven Selection. The seven operation families describe what a committed update does; they do not describe how the system chose that update. In the notation of Section 2, each $\delta \in \Delta_t$ can propose an add, abstract, activate, modify, forget, propagate, or migrate operation. The evolution policy ranks candidates through the observable utility proxy $\hat{u}_t$ and checks them against c_t before $\mathcal{U}_\psi$ changes the adaptive state. Proxy-driven selection names the failure-sensitive case in which that ranking favors locally useful candidates because relevant safety evidence is absent from the proxy, the candidate set, or the encoded checks. Selection remains orthogonal to the operation taxonomy: treating propose, evaluate, select, and commit as four more operations would mix the decision process with the state transformation it authorizes.

Constraint-Shaped Adaptation. The authorization condition c_t shapes which proxy-ranked candidates may be committed and which authority or scope they receive. It is not an eighth operation and does not establish that every relevant safety requirement has been encoded. Instead, it makes the currently represented safety boundary part of the selection record, so later analysis can distinguish a candidate that violated an explicit constraint from one that passed an incomplete constraint set.

Local-Success Generalization. Local success becomes safety-relevant when it is promoted into a durable role. Reflexion and Voyager establish the capability mechanics by turning feedback into strategy and successful traces into callable skills [177, 198]. Safety studies then expose three distinct failure modes. OEP consolidates a locally correct but non-transferable experience into a harmful rule; experience-driven safety work finds that accumulated benign-task experience can bias later agents toward acting rather than refusing; and EvoBreak deliberately acquires complementary benign experiences whose joint activation weakens the safety boundary [201, 228, 265]. The progression is from one promoted experience, to aggregate action bias, to adversarial composition over several experiences.

AgentPoison provides a fourth mechanism and clarifies why the corresponding success rates should not be ranked directly. It begins with an adversarial object written into memory or a knowledge base and tests trigger-selected retrieval [25]; OEP begins with locally useful experience and tests transfer after consolidation; the experience-driven study tests distributional change after benign accumulation; and EvoBreak tests whether a dependency set can be assembled and activated. These designs fail admission, selection, generalization, or composition gates under different opportunity sets. A comparable report must therefore identify the source set, operation, activation condition, and downstream scope rather than compare headline attack rates alone.

4.1 Add: Creating Reusable Influence

Add operations create a new state object or append a new version to an existing carrier. Their safety significance is that the object crosses from transient context into reusable influence. Adjacent capability systems show three mechanics: verbal critiques can be stored for later strategy choice [177], memory items can be retained across context boundaries [148], and skills can become reusable procedures [198]. Trajectory-Informed Memory Generation adds structured, provenance-bearing learnings extracted from execution traces, while FORGE turns failed trajectories into reusable natural-language memories that can be broadcast across a population [14, 41]. PASB makes the endogenous write path explicit: the system accepts an ordinary user claim and commits it with status promotion, attribution removal, or broader scope, so the unsafe transition can occur without a poisoned payload [131]. Adversarial paths reach the same operation through external email content saved by stealth memory injection or a hidden GhostWriter payload admitted by a tool-using personal agent [192, 259]. Privacy work adds the case in which useful personal information becomes retained profile state with sensitivity or scope obligations [196].

A feedback note such as “verify account scope before using external tools” is a state transition, not automatically a violation or a defense: the carrier is a stored note, the trigger is task completion, the safety attributes say advisory reminder, the authority says it may guide planning but cannot override current consent, and the rollback handle points to the episode that produced it. If the same operation stores an unconditional action shortcut, the future influence is stronger because the note now looks like a rule. SAVER treats both as add operations but distinguishes them by authority and downstream actionability.

Once a note, memory, skill, or observation is stored, later operations can summarize it, copy it, activate it, or revise it. Memory-poisoning work makes this descendant chain concrete: unsafe content can enter persistent memory or a knowledge base and become relevant only when a later operation reuses it [183]. The add operation is therefore a write event and the moment at which future influence becomes possible.

4.2 Abstract: Compressing State and Reassigning Safety Attributes

Once state has been added, abstraction can compress, summarize, cluster, or reframe it and thereby change its safety semantics [260]. The derived object may no longer carry the exact provenance, uncertainty, user boundary, or temporal scope of the source material. MemGPT makes movement between immediate context and managed memory concrete [148]; reflection can convert critiques into reusable heuristics [177]; and skill learning can convert successful traces into named routines [198]. These are adjacent operation mechanics; the safety claim begins when an abstraction drops source and derivation attributes [15] or privacy and revocation attributes [196] that later matter for retrieval, action, or recovery.

Summarization is not inherently unsafe; the problem is that abstraction can silently change authority [109]. Suppose three notes local to one session say that a user prefers short answers in a specific project. A summary that keeps the source scope intact remains advisory: “for this project, prefer concise replies.” A summary that drops the project boundary becomes a global preference. A summary that drops uncertainty becomes a rule. The content is similar, but the role has been rewritten from scoped evidence to general instruction, and the object’s future influence now spans tasks it was never granted [35].

AuthMem-Bench turns that role rewrite into a controlled abstraction test. Its paired histories hold the focal proposition and later task constant while varying whether the source permits the target use; the write-time module then asks whether consolidation preserves that distinction [248]. This differs from a poisoned-memory benchmark because the claim need not be false or malicious. It also differs from fragment relinking: authority collapse keeps the proposition but loses its use constraint, whereas relinking constructs an actionable instruction from several fragments. The two failures therefore require different post-abstraction checks: one over source authority and one over derived content.

Safe to Check, Unsafe to Use isolates a second abstraction failure at the compression boundary. A pre-compression filter can accept separated fragments because none contains a complete malicious instruction, yet the summarizer can relink them into compact, backend-actionable context [119]. The unsafe object is therefore derived by the abstract operation rather than copied from one source span. This case sharpens the record requirement: a compressed context needs links to its contributing fragments and a post-compression safety decision, because source-level checks do not establish that the derived instruction is safe to use.

If an unsafe or stale source item is later invalidated, its summaries, clusters, and distilled rules may keep the influence alive. We therefore require abstraction to emit a derived state pointer, a retained safety attribute record that includes permitted use, and a reversible relation to its sources [15]. Without that relation, a later forget operation can delete the original item while leaving the compressed influence in place. Abstraction belongs in the operation layer because the operation itself decides which safety attributes survive transformation.

4.3 Activate: Turning Stored State into Current Authority

Added or abstracted state remains latent until activation lets it affect a current reasoning step, tool call, plan, or response. Non-adversarial cases make the authority change especially clear: PS-Bench shows truthful retrieved memories making an unsafe request appear contextually legitimate, while PerMemSafe makes retained user-specific context part of the response-safety judgment [5, 53]. Attack studies reach

the same operation through stealthily injected personal-agent memory, GhostWriter payloads, or broader poisoned-memory reuse [183, 192, 259]. The evidence then extends to tool settings, where InjecAgent makes untrusted external content safety-relevant at a tool-using decision [246]. ReAct and Voyager supply adjacent mechanics for why this matters: reasoning traces can condition actions, and selected skills can become executable routines [198, 234]. Activation is therefore the operation that converts latent influence into current authority. The distinction is operational. When a stored preference is retrieved during a later task, the activation operation must record which item matched, why it matched, what authority it received, and whether the present task permits that authority. The same memory can be acceptable as background context, risky as a tool parameter, and invalid as permission to act. The promotion from stored preference to active parameter is legitimate only when the activation gate grants that role explicitly.

CaMeL and the Dynamic Rule-based Isolation Framework for Trustworthy agentic systems (DRIFT) make this gate concrete at the tool boundary. CaMeL derives control and data flows from the trusted user query, then uses capabilities to prevent untrusted data from changing program flow or authorizing disallowed flows at tool calls [32]. DRIFT constructs a minimal function trajectory and parameter checklist from the query, validates later deviations against user intent and privilege constraints, and masks conflicting instructions before they enter the active memory stream [95]. CaMeL therefore provides a design-time separation of control from untrusted data, whereas DRIFT adds runtime revision checks and current-stream isolation. ControlValve applies the same activation question to a multi-agent orchestrator: it generates a permitted control-flow graph and checks each agent invocation against that graph and a contextual rule [74]. All three treat activation authority as an explicit system decision rather than a property inherited from retrieved text or inter-agent messages.

Activation also creates a temporal bridge between benign adaptation and exposed behavior. In ReAct style loops, observations and tool results are not passive logs; they can shape the next thought and action [234]. In skill libraries, a stored routine does not carry authority to act until selected for execution [198]. In agents based on feedback, a critique becomes future influence when the agent uses it to choose a new strategy [177]. These examples are capability mechanics; the operation record should still separate retrieval from authority: selection says the item is relevant; activation says it may influence this decision under these safety attributes.

4.4 Modify: Revising State, Safety Attributes, and Permissions

Activation can expose stale scope, conflicting evidence, or unsafe authority that requires modification. Changing an existing object or its safety attributes can repair that influence or widen it further. MemGPT and memory mechanism surveys make editable memory concrete, while Reflexion and Voyager show adjacent editable carriers in feedback notes and skills [177, 198, 260]. Beyond Heuristics formulates memory reading, writing, and maintenance as sequential decisions under uncertainty, while EvolveMem revises retrieval configurations with diagnosis, revert-on-regression, and explore-on-stagnation steps [114, 182]. These systems make candidate selection and regression-sensitive modification observable; they do not by themselves establish that safety attributes are preserved. The operation can correct a memory, reconcile conflicting notes, revise a skill, change a workflow rule, or update activation metadata [8]. Memory-privacy work directly motivates preserving sensitivity and scope [196]; provenance retention and rollback metadata are additional design requirements introduced by this survey. Consider a feedback note that says “retry with a shorter plan.” After repeated failures, the agent modifies it to “when a task involves external accounts, ask for confirmation before action.” The role shifts from performance heuristic to safety constraint. Its authority increases over planning decisions but should remain below explicit user instruction and system policy. Its scope also widens: the rule now applies to a class of tasks rather than one episode. A SAVER trace records this as a modify operation rather than a new add because the old object, its descendants, and its activation history still matter.

Conflicts become operational when a memory says that a user prefers automatic reminders and a later interaction says reminders should pause during travel. A reconciliation step does more than choose one sentence. It changes temporal validity, scope, and activation priority. In safety attribute terms, the older memory may be downgraded to historical evidence, while the newer memory becomes active preference for a date range. If this safety attribute update is not explicit, later activation can resurrect stale authority.

4.5 Forget: Invalidating Influence and Descendants

When modification cannot restore a legitimate role, forgetting should withdraw the object’s influence. Removing, expiring, quarantining, or rolling back state is therefore an influence action rather than a storage action [196]. Later memory-poisoning and sleeper-memory work make persistent query-to-memory influence and delayed activation concrete [156, 183]. We infer that copies, summaries, and derived rules must be checked separately rather than assuming removal of the source also removes their influence.

Once retained state has produced summaries, rule edits, cached plans, or invoked skills, this survey treats those derived objects as additional activation sites. The recovery unit is therefore the discovered descendant set rather than only the original record, with memory-security work supplying the broader carrier and threat vocabulary [109].

For benign lifecycle changes, a time sensitive preference may expire, a temporary permission may be revoked, or a stale skill may be retired. In each case the object moves to an expired, revoked, or archived state; its authority over the current action drops to zero; and its future influence should be limited to audit or historical reasoning. SAVER therefore treats forgetting as more than deletion. The operation must answer whether the influence was actually withdrawn from the places where future decisions are made.

4.6 Propagate: Copying Influence Across Actors and Contexts

Before or after activation, propagation can copy state to another agent, memory segment, workflow stage, artifact, or context. Multi-agent surveys explain why shared state and delegated context are routine, while self-evolving-agent surveys place that movement inside a broader adaptive lifecycle [8]. Architectural analysis relates compromised memory to inter-agent exchange, and ACIArena measures cascading injection across trust channels [6, 191]. MAPLE-Guard then localizes the memory route as write, retrieval, promotion, and cross-agent reuse: a poisoned private memory can enter shared memory and influence an agent that never saw the source attack [223]. Propagation therefore changes both the number of activation sites and the receiver’s authority assignment; it cannot be measured only by inspecting visible inter-agent messages.

EVOMAL supplies a skill-to-skill propagation case. A retrieved malicious skill is imitated into an agent-authored descendant, and rolling library replacement lets that descendant become a later template. After the eight planted seeds are removed, Qwen3 reaches 68% ASPR at round five, Gemma4 reaches 29.4%, and Devstral remains infected, whereas DeepSeek-V4-Pro and MiniMax return immediately to baseline and GPT-OSS decays [217]. The result is therefore conditional evidence of self-sustaining propagation under the paper’s five-round replacement protocol, not a universal worm claim. It also makes source deletion an incomplete recovery test: the relevant unit is the retrievable descendant population after the seed disappears.

Recent multi-agent evidence separates propagation from control-flow hijacking. In the former, content or a practice crosses actors and is taken up by another agent; the live-laboratory cases in Agents of Chaos include cross-agent propagation among several other operational failures [173]. In the latter, untrusted content changes the active invocation edge. Triedman et al. show that an orchestrator can be induced to invoke unsafe agents or functions even when the individual agents resist the malicious request [193]. The transferred object is therefore not always a belief or memory item. It can be a routing decision that grants a new component authority. Recording both the content path and the invocation path prevents these two operations from collapsing into a generic prompt-injection label.

If a planner sends a worker the message “the customer may prefer email; verify before sending,” the propagated state remains advisory and scoped. If the same content is copied as “send email to the customer,” propagation has transformed authority even without changing much surface text. The receiver now sees an action instruction. SAVER records the sender, receiver, copied content, retained safety attributes, and permitted activation modes so that later exposure can be traced to the propagation path.

A state item copied into another context can outlive the original context’s governance decision, making propagation a multiplier for recovery. If a memory is later corrected, the correction must propagate too, or the system will carry divergent safety attributes. For multi-step agents, intermediate plans, observations, and

memory updates can pass through several stages before an external action occurs. The operation record should therefore name where state went and which future activation sites now need the updated safety attributes.

4.7 Migrate: Moving State Across Authority Regimes

Propagation multiplies copies; migration changes the regime under which a copy can be used. A migrate operation moves state into a substrate with different persistence, privacy, actionability, or governance semantics. MemGPT makes movement between memory regions visible [148], while self-evolving agent surveys ground broader substrate movement [8]. A feedback note moved into long-term memory becomes more persistent; a memory item converted into a skill or workflow rule gains authority to act; a local observation moved into a shared artifact gains a new audience [198]. Model updates are adjacent rather than identical, but self-evolution surveys establish substrate movement as a major axis of agent adaptation [8].

Context reconstruction is a migration when it changes the authority regime even if the text is unchanged. When Context Gets Root fixes each content item’s origin level and compares it with the level assigned by the harness at a later model call [62]. Tool-to-user and tool-to-system escalation occur when delegation, persistent-task replay, or custom-subagent loading raises that assigned level. Delegation also propagates content to another agent, and scheduled replay adds delayed activation, but the primary violation occurs at the migration step where provenance no longer constrains model-facing privilege.

Model operations are special cases of these primitives with larger activation scope, but they are treated as central only after the agent loop test is satisfied. Fine-tuning and adapter training are add or modify operations over model state; post-training and preference optimization modify reward or policy behavior from feedback; merge combines adapters or patches and may change scope, authority, and safety regression attributes; edit and unlearn modify or remove behavior whose residual influence can be hard to observe. Existing model safety papers show why these operations deserve attention: fine-tuning can erode safety behavior [159], PEFT updates create scoped but deployable risk objects [64], and dormant behavior can survive until the relevant activation context [50]. The minimal acceptance rule for this survey is stricter: a central model row should identify the update object, the activation route, the affected task family, a regression probe, and a rollback handle. In SAVER records, a central model operation should therefore name the training data provenance, adapter scope, deployment lineage, merge status, activation route, safety regression result, and rollback handle.

Two migration patterns matter most here: note $\rightarrow$ rule, where feedback from one episode becomes planning authority, and memory or trajectory $\rightarrow$ skill, where stored experience becomes executable capability. Voyager illustrates the capability mechanic. PoisonedEvolution isolates the preceding migration decision: attacker evidence must be included, attributed as reusable expertise, and realized in the generated skill even though the attacker cannot edit the skill bank [21, 198]. SkillJack then shows what the completed crossing can hide and preserve, because extraction can obscure malicious intent and leave the implanted behavior active after source-record removal [239]. SkillMisevo-Bench extends observation beyond the crossing: its clean reload separates authored skill state, later retrieval, and fresh-session execution, so successful migration and successful activation remain distinct events [132]. The migration gate must therefore check not only content correctness but also whether support is independent, whether the target substrate grants stronger influence, whether source constraints survive extraction, whether later activation is attributable, and whether rollback can still reach the migrated behavior.

Representation And Influence. Reflection, skill learning, and workflow compilation show that related guidance can be carried by a memory item, heuristic, callable skill, or gated policy [177, 198, 201]. Repairing one representation therefore establishes a scoped result: a summary, rule, or other carrier from the same lineage may still influence behavior. The literature establishes these transformation mechanics but does not yet provide a general test for functional equivalence across substrates. We treat cross-substrate reappearance as an evaluation question, using fixed downstream probes and lineage records rather than surface-form matching alone.

4.8 Composition Paths

A feedback note can be stored and later used to revise strategy [177]; managed memory can move information across context boundaries [148]; a behavior trace can become an invoked skill [198]; and an observation can shape the next action in an interleaved reasoning-and-acting trace [234]. The add–abstract–activate compositions used by SAVER are a comparative reading across these mechanics. Unsafe paths can follow the same structure, which is why operation records must carry safety attributes, authority, and future influence fields together.

Operations provide a comparable vocabulary for how reusable influence is created and changed. Add creates reusable influence; abstract changes what safety attributes survive compression; activate grants current authority; modify rewrites safety attributes and permissions; forget withdraws influence from objects and descendants; propagate multiplies activation sites; and migrate moves state into a new authority regime. The capability literature supplies the mechanics through reflection notes, managed memory, reusable skills, and thought-action traces [148, 177, 198, 234]. The safety literature then narrows the claim: memory poisoning anchors persistent unsafe reuse, privacy work anchors sensitive retained state, and prompt injection through tools anchors action-boundary activation [31, 183, 196]. Model update papers become direct anchors only when they document a reusable update that later affects agent behavior [159].

Takeaway. Substrates name *where* reusable influence lives; operations name *what changes it*. Removing one object does not establish influence-level repair when an operation has produced another representation. Add, abstract, activate, modify, and forget have dense local evidence, while propagation and migration are less often instrumented after state has moved. Section 5 asks which safety attribute failed to travel and which gate should have preserved it.

5 Violation: Safety Attribute Failures

$$S \times A \rightarrow \mathbf{V} \rightarrow E \rightarrow R$$

Violation Premise. Given the carrier-operation pair and attribute delta identified above, the next question is whether the resulting role remains legitimate. This is the first directed step after $S \times A$: a violation exists when provenance, user scope, sensitivity, freshness, activation permission, evidence basis, authority, budget, liveness, or an external-effect boundary becomes false, missing, too broad, or unusable. It is therefore the internal side of adaptive-state risk. Exposure is the later moment when this latent mismatch becomes observable.

The evidence enters through two complementary routes. GhostWriter and PASB separate storage from later activation through adversarial admission and ordinary claim commitment, making provenance and persistence failures visible [131, 192]. InjecAgent and programmable privilege control expose the authority question when state reaches a tool-using decision [176, 246]. These works are related by the failed attribute, even though their benchmark denominators differ. Together they produce a named violation and expected trigger; Section 6 uses that pair to identify diagnostic evidence. The detailed paper-level index appears in Appendix Table 9.

5.1 Provenance Loss and Safety Attribute Laundering

Provenance loss occurs when a state carrier loses the record of where it came from, who supplied it, which trust boundary it crossed, and what evidence justified its storage. In information flow terms, this is safety attribute laundering: an object crosses a transformation boundary and later appears with a cleaner or stronger role than its origin warrants. The carrier may be a long-term memory entry, a demonstration retrieved from a retrieval-augmented generation (RAG) store, a summary, a workflow note, a skill precondition, or a tool observation. The operation is usually add, abstract, activate, or migrate: untrusted content is admitted into reusable state, compressed into a shorter object, granted current influence after retrieval, or moved into a substrate with stronger persistence. The safety attribute failure is role opacity: future components cannot tell whether the content is user preference, environmental data, inference from the model, attacker controlled text, or verified policy. AgentPoison grounds the memory and knowledge base version of this problem: malicious demonstrations can be placed in a retrievable store and later influence agent behavior when a trigger query selects them [25]. GhostWriter adds the personal-agent version, where the attack succeeds

because memory subsystems do not apply security-focused governance before hidden payloads become later retrievable state [192]. PASB supplies the non-injection counterpart: status promotion and attribution removal turn a conversational claim into apparently established personal state, while scope broadening lets that state govern contexts beyond the original exchange [131].

A state item enters through a low-authority channel but later appears in a high-authority role because provenance was stripped or flattened during storage, summarization, or memory rewriting. That is the internal mechanism of safety attribute laundering. The exposure trigger is later retrieval or reuse under a task that gives the object current influence; until that trigger fires, the failure is latent. The response hook is provenance stamping at write, source retention during summary, distrust propagation during retrieval, and refusal at activation to let state with unknown source become task authority. These hooks are design requirements rather than proven universal defenses, but they follow directly from state safety attributes that GhostWriter’s memory-store setting [192] and InjecAgent’s indirect-injection setting [246] make critical to safety.

5.2 Authority, Actionability, and Boundary Escalation

Provenance loss becomes operationally dangerous when it enables authority, actionability, or boundary escalation. The carrier may be a memory item, retrieved instruction, plan fragment, tool result, workflow rule, learned skill, feedback summary, or privilege token. The operation is usually *activate*, *migrate*, or *modify*: descriptive state becomes a planning constraint, readable context becomes a tool argument, a local preference becomes a broader rule, or a stored fragment becomes part of an access control bypass. What fails is role preservation across time. AgentDojo gives the integrated-tool setting in which untrusted tool outputs can hijack behavior; Progent supplies the response vocabulary for explicit privilege control; and FragFuse shows the temporal memory variant, where low-authority fragments are stored across interactions and later fused through retrieval so the current query appears cleaner than the activated state actually is [31, 163, 176].

AuthMem-Bench isolates the same family without changing claim content. Matched histories contain the same proposition and lead to the same later task, but only one source condition permits the target use; authority collapse occurs when consolidation makes those histories operationally indistinguishable [248]. This design separates the internal violation from its exposure: the violation is the false authority upgrade during abstraction, while the exposure is a later prohibited tool action after the collapsed memory is activated. Authority is therefore not recoverable from proposition text alone once consolidation has erased the source-use constraint.

Instruction privilege escalation reaches the same invariant through context migration rather than memory consolidation. The harness carries unchanged tool-level content into a new call as an actual user or system-effective instruction, so the downstream model need not confuse roles or violate its visible instruction hierarchy [62]. AuthMem-Bench shows a source-use constraint disappearing during abstraction; When Context Gets Root shows the source level being replaced during reconstruction. Both are safety attribute laundering: the downstream component receives a cleaner authority label than the content’s origin warrants.

Authorization continuity exposes a complementary failure. AuthMem-Bench holds the proposition fixed and varies its source authority; the state-bound model holds the user grant fixed and asks whether changes to control state, capability, delegation, task phase, trust context, or enforcement have changed the subject or context to which that grant applies [261]. The model separates two independent invariants: the mutation must remain inside the grant’s transition envelope, and every realized effect must remain below its fixed ceiling. This is still an authority-preservation problem within the existing family, not a parallel taxonomy of mutation types.

This merged family keeps three previously separate labels under one invariant: lower authority state requires an explicit policy grant before it can become instruction, plan constraint, tool parameter, workflow rule, or boundary extension. A retrieved invoice may be evidence for an answer, while payment authority requires a separate authorization. A webpage may describe a task, while the user’s objective controls instruction authority. A short answer preference may guide formatting, while global workflow policy requires stronger evidence. ReAct, Voyager, and Reflexion explain why this escalation can occur in ordinary capability loops: observations condition later actions, reusable skills condition execution, and feedback can become future strategy [177, 198, 234]. The exposure trigger is action selection, tool invocation, workflow routing, skill

execution, or a later task where the shifted boundary changes a permission decision. The response hook is activation authorization plus periodic safety attribute reconciliation: check user, task, tool, external effect, state provenance, and original scope before granting influence. R-Judge and GuardAgent provide recognition and guard-reasoning signals, while Progent connects those signals to privilege reduction [220, 244].

Safe Promotion Invariant. Promotion is the special case in which δ_t^* turns episode evidence into a more persistent, general, or action-bearing object. The transition is valid only if $c_t(\delta_t^*)$ authorizes that role, the relevant ℓ attributes survive or are explicitly revised, counterevidence remains available, and lineage links the promoted object to its source and descendants. Task success is evidence of local effectiveness, not permission for general reuse. OEP makes single-experience transfer failure concrete, experience-driven safety work shows aggregate benign experience shifting the action-refusal boundary, and EvoBreak shows individually acceptable experiences becoming unsafe under composition [201, 228, 265]. Skill-oriented attacks then expose the cross-substrate step: PoisonedEvolution targets the attribution decision that grants recurring trajectory evidence procedural authority, whereas SkillJack tests the laundering and persistence of the resulting skill [21, 239]. SkillMisevo-Bench resolves the completed path into successive gates: all 21 evolved configurations author unsafe artifacts, 19 retrieve unsafe skills, 19 show benign-task contamination, and 15 retain fresh-session harm [132]. A configuration that stops before final harm is therefore not evidence of a safe promotion; the unsafe artifact may simply have failed to receive later activation authority. Utility-to-authority escalation names the local failure, while selection-induced boundary drift names the longitudinal diagnosis when individually admitted promotions cumulatively widen scope or authority. These mechanisms stay within the existing authority family rather than creating new top-level violation classes.

5.3 Privacy and Purpose Failure

Role widening can concern purpose and audience as well as action authority. Privacy can therefore fail before disclosure when sensitive state loses the attributes that made later use legitimate. The carrier may be explicit personal information in memory, inferred attributes, user-specific preferences, private tool outputs, conversation history, or summaries that merge several users or sessions. Add, abstract, activate, propagate, or migrate can store too much, broaden a profile, grant a sensitive item influence in an unrelated task, share it with another agent, or move it into a less protected artifact. The failed attributes are sensitivity, subject boundary, purpose limitation, retention constraint, or personalized risk context. Memory-privacy work grounds this state-level reading, while PerMemSafe shows how long-horizon user-specific memories can make an otherwise safe response unsafe in that user’s implicit context [5, 196]. Differential Harm Propensity isolates another personalized boundary: mental-health disclosure changes measured harm and refusal behavior, so the evaluator must treat the disclosure as risk-bearing context rather than a neutral profile field [237]. PS-Bench isolates the converse boundary failure: an inherently harmful request can be treated as safe because truthful personal memories bias the model toward a benign intent interpretation [53].

Privacy failure precedes exposure when a memory store retains sensitive state without purpose, expiry, scope, or contestability attributes, even before the information is printed or acted on. The exposure trigger is later retrieval, personalization across sessions, tool argument construction, multi-agent sharing, or generated output that reveals or uses the sensitive state. Personal-agent work identifies persistent user context, devices, services, and interaction history as future-use carriers [105], while memory-privacy work makes the sensitivity risk concrete [196]. Requiring the privacy attribute record at both update and activation time is this survey’s governance proposal.

5.4 Persistent Descendants and Incomplete Rollback

Once a provenance, authority, or privacy failure has been copied, repair has to follow its descendants. Persistent descendant failure occurs when unsafe, obsolete, or contested state remains influential because the system cannot withdraw it from every future activation site. The carrier can be a memory item, embedding index entry, summary, cached plan, skill, workflow rule, preference profile, or derived note. The operation is usually add followed by abstract, modify, propagate, or migrate; the later forget operation is incomplete. In ordinary systems language, this is a time-of-check/time-of-use repair problem: the original object may be corrected or deleted, but a later activation still sees a derived object as valid. Memory-poisoning and

sleeper-memory work make harmful persistence concrete, OEP extends it to consolidated experience, and SkillJack provides a direct cross-substrate case: the abstract reports that 80.0% of skill-mediated attacks persist after the original poisoned records are deleted [156, 183, 201, 239].

Persistence itself is useful when safety attributes remain correct and revocation paths work. The violation arises when persistence outruns governance. A poisoned demonstration may be deleted from one store while its summary remains in a profile. A stale user preference may be corrected in a conversation but left in a workflow template. A temporary permission may expire while a derived skill still assumes it. The exposure trigger is any future activation site that still sees the descendant as valid. The response hook is invalidation over descendants: every abstracted, copied, retrieved, or migrated object should carry a source pointer and expiry or revocation relation. Rollback therefore means influence withdrawal across descendants, with object deletion as one repair action. Memory poisoning work grounds the persistence side [183], while self-evolution surveys ground the broader adaptive lifecycle side [189]; broad claims about complete rollback across deployed adaptive agents would require stronger evidence than the present corpus supports.

5.5 Operational Integrity Failures

Persistent descendants broaden the consequence from semantic error to operational integrity failure. A workflow rule, retry policy, tool template, planner state, retrieved instruction, self-evaluation loop, skill, shared plan, or propagated artifact can affect resource, liveness, auditability, and accountability. Activate, modify, propagate, or migrate may turn a local retry heuristic into a general workflow rule, trigger repeated searches, loop over paid APIs, or leave causal state active after a visible action is blocked. The failed attributes are resource budget, rate limit, cost ceiling, timeout, stop condition, trace link, and repair ownership. OWASP frames excessive agency and unbounded consumption as application risks; AgentDojo moves observation beyond final text; Agent Security Bench and Agent-SafetyBench broaden the evaluation vocabulary [31, 147, 251, 262].

Resource harms and audit gaps belong together because both block reliable operation after state becomes influential. A poisoned memory can cause a wrong action, repeated paid calls, or an incident that cannot be attributed to a specific memory transition. A workflow rule can preserve task utility while multiplying cost. A risk-awareness benchmark can identify danger while leaving the causal state transition unnamed. Existing work covers different parts of that failure chain: personal-agent work supplies the device and user-context setting, GuardAgent and R-Judge supply guard-reasoning and safety-awareness signals, Progent supplies privilege control, and EDDOps adds the lifecycle monitoring and maintenance frame [105, 176, 219, 220, 244]. The response hook is a combined operation record: bind each state item or workflow edge to budget, liveness, source object, safety attributes before and after transformation, activation decision, tool authority, descendants, and repair owner. The current corpus supports the need for this attribute, while direct adversarial benchmarks for economic DoS, resource exhaustion repair, and lifecycle auditability remain thinner than tool injection and memory poisoning evidence.

5.6 Model Safety Regression

The same attribute logic extends from explicit state to model-side state, but the carrier is harder to inspect and often broader in scope. Model safety regression enters this survey when an update to parameters, adapters, reward models, edited components, policy prompts, or unlearning artifacts weakens an attribute that a later agent relies on. The operation is central when the agent loop or deployment controller fine-tunes, trains adapters, post-trains, merges, edits, distills, unlearns, selects, loads, or rolls back that object for later behavior. The failure can be alignment erosion, adapter authority escalation, reward drift, persistent backdoors, incomplete unlearning, or lost deployment lineage. Fine-tuning safety, PEFT risk, feedback poisoning, and dormant behavior mark successive parts of this path [50, 64, 159, 162]. They support central SAVER claims only when the carrier, activation route, metric, and agent-loop scope are explicit.

An adapter that looks benign may gain broader authority when loaded into a production agent. A task-local fine-tuning update may change refusal behavior outside the intended scope. A merged adapter stack may erase the safety assumptions under which each component was evaluated. An unlearning patch may remove a visible behavior while leaving residual influence under paraphrase or downstream fine-tuning. The same

transition logic therefore applies to the model substrate. The exposure trigger is a later inference, tool use decision, refusal boundary, policy check, or generalization failure across tasks. Safe LoRA provides a bounded LoRA-weight intervention for reducing fine-tuning safety loss [65]; broad LLM security surveys catalogue model-level security and privacy risks [205], and backdoor surveys frame triggerable residual risk [267]. The adapter registry, deployment lineage, version rollback, and residual probes proposed here are governance fields needed to connect those local results to an agent loop.

Takeaway. These violations share one structure: a state carrier crosses an operation, safety attributes that should have been preserved become missing or too broad, a later trigger can expose the failure, and a response hook must act before or during activation, with response after harm as a fallback. Violations are therefore *latent*: they exist before any user sees a bad output and may sit dormant across many transitions. The six families above name what has gone wrong inside the state ledger, independent of where the failure later manifests. Section 6 asks where they surface; Section 7 asks what governance trace can repair them.

6 Exposure: Surfaces and Manifestations

$$S \times A \rightarrow V \rightarrow \mathbf{E} \rightarrow R$$

Exposure Premise. Once a failed attribute and expected trigger are known, the analysis turns to where the latent mismatch first becomes observable and what record connects that event to the earlier carrier and operation. Exposure is diagnostic evidence, not a second violation taxonomy. The distinction prevents latent unsafe state from being equated with realized harm and prevents visible harm from being attributed only to the final output step.

The recurring bridge cases are memory bypass, prompt injection, economic denial of service, model update, multi-agent uptake, persistent privacy, and boundary drift. Each starts from an internal safety attribute failure, but becomes measurable at a different boundary: delayed retrieval, fused query, tool execution, budget consuming loop, uptake across agents, stale personalization, or mismatched governance signal.

Surface Reading. For each surface, the relevant question is what a monitor, evaluator, or incident record can observe. A useful record names the trigger, visible symptom, activated state object, and response handoff; the failed attribute remains the responsibility of Section 5. The chapter outputs this diagnostic tuple to Section 7, where the implicated object and lineage determine the scope of intervention. Appendix Table 10 retains the detailed evidence-role index.

6.1 Memory Retrieval Exposure

Memory retrieval exposure is the diagnostic surface for latent failures over stored or indexed state. The visible event is a later task selecting a memory entry, retrieved demonstration, profile note, summary, or derived preference and granting it influence. Direct attacks reach that event through different paths. FragFuse stores fragments whose prohibited meaning emerges only after fusion; stealth memory injection and GhostWriter target external-content admission before later activation; AgentPoison poisons retrievable examples selected by a trigger [25, 163, 192, 259]. Their common surface is delayed retrieval, but their causal operations and suitable controls differ.

Adjacent work broadens the diagnostic object. MemoryGraft studies poisoned experience retrieval, OEP studies harmful transfer from a consolidated local experience, and Tool-Drift follows remembered state into routing [30, 181, 201]. GLOVE examines disagreement between memory and environment, whereas MemPot treats extraction as the relevant defensive surface [210, 238]. Privacy work adds retrieval that is semantically correct but inappropriate for the current user or purpose [196]. PASB clears the originating conversation before a neutral query, which makes later sycophantic behavior evidence of durable-state activation rather than residual dialogue context [131]. PS-Bench adds a different exposure: benign memory retrieval changes inferred intent and the visible symptom is unsafe compliance rather than a corrupted memory record [53]. These studies should be compared by the state selected and the authority it receives, not by treating every later-memory failure as the same attack.

A memory can sit inert until a query, user profile, workflow stage, or tool planning step matches it and activates it. The exposure trigger is that later match, not the original write. At that point the manifestation may look like a wrong answer, an unsafe recommendation, a personalized disclosure, or a tool parameter that seems to come from the agent’s own reasoning. Memory retrieval exposure is therefore easy to misdiagnose as model failure. The visible behavior occurs at generation time, but the root cause may be an earlier state update whose safety attributes were not preserved. A retrieval decision should therefore filter or downgrade memories by provenance, sensitivity, expiry, and allowed use before they enter current context; a separate activation decision should determine whether the retrieved item may act as evidence, preference, plan constraint, or tool argument. Retrieval alone should not imply authority.

Retrieval Record. Managed memory supports context across interactions [148], while feedback reuse creates another recurrent-state mechanism [177]. Carrying safety attributes and checking activation scope are additional governance requirements introduced by this survey. PerMemSafe adds a personalized safety variant: the evaluator must know which remembered user-specific context made the later response safe or unsafe [5]. Because retrieval mediates authority, the diagnostic record must capture the retrieval decision and the prompt assembly that follows it.

6.2 Tool and Action Exposure

Retrieval becomes externally consequential at the tool and action boundary, where authority, actionability, and resource failures produce an API call, file write, email, purchase, database update, code execution, skill invocation, or repeated high-cost path. Lower-authority state now reaches a tool, skill, or API surface where it can change the world. ReAct gives the general mechanism in which observations and reasoning steps condition actions, while early prompt-injection work explains how external content can pressure that path [118, 234]. AgentDojo provides the clearest direct anchor for untrusted data reaching integrated agents, and InjecAgent extends the pressure to benchmarked agent settings [31, 246]. AuthMem-Bench supplies a controlled bridge from the earlier memory violation to this surface. Its action module keeps the claim, tool schema, and target argument fixed while changing whether the stored representation retains source authority; prohibited tool execution then measures the external consequence of an authority-collapsed memory [248]. The benchmark therefore localizes **V** at consolidation and **E** at action rather than treating the final tool call as an unexplained model failure. HarnessSafe generalizes this evidence rule across persistent carriers. A run reaches its violation stage only when a case-specific oracle confirms an executed behavior, committed state change, propagated artifact, or external effect; suspicious text, unsafe planning, or stated intent is insufficient [254]. Exposure is thereby tied to both the carrier path and observable consequence, while the furthest preceding trace stage shows whether the chain failed before or at the action boundary. Newer work expands this surface along three axes rather than forming one leaderboard. Adaptive attacks and LivePI test whether defenses survive changing or live interaction patterns, while SnapGuard isolates screenshot-mediated injection in web agents [39, 247, 263]. Task Shield and SafeSearch focus on task alignment in action or search settings; Tool-Drift and prompt tool selection examine how remembered state or connector descriptions alter routing [30, 36, 75, 175]. The shared outcome is an action-boundary event, but each line of work perturbs a different upstream object. The response side has a different role. Signed-Prompt makes source authenticity explicit, design pattern work frames defense composition, LlamaFirewall anchors runtime guardrail placement, Progent anchors source authorization and privilege scope, and formal agent security work frames data isolation and authority boundaries [12, 27, 176, 179, 184]. AGENTVIGIL is retained as a red-team expansion entry, while IPIGuard is retained as a dependency-structure expansion entry; neither is treated here as precise evidence for defense effectiveness [4, 212].

The safety attribute failure is usually authority escalation, actionability escalation, or resource boundary loss. A webpage, email, retrieved document, or memory item is treated not just as information but as a command or parameter. The exposure trigger is the point at which the agent selects a tool, fills an argument, invokes a skill, commits an external effect, or opens a repeated high-cost call path. This trigger can occur without a dangerous final answer. The user may never see the injected instruction, and the model may never print the unsafe state, but the action can still expose the violation through an irreversible action, a denial-of-service loop, or economic waste. Authorization before execution should bind each action to current user intent,

allowed tools, argument provenance, external effect, privilege state, and budget envelope; block, downgrade, or request confirmation when lower-authority state tries to become action authority or consume resources.

Risk awareness benchmarks can identify situations relevant to safety [244], and guard agents can reason about local responses [220], but the tool/action surface is governed only if the signal changes execution: remove suspect state, reduce privileges, force a clean plan, route to human approval, or record the failed authority transition for later repair. Guard signals alone are not enough.

Context-Horizon Exposure. Multi-turn attacks make the accumulated interaction, rather than the latest user message, the diagnostic object. Trojan Knowledge composes knowledge gathered through benign subqueries, while One Turn Too Late identifies the earliest response whose delivery closes a hidden harmful intent [174, 215]. MT-AgentRisk carries this decomposition into tool use [103]. Relinking reaches a comparable closure through summarization rather than dialogue order: distributed fragments become actionable only after compression [119]. Across these mechanisms, exposure begins when the combined state first supports a harmful response or external action. A turn-local refusal score cannot localize that point without the interaction prefix, any compression step, the candidate response, and the tool decision that follows.

6.3 External Artifact Exposure

Some outputs of tools and workflows persist beyond the episode, shifting exposure to an external artifact. A generated document, code patch, configuration file, database row, ticket, report, memory export, workflow template, shared prompt, or skill entry can later influence users, agents, or workflows. Self-evolution and multi-agent work establish why durable artifacts and shared environments belong in scope, while MemGPT and Voyager supply adjacent mechanics for memory movement and reusable skills [8, 148, 198]. Across these forms, scoped or descriptive information becomes durable evidence or procedure in another context.

A human reads the generated document, another agent retrieves it, a workflow uses the template, a developer executes the code, or a future run imports the artifact as trusted context. Artifact consumption is the exposure trigger, so the harm may occur after the original episode has ended. A stale summary can become a project rule. A private detail can become part of a report. A generated skill can encode an unsafe shortcut. A workflow template can preserve a boundary drift that later users never see as a choice. This surface is especially important for self-evolving agents because evolution often works by accumulating artifacts: reflections, skills, memories, notes, plans, and reusable procedures.

Release control for external artifacts starts with safety attributes rather than with final output screening. External artifacts should carry provenance, sensitivity, intended audience, allowed reuse, expiry, and derivation links. When the artifact can authorize action, it also needs an execution precondition and revocation path. When it is imported back into memory or a workflow, the original safety attributes should travel with it rather than being erased by the file boundary. Reflection systems provide reusable feedback mechanics [177], skill-library systems provide reusable procedure mechanics [198], and the self-evolution survey catalogues iterative capability-update settings [189]. Treating the resulting artifacts as governed, provenance-bearing objects is our extension. SAVER turns artifact-specific attack prevalence and complete artifact governance into explicit governance gaps rather than leaving them as implicit assumptions.

6.4 Propagation Across Agents

When state crosses actors, the visible event is one agent’s unsafe state becoming another agent’s context, belief, instruction, or action precondition. Self-evolution and trustworthy-agent surveys define why reusable shared state belongs in the safety setting; multi-agent surveys and coordination benchmarks identify messages, delegated tasks, shared memories, plans, and summaries as transfer channels [8, 240].

ACIArena turns those architectural channels into a cascading-injection benchmark over external inputs, profiles, and inter-agent messages [6]. Agents of Chaos supplies a deployment-oriented counterpart: its two-week live-laboratory study reports eleven representative cases that include cross-agent propagation of unsafe practices, information disclosure, destructive actions, identity spoofing, and partial takeover [173].

These cases establish that the transfer channels matter under realistic combinations of persistent memory, communication accounts, files, and shell access, but they are case evidence rather than prevalence estimates.

Control-flow hijacking exposes a different part of the same multi-agent boundary. Triedman et al. show that adversarial external content can redirect orchestration and communication toward unsafe agents or functions. With GPT-4o agents, the evaluated Web attacks achieve arbitrary-code-execution rates of 58–90% across orchestrators, and some model-orchestrator configurations reach 100% [193]. The attack can succeed even when individual agents resist direct and indirect prompt injection or refuse the harmful action. Thus, receiver compliance is not the only failure condition: the orchestrator can assemble a harmful execution from components whose local behavior appears safe.

The literature now exposes three related but distinct multi-agent attack forms. Message injection gives untrusted content instruction status; state or practice propagation changes what later agents retain or repeat; control-flow hijacking changes which agent or function receives authority. Embedding-defense and MAS-security work locate weak shared-memory and message boundaries, whereas MADRA, Devil’s Moltbook, and collective misremembering examine uptake into plans, repeated exposure, or consensus [144, 197, 199, 225, 252]. A complete exposure record must preserve both content lineage and invocation lineage: source agent, recipient authority, selected edge, changed safety attributes, activation condition, and repair path.

Uptake occurs when a planner sends a worker a contaminated instruction, a summarizer converts a disputed memory into a shared fact, a debate participant repeats a wrong claim that sounds authoritative, or a coordinator treats another agent’s uncertain output as verified. Once that happens, the original safety attribute failure can multiply: more agents can activate the state, more artifacts can derive from it, and recovery must locate every copy or descendant. The visible manifestation can be wrong consensus, repeated leakage, inconsistent authority decisions, or a group plan that no single agent would have produced alone. This surface therefore names the quantities a serious evaluation must recover: propagation path, recipient authority, uncertainty retention, source attribution, and repair completeness.

At the transfer boundary, messages and shared artifacts should carry sender, source state, confidence, sensitivity, allowed recipients, and activation permissions. Receivers should not automatically upgrade received content to local authority. Shared memories should support quarantine, correction propagation, and descendant repair. A multi-agent monitor should ask whether an individual output is safe and whether a safety attribute state item has crossed a role boundary or created inconsistent authority across agents. The requirement remains architectural, with limited validation across the field.

6.5 User-Facing Misattribution

After state has crossed retrieval, action, artifact, or agent boundaries, the visible failure can be attributed to the wrong causal layer. The user sees a generated answer, rationale, refusal, recommendation, tool result, or explanation, while the agent has composed that surface from model priors, retrieved memory, task context, tool outputs, and policy state. A LangGraph plus ChromaDB study makes the memory-side attribution problem explicit by showing poisoned shared memory mistaken for model misalignment [2]. InjecAgent supplies the action-side counterpart: untrusted external content can alter tool-mediated behavior [246]. Misattribution arises when the incident trace reports only the final answer or action rather than the state and source that shaped it.

A user asks why the agent acted, and the system provides a fluent answer that does not name the activated memory, retrieved document, tool output, or privilege decision. Explanation without trace is the exposure trigger. The harm includes misdirected incident response, not just epistemic confusion. Operators may retrain a model when they need to quarantine memory. They may strengthen an output filter when they need a retrieval gate. They may blame a user instruction when the decisive input was untrusted tool data. Causal attribution requires every answer shown to a user that depends on adaptive state to expose, at least to an auditor, which state carriers were activated and what authority they received. Public explanations can preserve privacy and stay summarized, but internal incident traces need enough detail to support repair.

A user cannot correct a stale preference if the agent presents the behavior as a general model tendency. An operator cannot revoke a poisoned descendant if the trace stops at the final output. A safety team cannot compare guard failures if the system records only blocked or allowed, not why the state was allowed to influence the decision. Misattribution is therefore also a contestability problem. Risk awareness work can support diagnosis signals [244], and guard agent work can support local response reasoning [220], but accountability to users requires that these signals connect to state traces and repair handles. SAVER’s contribution is to make the causal chain auditable rather than treating misattribution as only a user interface problem.

6.6 Governance Blind Spots

Misattribution becomes a governance blind spot when a safety mechanism observes the wrong layer, acts at the wrong time, or cannot repair the causal state. The relevant carriers are now monitor signals, guard decisions, privilege policies, audit logs, and recovery records. R-Judge supplies safety recognition over trajectories, and TrustAgent performs trust-aware planning [68, 244]; AgentDojo supplies adversarial tool-use episodes at the exposure boundary [31]. GuardAgent and LlamaFirewall move from recognition toward runtime intervention, Progent constrains privilege at tool boundaries, and EDDOps asks whether these controls remain connected to deployment practice [27, 176, 219, 220]. The blind spot persists when a correct signal stops at the visible event and leaves the causal state active.

A benchmark records that an agent recognizes risk but the agent still lacks a rollback path. A guard blocks one response but the poisoned memory remains retrievable. A privilege policy prevents a high-risk API call but does not invalidate the workflow rule that requested it. An output filter catches a leaked string but does not update the privacy attributes on the underlying memory. The exposure trigger is a mismatch between signal and control: governance sees an event, not the adaptive state chain. Effective monitoring has to connect signals back to state operations: quarantine the activated item, update safety attributes, revoke descendants, reduce privileges, require human review, and emit an audit record that future gates can read. These hooks define the lifecycle response target. Existing guard, benchmark, and privilege control papers support local mechanisms and evaluation signals, while complete longitudinal governance remains a major open governance problem. A fuller response would need update admission, migration gates, activation authorization, containment, rollback completeness, deletion verification, contestability, and multi-agent repair.

6.7 Model Deployment Exposure

The same diagnostic problem becomes broader for model-side state because one loaded update can affect many tasks. Exposure occurs when an adapted model, adapter, policy component, reward model, edited checkpoint, unlearning patch, or safety classifier changes behavior after entering the inference or routing path. Fine-tuning safety identifies erosion after tuning; PEFT, feedback-poisoning, and dormant-behavior work add deployable update objects and delayed triggers [50, 159, 162]. Text-encoder and plugin attacks show related compromise of shared model artifacts [20, 102]. Stronger agent-loop comparisons still require direct activation and deployment-lineage evidence.

A memory item may expose only when a matching task retrieves it; a deployed adapter can expose whenever a compatible prompt, tool route, or downstream task hits the changed behavior. The model-side trigger is often broad rather than local. The manifestation can be a global shift in refusal boundary, unsafe compliance across tasks, persistent over refusal, systematic bias in tool selection, or policy activation across deployment. This makes misattribution especially likely: the visible behavior may look like an ordinary model limitation unless the system records which adapter, checkpoint, policy prompt, safety classifier, and reward model were active. Deployment governance therefore needs an adapter registry, model card or safety card, training data lineage, model difference review, canary safety evaluation, task scope restrictions, version rollback, and residual influence tests. The response literature again splits by layer: Safe LoRA and LoRASHield intervene at the LoRA or adaptation layer, AEIOU shows prompt-moderation-style intervention in a related generative setting, and broader security and backdoor surveys supply protocol and benchmark vocabulary for persistent triggerable behavior [18, 65, 208, 267].

Across these surfaces, memory work most clearly measures delayed activation after a write; tool and prompt injection work most clearly measures the external effect boundary where information becomes action; model update work measures broader behavior shifts after an update is deployed into the agent loop; and multi-agent work exposes propagation and uptake, but still lacks mature evidence for descendant repair. User-facing misattribution and governance blind spots explain why exposure diagnosis must include causal state traces rather than final outputs alone.

Takeaway. Exposure surfaces are where latent violations become measurable events. They are not additional failure classes: the same violation can surface through many surfaces, and the same surface can reveal many violations. Tool and memory surfaces are best instrumented; multi-agent propagation, model exposure across deployments, and user-facing misattribution remain less mature. The diagnostic output is the visible event, activated object, failed attribute, and known lineage. Section 7 asks how an intervention uses that tuple to contain the event and repair the responsible state.

7 Response: Governance, Monitoring, and Recovery

$$S \times A \rightarrow V \rightarrow E \rightarrow \mathbf{R}$$

Response Premise. Exposure analysis produces a diagnostic tuple: the visible event, activated object, failed attribute, and known lineage. Response quality depends on how much of that tuple the intervention addresses. A guard may block the current event; a recovery claim must also show that the implicated reusable state and reachable descendants no longer reproduce the same influence. GhostWriter’s separated injection and activation phases and PASB’s cleared-session reuse make the temporal interval concrete, privacy work adds purpose and deletion obligations, and InjecAgent identifies the action boundary at which local containment becomes measurable [131, 192, 196, 246].

Pipeline Reading. Response mechanisms separate by the object they control and by when they intervene. IPIGuard and SEVerA inspect dependency or synthesis conditions before reusable behavior is committed [4, 11]. Memory-oriented controls act on writes, retrieval, session risk, environmental mismatch, or policy-aware recall; SRM and AM-Sentry focus on admission and retrieval, whereas GLOVE, Governed Memory, and MemGovern extend the control object toward environment consistency and policy-governed memory use [187, 192, 238]. SSGM and formal agent security move closer to constraints over stored-state use and action authority [85, 179]. These mechanisms are complementary because success at one intervention point does not establish closure at the next.

The resulting pipeline asks why an update was selected, whether admission and migration preserved its attributes, what authority it received at activation, whether a local guard contained the visible event, and whether recovery reached descendants. The early stages have concrete memory, verification, privilege, and runtime mechanisms. Rollback, deletion verification, and contestable closure remain less mature because they require an incident identity and denominator that survive across stores and sessions.

Recent runtime protection and guardrail papers support local response metrics without yet addressing lifecycle recovery. Appendix Table 11 indexes representative response works by stage, role, controlled object, and control focus.

Recovery Chain. After the first harmful behavior is observed, the field can already replay bounded admission, activation, and even misdiagnosis, but it still rarely carries one incident id through descendant inventory, rollback completeness, deletion verification, and contestability for users. The reason is partly timing, since response research is newer, and partly the proof burden: repair across sessions needs evidence that the same influence no longer survives through summaries, embeddings, workflow rules, shared artifacts, or model updates. Inventory comes first because rollback cannot be complete over an unknown set; rollback completeness is then judged over the discovered descendants; deletion verification asks whether repaired descendants still leave residual influence; and contestability turns that repair status into a closure record for a user or operator. This later recovery chain is where governance either becomes a state level accountability chain or collapses into disconnected local mitigations.

Smallest Governance Slice the Literature Already Implies. Across the literature, lightweight governance implies six artifacts: a candidate and promotion record, a state registry at admission, a migration ledger, an activation gate, an incident and descendant table, and residual probes after repair. These artifacts answer a deployment question: why an update was chosen, what state may persist, what may change substrate, what may become action authority, how exposed descendants are inventoried, and how a user or operator can verify closure.

7.1 Preventive Governance

Admission control. Admission control is the first intervention point where transient context becomes reusable influence. A self-evolving agent may propose a memory item, preference, reflection note, workflow rule, skill, policy update, or profile fact after an interaction. The gate acts before persistence and asks a stricter question than relevance: may this source, actor, and proposed durable form influence future contexts? A relevant instruction can still be outside the user’s authority, outside the task scope, sensitive to privacy, adversarially supplied, or unsafe when stored for later reuse. GhostWriter’s AM-Sentry turns write-time memory saving into an explicit response surface, while memory privacy work adds the sensitivity and scope classification that admission has to preserve [192, 196]. PASB shows why the same gate is needed for ordinary user content: the commit decision should preserve whether a claim is attributed, tentative, local to one exchange, or authorized as a durable preference or procedure [131]. The benchmark motivates this control point but does not evaluate rollback or deletion verification.

Skill admission is relational because adding one locally plausible package can complete a harmful capability already distributed across the installed set. ChainGuard operationalizes this check by reconstructing candidate-centered producer–consumer, artifact, state, context, and execution paths between a candidate skill and previously installed skills [245]. This is direct evidence for context-aware admission rather than isolated package scanning. The mechanism blocks formation of a detected chain at installation time; it does not repair artifacts created before detection or establish descendant rollback, deletion verification, or contestable closure.

What stronger admission claims add to the literature is not a vague approval, but a readable admission record: source object, durable form, source trust, scope, sensitivity, freshness, allowed task family, allowed tool influence, and revocation handle. Those fields are what let a system reject the item, admit it unchanged, admit a redacted version, store it as context without authority, or require human confirmation before future use.

AuthMem-Bench provides direct evidence for persisting and enforcing the authority field in that record. Natural-language source attribution reduces but does not eliminate prohibited actions, whereas structured authority labels provide a stronger signal; in the selected end-to-end pipeline, predicted labels reduce the observed unauthorized-action rate from 16.9% to 0.0% over 350 pairs while authorized task success changes from 39.7% to 40.0% [248]. This supports a write-to-activation contract in which the label is stored before retrieval and checked at use. It does not establish universal prevention or lifecycle recovery: labels cannot restore omitted memories, and the study does not test descendant rollback, deletion verification, or contestability.

Before a dataset, feedback trace, adapter, edited checkpoint, or unlearning patch can support a stronger governance claim, model-side admission needs the same review fields: provenance, adapter scope, base model lineage, intended task population, allowed tool contexts, safety regression suite, merge constraints, and rollback path. These are the fields that let a system refuse an update with unclear provenance, quarantine an adapter outside its intended scope, require canary safety evaluation before merge, or prevent a task-local update from becoming policy across deployment. Model-side sources fill those fields in stages: fine-tuning safety identifies the admission-time regression risk, dormant-behavior work makes activation context explicit, PEFT and feedback-poisoning work tie the risk to adapter or update provenance, and Safe LoRA gives a bounded mitigation handle for LoRA-style updates [50, 65, 159, 162]. The artifact that matters is an adaptation admission record, not a generic statement that the model was fine-tuned.

Promotion gates and longitudinal reconciliation. A promotion gate acts between candidate generation and admission. It first applies $c_t(\delta)$ as an eligibility condition, then lets $\hat{u}_t$ rank only candidates whose provenance, scope, authority, privacy, budget, and rollback requirements are satisfied. This ordering prevents task utility

from acting as implicit permission. The gate records the candidates considered, the evidence used to accept or reject them, the promotion target, the authority and scope delta, and any failure or counterexample omitted from the decision. OEP directly motivates transferability checks and active counterexample search before experience consolidation [201]; evaluator coevolution motivates recording the evaluator, objective, and epoch that produced the choice [70].

PoisonedEvolution makes the evidence side of this gate measurable. Its Inclusion–Attribution–Realization decomposition separates whether attacker records enter the stream, receive reusable credit, and survive synthesis into a skill [21]. In one $n = 30$, $k = 3$ F1 setting, a preliminary provenance-diversity gate blocks all 25 single-cluster candidates, reducing post-gate SER from 25/25 to 0/25. This is bounded evidence rather than a complete defense: candidate grouping is assumed, the benign control is small, and coordinated contributors may imitate diversity. The durable requirement is therefore auditable support and counterevidence, not source count alone.

SafeEvolve governs both sides of the resulting skill lifecycle. At write time, a critic and paired deleter localize and narrow unsafe reusable instructions; at reuse time, lineage risk, retrieved-skill outcome attribution, and safety-aware retirement change which versions remain active [132]. Its component tests report no later retrieval of threshold-crossing skills after retirement in the evaluated AutoSkill and EvoSkill runs. This is direct evidence for repair, attribution, and influence withdrawal over the active skill object, but not descendant-wide rollback, deletion verification across derived artifacts, or user-facing contestability.

EVOMAL moves the corresponding gate from submission screening to the create path. Its counter-prompt acts when the agent reads and re-authors a skill, reducing the six headline-model ASPRs from 20.3–41.8% to at most 1.8% with zero callbacks; the worst reported extended cell is 6.7% [217]. The utility claim is narrower than solution correctness: the paper’s Submitted rate records whether an agent emits a solution within 15 steps, while benign helper copying falls by as much as 11.7 percentage points. Its signed-quarantine gate is a stronger structural proposal, but the extinction result assumes unforgeable curator signatures and an admission log, prevents re-retrieval rather than repairing existing descendants, and does not stop a persistent attacker’s initial seed from firing.

Counterfactual testing asks what the candidate changes before the system commits it. For a fixed probe set, compare behavior under $\pi_{\mathcal{A}_t}$ with behavior under $\pi_{\mathcal{U}_\psi(\mathcal{A}_t, \delta)}$. A candidate that preserves task utility but widens tool authority, suppresses required confirmation, weakens refusal, or overrides newer evidence should be narrowed, quarantined, or sent for review. Scope-preserving abstraction carries the same rule into summaries and skills: retain preconditions, user and task scope, temporal validity, authority assumptions, safety constraints, invalidation conditions, and source lineage.

Longitudinal reconciliation repeats those probes after later updates because a sequence of locally accepted promotions may change the boundary even when no single record appears decisive. The experience bank should keep relevant refusals, safe aborts, failures, corrections, and near misses alongside successful trajectories so later candidate sets do not erase the evidence defining the safety boundary. Reconciliation then compares the current $\mathcal{A}_t$ with the recorded baseline, inspects accumulated ℓ changes, freezes suspect descendants, and routes confirmed drift into the rollback and deletion-verification stages below. These controls are proposed governance requirements assembled from current mechanisms; the literature does not yet establish them as a mature end-to-end defense.

7.2 Transition and Activation Gates

Migration and attribute preservation. Once state has been admitted, migration control governs crossings between substrate, authority, privacy, or actionability regimes. The same content can be harmless as a private note, risky as a generalized summary, and unacceptable as a tool parameter or workflow rule. A migration gate acts when memory becomes profile, retrieval context becomes plan step, plan step becomes tool argument, or a local rule is shared with another agent. Provenance, sensitivity, scope, authority, freshness, and reversibility must either survive the transformation or force a downgrade, redaction, quarantine, or explicit approval. Personal-agent work defines the user context, and MemGPT makes memory movement visible [105, 148]. Multi-agent work shows the shared-state version, Voyager supplies the skill-library case,

and self-evolution surveys provide the broader movement axis [198]. These papers establish what moves; evaluated controls for preserving attributes during the move remain sparse.

The migration trace prevents a common governance collapse: once state is compressed into a summary or policy like rule, the system may treat it as trusted even if the original source was narrow, private, or adversarially exposed. The trace that matters identifies the source, source safety attributes, target substrate, operation, actor, authorization context, environment context, safety attributes inherited, safety attributes strengthened, safety attributes dropped, and the reason for any downgrade or override. Migration control can prevent unauthorized authority upgrades and privacy scope drift, and it can detect when the target substrate cannot carry required safety attributes. Even with such a trace, semantic drift, hidden descendants, and later misuse remain hard to rule out.

Activation authority. After admission and migration determine where an object may reside, activation authority determines whether it may influence the current decision. The gate acts before state guides reasoning, appears in the prompt, selects a tool, fills an argument, invokes a skill, or constrains a plan. Read access and influence authority are separate decisions. A memory may be visible for background context but barred from authorizing an external effect; a retrieved instruction may guide a summary but not override policy; a skill may be callable for local computation but blocked from account-changing tools. Progent makes tool privilege explicit, and Traversal-as-Policy applies global and node-local gates to reusable workflow policy [99, 176]. CaMeL offers a complementary architectural control: it derives control and data flows from the trusted query and enforces capability policies when tools are called, so untrusted data cannot redirect program flow or authorize a disallowed flow [32]. These mechanisms turn action authority into a checked decision rather than an implication of retrieval.

DRIFT addresses the point at which a fixed plan must accommodate legitimate runtime change. Its Secure Planner produces a minimal function trajectory and parameter checklist; the Dynamic Validator checks deviations against user intent and read, write, or execute privilege constraints; and the Injection Isolator masks conflicting instructions before they enter the memory stream [95]. In SAVER terms, CaMeL primarily controls activation at the tool boundary, while DRIFT combines activation-time validation with isolation at the current-stream admission boundary. GuardAgent, TrustAgent, and LlamaFirewall add guard and trust-control machinery around related decisions [27, 220]. Together, these systems provide strong local authorization and containment evidence. They do not, however, show that previously stored cross-session state or its descendants have been rolled back, deleted, or made contestable.

Downstream review cannot reconstruct authority after the harness has already replaced its provenance. In When Context Gets Root, automatic permission reviewers can recognize that a proposed action is high risk yet approve it because reconstructed context presents attacker-derived content as explicit user intent [62]. The missing gate therefore precedes action review: context construction must carry the content’s origin level, reject unauthorized privilege increases, and expose the same lineage to both the worker and reviewer. This is a control requirement derived from the demonstrated failure; the paper does not evaluate an origin-preserving defense or recovery after elevated state has persisted.

Local permission is insufficient if the authorized agent or context later changes. State-bound authorization fixes both a transition envelope and an immutable effect ceiling when the user grants authority [261]. Authority may contract below that ceiling; expansion or an incomparable change requires specified evidence, and agent-produced evidence cannot raise the ceiling. The paper proves non-amplification only under complete mediation, sound effect abstraction, attenuating delegation, and monitor integrity. It does not empirically establish how often grants become stale, whether the current principal still serves user intent, or how a deployment should choose its envelope and ceiling.

Once a grant survives that continuity check, its authority must also survive the interval between an accepted call and its external effect. AID-Guard keeps one durable authorization and reservation lineage through provider commit, an ambiguous outcome, retry, and recovery: it revalidates the immutable request and current provider state at commit, retains the reservation while the outcome is unknown, and permits release or one successor only after certified no effect and a delivery fence [190]. The two works therefore control

consecutive boundaries: state-bound authorization decides whether an old grant survives agent or context mutation, while AID-Guard preserves one admitted grant through the provider-effect lifecycle. AID-Guard’s recovery prevents uncertain delivery from becoming a second delegation; it does not establish rollback of adaptive-state descendants, deletion verification, or contestable closure.

Multi-agent activation adds the choice of which agent may run. Control-flow hijacking can evade semantic alignment checks over inter-agent messages because a checker may use a brittle notion of task relevance or lack the execution context needed to judge the invocation [74]. ControlValve instead generates a permitted control-flow graph and enforces the graph together with a contextual least-privilege rule for each invocation. This moves the decision from message-level relatedness to structural transition authorization. The generated graph and contextual rules are themselves policy state, however, so runtime agent additions or legitimate replanning still require provenance and governed revision. The mechanism blocks an unauthorized edge; it does not repair state or artifacts produced before the block.

MAPLE-Guard addresses the complementary path in which no malicious message needs to cross the active communication edge. It gates memory writes, retrievals, promotion from private to shared memory, and later cross-agent reuse [223]. In its two reported benchmark settings, the abstract gives within-paper ASR reductions from 38.2% to 0.9% on LongMemEval and from 34.7% to 0.2% on AppWorld, with corresponding multi-agent defense success rate increases from 54.0% to 74.3% and from 42.5% to 99.8%. This supports link-level gating across the memory lifecycle; it does not establish descendant-complete rollback or contestable closure after poisoned state has already propagated.

Activation is better represented as influence level than as a binary allow list. A state item may be visible only, summarizable, advisory for planning, eligible as a tool argument, authorized for action, confirmation required, or blocked. This prevents stale or low-authority state from becoming executable authority simply because retrieval selected it. Risk recognition, injection benchmarks, and memory screens operate at different points in this decision. R-Judge tests whether risky situations are recognized; AgentDojo and InjecAgent supply tool-use and indirect-injection contexts where recognition must affect execution; GhostWriter’s retrieval screen applies the same question to poisoned memory activation; and adaptive attack studies ask whether the boundary survives attacker revision [31, 192, 244, 246, 247]. PS-Bench supplies a bounded context-sensitive response: its detection-reflection intervention checks whether personal context is legitimizing a harmful request before generation and, in the GPT-4o-mini experiment, reduces average attack success rate across the evaluated memory frameworks by about 27.4% [53]. It does not test tool authorization or stored-state repair. These signals become governance only when connected to authorization, blocking, escalation, or recovery. The activation decision log names the object, context of use, influence level granted, policy reason, and authority withheld; authorization alone does not establish downstream recovery.

Multi-turn and compression defenses intervene at different activation points. TurnGate is a response-aware temporal gate that asks whether a candidate response makes the accumulated dialogue harm-enabling [174]. KBRA instead protects the transformation boundary at which separated context becomes a new instruction [119]. ToolShield acts farther downstream by exploring a tool’s effects and distilling safety experience for later use [103]. Treating these mechanisms as interchangeable output filters would hide which state transition each defense actually controls.

7.3 Monitoring and Containment

If activation proceeds or exposure begins, governance moves from authorization to the live episode. Monitoring detects suspicious activation, instruction conflict, privilege escalation, privacy leakage, or unexpected tool arguments; containment limits impact by freezing suspect state, removing it from retrieval, downgrading authority, pausing external calls, forcing a clean replan, or escalating to human review. GuardAgent and TrustAgent supply guard and trust-control vocabulary, while R-Judge and AgentDojo provide recognition and injection settings in which those controls can be evaluated [31, 220, 244]. MAGE brings persistent safety context into the pre-action loop, and EDDOps addresses maintenance after deployment [211, 219]. Together they connect detection, bounded containment, memory-aware context, and operations, but not descendant repair.

A containment trace that identifies the activated object, any descendant object, monitor signal, tool or output at risk, containment action, state removed from the active context, and whether future activations are frozen pending review can distinguish the visible incident from the durable cause. This prevents a guard from being credited with more than it did: blocking one output is a runtime response, while repairing the memory, summary, workflow rule, or skill that caused the exposure is a rollback claim.

HarnessSafe operationalizes the first half of that distinction. Its stage distribution records how far a chain progresses across seven persistent-carrier families, and configurations with similar endpoint attack-success rates can stop at different stages [254]. This is stronger evidence for containment location than a binary block rate, but it remains a trace of where progression ended. It does not show that implicated descendants were repaired, that deletion was verified, or that the failure stays absent after adaptation resumes.

Connected traces also require an independently governed control and evaluation plane. The task-facing agent may need to see the declared environment, permitted tools, and escalation or abort contract, while enforcement credentials, tamper-resistant audit state, and recovery records remain outside its write authority. Complete invisibility is neither required nor sufficient. The OpenAI–Hugging Face incident supports strict evaluation isolation, narrow trust boundaries, short-lived credentials, and cross-system correlation as infrastructure controls [87, 146]; it does not establish that every governance function must be external to or hidden from the agent.

For model state, monitoring includes safety regression and canary evaluation rather than only runtime guard signals. A deployment should test whether an adapter, merged model, post-trained checkpoint, edited component, or runtime cache changes refusal boundaries, unsafe compliance rates, excessive refusal, privacy leakage, tool selection behavior, or trigger sensitivity before and after activation. Safe LoRA illustrates a response at the weight level to LoRA fine-tuning risk [65], while KVERaser illustrates a response at the cache level when a harmful or stale span is identified only after prefill [97]. A deployed governance chain still needs versioned comparisons, cache-local residual probes, and rollback records tied to a specific model lineage, adapter manifest, or cached context.

7.4 Recovery and Contestable Closure

Rollback and descendant recovery. When containment confirms that a transition is unsafe, stale, contested, or outside its authority, response shifts from the live episode to rollback over lineage. Later memory-poisoning and sleeper-memory studies establish persistent influence and delayed activation [156, 183], while GuardAgent establishes local recognition and blocking [220]. Their separation motivates, but does not itself prove, the survey’s rollback requirement: the target should include original memory, summaries, retrieval indices, workflow rules, skills, permissions, cached plans, shared artifacts, and external outputs. Quarantine and lineage capture should precede deletion so repair does not erase the evidence needed to reconstruct the incident.

The denominator for rollback completeness is the discovered descendant set, not the single object first reported. A stronger rollback claim therefore reads as a repair sweep: enumerate reachable descendants, classify their substrate and authority, apply repair or downgrade actions, and run residual probes before reopening scope. This denominator is operational rather than ontological. It is reconstructed from reachable records, caches, retrieval stores, workflow edges, deployment manifests, and tested activation paths, so strong reports should state discovered scope, inaccessible scope, and residual uncertainty.

Removing an unsafe adapter, reverting a merged checkpoint, undoing a model edit, erasing a harmful cached span, or rolling back an unlearning patch follows the same rollback structure with different handles: deployment lineage, active adapter or cache state, evaluation deltas, affected endpoints, and probes after rollback inside the agent loop. A rollback is scoped, not global, unless the reverted deployment, cached policies, inherited adapters, surviving cache influence, and downstream agents all pass the relevant residual tests.

Deletion verification. After descendants have been inventoried and repaired, deletion verification tests whether removal eliminated reusable influence rather than one visible record. It is the receipt that follows rollback completeness: the system records what was checked, what remains outside control, and whether the incident path still reactivates through copies, abstractions, or synchronization surfaces. Many local defenses stop before

this point because blocking or deleting one object does not show that summaries, retrieval indices, caches, workflow rules, skills, permissions, shared artifacts, or external systems stopped carrying the influence.

A convincing deletion verification report therefore inherits the rollback denominator rather than starting from a fresh object list. It identifies the incident id, target object, deletion reason, substrates checked, descendants checked, tests performed, inaccessible substrates, residual matches, and residual activations. For adaptive agents, this report has to follow descendants because the original object may no longer be the only source of influence. A summary may retain the private fact, a skill may encode a derived behavior, a profile abstraction may preserve the sensitive attributes, or a shared artifact may reintroduce the state during a future session. The result is a scoped verification report, not a global forgetting guarantee.

Contestability. The recovery chain closes only when a user, operator, or auditor can contest the state and change its future influence. Contestability must expose enough of the object’s origin, safety attributes, migrations, activations, descendants, rollback status, and deletion-verification status to support a meaningful challenge. Memory-privacy work motivates sensitivity and scope review, PASB shows retained influence surviving the original exchange, and GuardAgent supplies a local reasoning anchor rather than a full contestability mechanism [131, 196, 220]. Fully realized contestability remains rare.

Contestability prevents adaptive state from becoming uninspectable authority after an internal repair sweep claims success. A useful dispute flow freezes or downgrades activation, attaches correction evidence, inspects the rollback inventory, reviews deletion verification receipts, and emits a resolution record visible to the user. Without that flow, rollback remains an internal maintenance action rather than an accountable governance process.

7.5 Synthesis: Governance Follows State Descendants

The response component closes the SAVER chain by relating mechanisms that otherwise appear as separate defense families. Memory-oriented controls address admission, retrieval, sensitivity, and purpose through write-time screening, privacy governance, and personalized risk-aware memory [5, 192, 196]. Action-oriented controls first address source authenticity and task alignment at the effect boundary [75, 184]. Risk judges and guards add recognition or local blocking, while Progent, CaMeL, and DRIFT connect those signals to privilege, control flow, and runtime deviation checks; EDDOps addresses their operational maintenance [27, 32, 95, 176, 219, 220, 244]. Model-side work follows the same logic with different handles: fine-tuning safety defines a regression surface and Safe LoRA supplies a bounded adapter-level intervention [65, 159].

Response Feedback. Response also creates state for later rounds. Evaluation evidence can revise operational memory, prompts, routing, guardrails, test cases, safety cases, or rollback records; EDDOps treats these as governed runtime or redevelopment changes [219]. DARWIN and BraveGuard instantiate two complementary guardrail-update loops. DARWIN evolves jailbreak strategies through discovery, mutation, selection, and feedback-driven composition, then trains successive guard checkpoints on the resulting adversarial samples [158]. BraveGuard instead converts open-world threat discovery into executable computer-use tasks, agent trajectories, and supervision for repeated guard training [44]. The former couples guard evolution to an adaptive adversary; the latter couples it to an expanding trajectory corpus.

For SAVER, these systems make the response itself an adaptation producer. In the model from Section 2, a guard update may alter c_{t+1} , the evaluator or objective behind $\hat{u}_{t+1}$, or the candidates available in Δ_{t+1} . Its transition record should therefore connect the discovered threat, generated task or attack, selected training example, guard checkpoint, and later deployment decision. DARWIN and BraveGuard establish that this feedback can improve guard performance under their respective benchmark settings, but they do not establish descendant-complete rollback, resistance to poisoned update evidence, or contestable closure across guard versions. EDDOps supplies the broader process edge; the guardrail studies make the evolving response state concrete.

These mechanisms cover successive response points, but they do not yet form a verified recovery chain. A connected claim must identify the implicated state, enumerate reachable descendants, report rollback over

that discovered set, test residual activation after deletion or repair, and expose a contestable closure record. Without those links, a blocked action remains evidence of local containment rather than lifecycle recovery.

Takeaway. Response evidence is least connected across the lifecycle. Local mitigations cover selection, admission, migration, activation, monitoring, rollback, deletion verification, and contestability, yet few systems bind these stages to one incident handle and denominator. A response may also write state that shapes the next selection round, so closure includes descendant repair and the later consequences of the response itself. Section 8 asks which traces make that recovery claim checkable across sessions.

8 Evaluation and Benchmarks

$$\text{S} \times \text{A} \rightarrow \text{V} \rightarrow \text{E} \rightarrow \text{R}$$

Evaluation Premise. Existing benchmarks usually score task success, immediate safety behavior, attack success, or local intervention. These endpoints remain useful, but self-evolution creates temporal and spatial dependencies: a memory write, summary, skill edit, delegated message, or model update can remain dormant until a later context activates it, or move to another carrier or agent before activation. Evaluation must therefore identify the transition that produced the visible behavior, not only score the behavior itself.

The measurement object is a lifecycle trace. It binds the initial carrier and authority regime to the operation applied to them, records the safety attribute that failed, locates the first observable consequence, and tests the response against the implicated state and descendants. Without that trace, the same harmful outcome may be attributed to prompt admission, summarization, retrieval scope, tool authorization, or failed rollback. A local block also remains indistinguishable from repair if the enabling state can reactivate later or elsewhere.

The temporal dimension does not make all multi-turn scores comparable. MTID records the earliest response that makes accumulated dialogue harm-enabling, MT-AgentRisk constructs tool-realistic attack sequences, and Relink measures compression-induced recombination against clean-split controls [103, 119, 174]. Their denominators differ because they count response closure, tool-using attack sequences, and relinking events. They nevertheless show why interaction length should be analyzed through the operation that combines context rather than treated as a scalar risk factor. A3S-Bench complements this evidence by placing payloads in workspace artifacts outside conversation-level monitoring and evaluating the resulting action traces [128]. This directly tests a spatial entry path through files, skills, and configuration, but not general propagation across agents or deployments.

Selection adds a second temporal dependency before the measured transition. The Red Queen Gödel Machine treats the utility function and evaluator signal as epoch-scoped parts of the self-improvement loop rather than a fixed verifier outside it [70]. A lifecycle record should therefore name the agent, evaluator, objective, and epoch versions before comparing improvement, degradation, or repair. The section first audits selection and promotion, then reads existing benchmarks by the transition segment they observe, separates incompatible denominators, and closes with the traces needed for longitudinal recovery evaluation.

8.1 Selection and Promotion Evaluation

A selection audit observes the stage that ordinary transition benchmarks usually hide. At each evolution step it records Δ_t , the source of each candidate, the evaluator and objective version behind $\hat{u}_t$, the evidence checked by c_t , the selected update δ_t^* , rejected candidates, and the resulting change to ℓ . The benchmark then delays activation, runs the updated state on tasks outside the source episode, and records whether the consequence enters the next candidate set. OEP supplies a direct test of harmful transfer from accepted experience updates, while the Red Queen Gödel Machine supplies the denominator rule for changing evaluators and objectives [70, 201].

The protocol needs separate adversarial and endogenous tracks. The adversarial track inserts crafted updates and measures whether selection, activation, and recovery reject or contain them. The endogenous track uses ordinary successes, failures, refusals, and corrections without a malicious writer. It holds the probe suite and authority policy fixed while the agent evolves, then asks whether selection alone changes refusal boundaries,

confirmation requirements, tool authority, privacy scope, or preference priority. This design does not assume that benign adaptation drifts. It makes the possibility falsifiable and separates it from attack success.

Three proposed metrics cover the first usable longitudinal denominators. Let $\text{Promote}(\delta)$ mean that candidate δ enters a more persistent, general, or action-bearing role, and let $\text{SafePromote}(\delta)$ mean that the promotion satisfies the invariant in Section 5. Safe Promotion Rate is

$$\text{SPR}_{0:T} = \frac{\#\{t \leq T : \text{SafePromote}(\delta_t^*)\}}{\#\{t \leq T : \text{Promote}(\delta_t^*)\}}.$$

The denominator is all committed promotions; a run with no promotions reports the metric as undefined rather than perfectly safe. For Boundary Drift, let $\mathbf{b}(\mathcal{A}_t)$ be the decision vector produced by a fixed set of paired safety probes and let d be a preregistered distance:

$$\text{BD}_t = d(\mathbf{b}(\mathcal{A}_0), \mathbf{b}(\mathcal{A}_t)).$$

For Negative-Evidence Coverage, let $\mathcal{N}_t$ contain safety-relevant non-success experience observed at step t , including correct refusals, safe aborts, failures, corrections, and near misses, and let $\mathcal{N}_t^{\text{keep}}$ be the subset retained or consulted during selection:

$$\text{NEC}_{0:T} = \frac{\sum_{t=0}^T |\mathcal{N}_t^{\text{keep}}|}{\sum_{t=0}^T |\mathcal{N}_t|}.$$

NEC is conditional on negative evidence that was actually observed. A report should therefore separate *observed but discarded* refusals, aborts, failures, corrections, or near misses from evidence that was *never generated or unavailable* because the action space, environment, or reward design supplied no meaningful safe alternative. The first case diagnoses selection or retention; the second diagnoses the opportunity set. Treating both as low NEC would hide which mechanism needs repair.

Satisfiability and Abort Calibration. Benchmark tasks should cross two independent labels. Feasibility records whether the declared goal is satisfiable in the provided environment, blocked by a missing dependency, or infeasible under the declared conditions. Action authority records whether a candidate path is permitted, requires explicit escalation, or is prohibited. A blocked path does not authorize an out-of-scope workaround. Reports should pair the true safe-abort rate on infeasible tasks with the false-abort rate on tasks feasible within current authority; cases feasible only after authorized escalation should be scored separately for correct escalation. Both rates should be reported with task utility so an agent cannot appear safe merely by abandoning useful work. The OpenAI-Hugging Face incident motivates this separation: the ExploitGym objective was narrow, but task completion alone did not bound which infrastructure paths were acceptable [87, 146, 213]. The incident does not establish that an abort option would have prevented the intrusion; it shows why outcome success and path authorization need separate denominators.

These are measurement contracts proposed by this survey, not established cross-paper metrics. Reports should publish the probe set, epoch boundaries, missing candidate logs, task-feasibility labels, abort and escalation options, and denominator exclusions. The utility-risk covariance identity in Appendix B is a conditional diagnostic for the same protocol; it does not replace these observable counts.

Post-Response Continuation. An immediate residual probe does not settle what happens after adaptation resumes. After recording the response, a benchmark should continue ordinary evolution for a preregistered number of update opportunities and rerun the fixed safety probes. The report should state whether the incident lineage, a known descendant, or an equivalent failure under the fixed probe suite reappears, then name the substrate, the first reactivation point, and any inaccessible scope. Adaptive prompt-injection studies show why an action-boundary defense must be retested after attacker revision, while EDDOps motivates re-evaluation after governed system changes [219, 247]. These sources do not establish cross-substrate response-induced migration; continuation after response is a measurement contract proposed here. Endogenous and adversarial tracks can share the same logs, but candidate-source control remains part of the threat model.

8.2 Reading Existing Benchmarks Through SAVER

A lifecycle reading asks what part of SAVER a benchmark already observes and what remains hidden. AgentDojo-style indirect prompt injection evaluation is a clear example. It can show whether untrusted environmental content changes an outcome mediated by tools, so it is strong at the exposure surface: low-authority data reaches a tool-using decision and causes an unsafe or task-inconsistent action [31]. This is a recognizable access control and privilege escalation problem in an agent setting. The missing evidence is temporal: unless the test records state lineage, it cannot diagnose whether the injected instruction was merely present in the current prompt, was written into long-term memory, was summarized into a user profile, was promoted into a workflow rule, or was later retrieved as trusted guidance. A transition reading keeps the original exposure test, then asks for a write log, source safety attributes, transformation record, activation context, and recovery probe after the episode.

R-Judge evaluates whether an agent recognizes risky situations, which is valuable for the violation and response vocabulary [244], but it illustrates a different boundary. Risk recognition and transition localization are separate measurement objects. A lifecycle reading asks whether the evaluator can connect the risk judgment to a state carrier and operation: for example, a stale memory, a low-authority retrieved instruction, a broadened tool permission, or a policy update loaded outside its intended scope. Agent Security Bench helps map agent security tasks [251], and Agent-SafetyBench maps broader agent safety behaviors [262]; they become transition safety benchmarks only when they record how state entered, changed safety attributes, activated, and was repaired.

Under this reading, attack success rate, risk recognition accuracy, and task success remain useful but stay surface scores. A transition report records the missing trace: which state carrier was involved, which operation changed its role, which safety attributes were supposed to travel, which gate saw it, which exposure occurred, and which recovery target was tested. The compact benchmark family coverage pattern in Table 1 lets the reader recover the benchmark logic without pausing on one paper at a time: what latent violation is planted, where it is activated, where it becomes exposed, and which response denominator is actually being measured. The matrix is deliberately qualitative so the field level asymmetry stays visible without turning incompatible scores into a leaderboard. Existing suites most often cover exposure, activation, recognition, and local response; they much less often show migration lineage, descendant rollback, deletion verification, or contestability. The appendix records the denominator rule behind this choice instead of reproducing a score list. Appendix Table 12 retains the corresponding paper-level benchmark index, including target system, observed lifecycle slice, and denominator.

8.3 Denominator Discipline Without a Leaderboard

Current quantitative evidence supports a narrow synthesis: benchmarks measure local transition slices with separate safety frontiers. Action-boundary attack success rate (ASR), safety-recognition F1 score, memory-poisoning success, runtime-intervention rate, and governance-routing score each answer a different question about the adaptive-state chain. Quantitative comparisons are therefore used to identify measurement objects and opportunity sets, not to reproduce paper-specific percentages; a score list would imply comparability across benchmarks that the heterogeneous denominators here do not support.

Safe comparison is possible inside a shared benchmark denominator, transition slice, and measurement object. InjecAgent-style cases support exposure comparisons inside one indirect-injection benchmark; R-Judge-style annotated records support recognition comparisons; PASB-style contrasts support session-only versus committed-state comparisons; and SafeAgent- or Progent-style runtime rows support one controller family inside one harmful-task benchmark. Once those denominator families are mixed, several local evaluation panels collapse into a category error rather than a meta-analysis.

The denominator families become clearer when the non-adversarial and adversarial cases are interleaved. PerMemSafe asks whether user-specific memories make a later response unsafe under implicit personalized context [5]; PS-Bench compares harmful queries with and without benign personalized memories to test intent legitimation [53]; FragFuse tests access-control bypass after fragments are stored across memory interactions and reconstructed at retrieval [163]; GhostWriter separates hidden memory injection from later activation in

Table 1 Benchmark family coverage along the reusable-state lifecycle, ✓ direct, ● partial, ✗ mostly absent.

Family	Admit	Move	Act.	Expose	Contain	Rollback	Delete	Contest	Boundary
AgentDojo [31] / InjecAgent [246]	●	✗	✓	✓	✗	✗	✗	✗	Tool-mediated injection and exposure; no response or cleanup denominator.
ACIArena [6]	●	●	✓	✓	✗	✗	✗	✗	Cascading injection across profiles, messages, and agents; response and repair unmeasured.
HarnessSafe [254]	✓	✓	✓	✓	●	✗	✗	✗	Connected carrier-to-violation trace and stopping stage; no selection or post-incident recovery.
SkillMisevo-Bench [132]	✓	✓	✓	✓	✓	●	✗	✗	Skill authoring-to-carryover trace plus local repair and retirement; no descendant or deletion proof.
EVOMAL [217]	●	✓	✓	✓	✓	●	●	✗	Create-path and five-round source-removal trace; no natural-stream or descendant-deletion proof.
R-Judge [244] / ASB [251] / Agent-SafetyBench [262]	✗	✗	●	✓	✗	✗	✗	✗	Recognition, security tasks, and unsafe-choice scoring; no lifecycle response.
OEP [201] / PASB [131] / MemEvoBench [221]	✓	●	✓	●	✗	✗	✗	✗	Experience or memory updates and later reuse; no recovery denominator.
PerMemSafe [5] / PS-Bench [53] / RBI-Eval [224]	●	✗	✓	✓	●	✗	✗	✗	Personalized context and warranted-memory-use evaluation.
PersistBench [155]	●	✗	✓	✓	✗	✗	●	✗	Forgetting and use boundaries; no descendant-wide repair.
SafeAgent [112] / Progent [176] / CaMeL [32] / DRIFT [95] / LlamaFire-wall [27]	✗	●	✓	✓	✓	✗	✗	✗	Tool authorization and local containment, not descendant repair.
Fine-tuning safety [159] / PEFT safety [64] / SEVerA [11] / Dormant tests [50]	✓	●	✓	●	✗	✗	✗	✗	Model update, activation, and verification slices; no agent-loop recovery trace.
Safe LoRA [65]	✓	✗	✓	●	●	✗	✗	✗	LoRA update and bounded mitigation; no lifecycle rollback evidence.

tool-using personal agents [192]; AgentPoison tests trigger-selected poisoned retrieval [25]; OEP tests accepted experience updates and harmful transfer [201]; and EvoBreak tests adaptive acquisition followed by joint activation of complementary experiences [228]. These designs can all report ASR-like or safety-rate values, but their opportunity sets differ. The useful comparison is the denominator family: personalized context followed by response safety, fragmented injection followed by fusion, hidden memory adoption followed by later activation, trigger-selected retrieval, one accepted update followed by transfer, or an acquired experience set followed by composition. The mechanism difference matters more than the magnitude.

PASB supplies a particularly clean within-paper commit contrast. Its 1,600 tasks separate a five-turn persist stage from a cleared three-turn query stage and let the evaluated agents decide what to store. Across twelve models, the reported downstream failure rises from 45.0% in session-only episodes to 71.9% after commitment, a 27.0 percentage-point increase [131]. This result localizes risk at the transition from accepted conversational content to durable state; it is not a cross-benchmark ranking against poisoning or activation-only studies. The session-only baseline also removes the persistence mechanism whose personalization utility motivates the agent, so its lower failure rate is not evidence of a cost-free safety improvement. PASB-style commit contrasts should therefore pair safety outcomes with matched utility measures; PerMemSafe illustrates the coupled objective by evaluating implicit personalized safety together with helpfulness under long-horizon memory [5, 131].

Three recent studies add different within-paper transition contrasts. PoisonedEvolution measures evidence-to-artifact promotion: at the same 10% attacker-support ratio, it reports 546/600 embedded skills in SkillClaw and 369/600 in the structurally different Trace2Skill pipeline [21]. Its SER denominator is completed evolution trials and stops before runtime activation. SkillJack instead compares poisoned trajectories with extracted skills: in SkillIX, the abstract reports safety detection falling from 98.5% to 11.4%, while 80.0% of skill-mediated attacks persist after source-record deletion [239]. MAPLE-Guard compares ungarded and guarded memory-link opportunities at write, retrieval, promotion, and cross-agent reuse on LongMemEval and AppWorld [223]. These are, respectively, artifact-promotion, representation-and-deletion, and guarded-lifecycle-link denominators. None should be ranked against PASB’s commit effect, EvoBreak’s experience-set attack, or runtime ASR.

SkillMisevo-Bench connects artifact promotion to later execution under fixed denominators. Each agent-method condition contains 25 episodes and 525 tasks: 225 malicious learning tasks, 225 benign tasks, and 75 clean-session persistence tasks. Across the 21 evolved configurations, unsafe artifacts appear in 21, unsafe retrieval and benign-task contamination in 19, and fresh-session harm in 15 [132]. These are successive gates rather than interchangeable rates. In its two-configuration exposure sweep, pooled carryover ASR rises from 16.0% at $K = 0$ to 35.3% after three malicious tasks and 41.3% after nine. Because the no-exposure baseline is already nonzero, the defensible result is the within-paper increase after bounded exposure, not that all observed carryover harm originates in the malicious tasks.

The same study makes the response tradeoff more precise than its abstract-level mean. Across OpenClaw with AutoSkill and EvoSkill, SafeEvolve reduces mean unsafe retrieval from 35.33% to 8.67% and carryover ASR from 21.33% to 4.00% [132]. Online benign utility changes from 58.44% to 58.00%, which yields the reported 0.4-point cost, but carryover utility changes from 53.33% to 40.67%. The result therefore supports a low immediate utility cost under the paper’s online denominator while also showing a larger loss of useful persistent reuse. It is not evidence of a cost-free or universal safety-utility frontier.

EVOMAL adds a create-path and descendant denominator. ASPR counts tasks that store a newly authored skill reproducing a planted pattern; it is distinct from the callback rate that records payload execution. Across six models on 153 tool-relevant SWE-bench Verified tasks per model, headline ASPR ranges from 20.3% to 41.8%, producing 31–64 new malicious skills from eight planted seeds and leaving 4.9–9.0 times as many malicious skills as were planted [217]. The reported 86.7% targeted maximum is the Qwen3 pytest-family cell with $N = 15$, not the full task distribution. The five-round removal experiment is a separate rolling-replacement denominator and should not be collapsed with single-round ASPR or callback success.

ColluSkill adds a skill-set denominator. Its 200 three-skill attack chains reach 96.0% average ASR against six evaluated scanners; adding full installed-skill context to ChainGuard reduces the attack ASR from 69.0% to 22.5% while changing the reported benign-workflow pass rate from 99.7% to 99.5% [245]. These values measure candidate-with-context scanning on the paper’s generated chains and benign workflows. Because the attack iteratively uses feedback from the target scanners that are later evaluated, the 96.0% result is evidence of adaptive scanner evasion under that feedback setting, not unseen-scanner generalization or ecosystem prevalence.

AID-Guard adds an authorization-to-effect denominator rather than another attack-success leaderboard. Its 210 frozen Stripe contract trials matched their predeclared outcomes; separate campaigns exercised 40 terminalize-then-successor schedules across Stripe and Resend, 30 overlapping Stripe confirm/cancel races, and 10 Stripe crash-recovery schedules without an observed duplicate effect [190]. The same paper makes the safety-utility cost visible: over 48 benign AgentDojo episodes per model and condition, the strict exact-manifest profile reduced benign utility by 43.8 percentage points for DeepSeek and 35.4 points for Qwen. In a separate no-seed rerun, a typed three-call authority form completed 10 and 9 more benign tasks than exact single use, while all 864 attack episodes produced no unsafe effect. These results remain paper-local: the provider claims depend on the declared effect-path inventory and supported contracts, and the no-seed frontier does not identify a causal effect size or an optimal policy.

The practical path for meta-analysis is a normalization layer over opportunity set, transition slice, and measurement object. Synthesis across papers becomes safer once papers report conversions such as activation opportunities to realized exposures, discovered descendants to repaired descendants, or seeded latent violations to residual activations after repair. Until then, exact values belong in source-specific extraction records, while each section records what its metric measures and which lifecycle stages remain unmeasured.

Admission filters, task alignment mechanisms, activation gates, runtime guards, memory governance controls, model mitigations, and process audits all support useful local claims, but they observe different opportunity sets. The same denominator rule applies in prose: a block rate near a tool boundary, a memory write defense, an adapter safety test, and a process audit for governance become comparable only after the same transition slice and denominator are visible.

Table 2 Lifecycle evaluation loop. A benchmark should seed state, transform it, trigger activation, measure exposure, verify repair, and continue adaptation before closure.

Lifecycle Evaluation Contract		state + adaptation + challenge + response						
		Seed State	→	Adapt / Move	→	Trigger / Expose	→	Respond / Resume
1 Seed	Initialize memory, workflow, tool, shared agent, or model carrier. <i>Trace</i> : id, source, substrate, environment, intended assets, feasibility label, authorization boundary, expiry, rollback handle.							
2 Adapt	Add, abstract, activate, modify, forget, propagate, or migrate. <i>Trace</i> : changed safety attributes, gate decision, dropped fields, descendant link.							
3 Delay	Insert distractor tasks or time gaps. <i>Trace</i> : order, access log, retrieval attempts, background updates.							
4 Trigger	Challenge stored state and test whether it receives authority. <i>Trace</i> : task, feasibility status, policy, action authorization, retrieved item, tool, workflow, adapter.							
5 Expose	Measure output, tool call, disclosure, delegation, artifact, monitor, or model behavior. <i>Trace</i> : transition attribution, not aggregate success alone.							
6 Repair	Run containment, quarantine, rollback, user challenge, or operator review. <i>Trace</i> : affected state and descendants.							
7 Probe	Probe residual influence after repair. <i>Trace</i> : rollback ledger, descendant checks, reactivation probes, deletion/unload evidence.							
8 Continue	Resume adaptation for fixed update opportunities, then repeat the safety probes. <i>Trace</i> : response-generated state, first reactivation, substrate, lineage or equivalent probe failure, inaccessible scope.							
Outputs: transition trace; exposure class; response decision; post-response reactivation result. Stop claim at: no lineage, no denominator, or reactivation after resumed adaptation.								

8.4 Benchmark Patterns and Missing Traces

Most benchmarks remain locally strong but observe only one transition slice. Three connected designs now cover complementary scopes. HarnessSafe spans seven persistent-carrier families and assigns each executable case to its furthest evidence-supported stage; SkillMisevo-Bench follows one skill carrier from authoring through clean reload, retrieval, contamination, and carryover harm; and EVOMAL isolates create-path copying, callback exposure, and descendant propagation after seed removal [132, 217, 254]. HarnessSafe provides cross-carrier breadth, SkillMisevo provides skill-evolution depth, and EVOMAL adds an explicit descendant process. They still leave rejected candidates, natural deployment streams, verified descendant deletion, contestability, and reactivation after broader adaptation incomplete.

Different benchmark families make different slices observable. FragFuse and GhostWriter cover delayed memory reuse through fragment fusion and separated admission/activation [163, 192]. PerMemSafe and OEP move the slice to long-horizon personalized safety and experience-derived rule formation [5, 201]. InjecAgent and ACIArena push exposure outward to indirect tool injection and cascading injection across multi-agent trust channels [6, 246]. The repair probe remains much thinner than the earlier trace segments.

A response system can report a high block rate over visible attacks while missing the descendants that matter for recovery; a migration gate can look accurate while never seeing hidden boundary crossings; and a model update can pass static safety checks while failing inside the agent loop. The metric denominator is therefore as important as the score. The useful score asks whether the evaluator can reconstruct lineage and verify repair after a final action is blocked. Table 2 spells out the same trace more concretely as an evaluation contract.

Family Level Reading. Capability benchmarks establish where useful state is created and reused; attack benchmarks create violation and exposure pressure; state transition benchmarks localize attribute failure; and recovery benchmarks ask whether containment, rollback, audit, and contestability are measured under explicit opportunity sets.

Capability Benchmarks. Capability benchmarks establish the baseline that safety controls must preserve. Benchmark surveys map task success and interaction quality, while long-context and memory tasks make state creation and reuse visible [130, 138]. PERMA adds evolving preferences and cross-domain interference, while MemEye varies evidence granularity and evolutionary synthesis in multimodal memory [54, 115].

Coordination benchmarks extend the observation point to shared state, and evaluator-coevolution work makes the utility signal itself part of the evolving loop [70]. These benchmarks do not directly test transition safety. They identify where memory writes, summaries, workflow edits, delegated state, evaluator updates, or profile changes should be instrumented without disabling the capabilities under review.

Attack Benchmarks. Attack benchmarks become comparable only after their pressure point is identified. Prompt-injection studies perturb assembled context; AgentDojo and InjecAgent carry that perturbation to tool-mediated decisions; adaptive attacks then test whether the action boundary survives attack revision [31, 118, 246, 247]. This family is strongest at exposure, because the benchmark observes whether untrusted context changes a current action.

When Context Gets Root shifts the perturbation from malicious wording to the harness operation that assigns instruction privilege [62]. Its evaluation covers 13 confidentiality, integrity, availability, and remote-code-execution objectives across six coding-agent harnesses. Delegation-based escalation reaches every objective under full-access execution and on all three harnesses evaluated with automatic permission review; these are objective-coverage results under attempt-capped cases, not a 100% per-attempt success rate. Four persistent-goal or scheduled-task paths across three harnesses reproduce the same tool-to-user transition. The study therefore localizes a migration-to-authority failure, but it does not test an origin-preserving defense, descendant repair, or resumed adaptation after containment.

Persistent-state benchmarks move the pressure earlier in the chain. AgentPoison measures trigger-selected reuse from a poisoned store, whereas GhostWriter separates hidden memory admission from later activation; broader memory-poisoning work studies the same persistence problem across additional settings [25, 183, 192]. HaluMem decomposes extraction, update, and question answering, while MemConflict varies memory-conflict types [17, 188]. PASB observes the agent’s own commit decision, and PS-Bench and PerMemSafe test later use of benign personal state [5, 53, 131]. EvoBreak adds composition over an acquired experience set, SkillJack adds trajectory-to-skill migration and source-deletion persistence, and MAPLE-Guard adds lifecycle-link enforcement across private and shared memory [223, 228, 239]. These studies belong to different opportunity sets from prompt injection because the benchmark must observe a write, transformation, promotion, or retained context before later reuse.

AuthMem-Bench is unusually close to a connected $S \times A \rightarrow V \rightarrow E$ trace. Its three modules measure write-time authority collapse, the action consequence of a controlled memory representation, and end-to-end authority-label preservation while keeping paired claim content and later tasks fixed [248]. The modules should still retain separate denominators: write-time upgrade, controlled unauthorized action, and end-to-end action after omission and retrieval are not interchangeable rates. The benchmark ends after one write-to-action cycle, so it does not measure descendant repair or post-response reactivation.

HarnessSafe broadens that connected trace from one memory-consolidation boundary to seven carrier families and seven agent harnesses [254]. Its carrier path and persistence boundary identify $S \times A$; case-specific oracles connect V to observable E ; and the furthest-stage distribution supplies containment evidence for R . This mapping is more informative than endpoint ASR: OpenCode and Hermes Agent report similar conditional ASRs in the paper but stop at different stages. The Chain-Stage Score remains paper-specific, measures progression rather than harm severity, and excludes unsupported or incomplete workflows rather than counting them as safe.

Propagation and recognition studies expose still later or orthogonal slices. ACIArena follows malicious instructions across multi-agent trust channels, malfunction amplification measures local escalation, and R-Judge asks whether the resulting risk is recognized [6, 244, 249]. Agents of Chaos uses eleven live-laboratory cases rather than a fixed benchmark denominator, while Triedman et al. report attack trials indexed by model, orchestrator, input channel, and payload [173, 193]. ControlValve evaluates a response at the agent-invocation boundary by asking whether execution stays inside a permitted control-flow graph and its contextual rules [74]. These denominators should remain separate: case-study diversity, per-configuration attack success, and invocation-policy enforcement do not form one leaderboard. Together they expose concrete violations and controls, but attack success remains a surface metric unless an evaluation also identifies source content, selected edge, recipient authority, downstream effects, and response.

Transition and Recovery Benchmarks. Trajectory assurance supplies the high-level target: verify a stateful sequence against system-level constraints rather than assume that individually permitted actions compose safely [121]. The narrower measurement question for this survey is which reusable state transition changed that sequence and whether its influence can later be contained or repaired. State-transition benchmarks are closest to this unit because they ask whether the enabling carrier, operation, source, safety-attribute change, and activation context can be identified. Agent Security Bench (ASB) provides partial ingredients for agent-security task coverage [251], while Agent-SafetyBench provides adjacent agent-safety behavior coverage [262]. HarnessSafe advances this line by preserving carrier, boundary, benign trigger, and violation roles across harness-native implementations, then using matched controls to test whether progression depends on the declared lifecycle [254]. Transition evaluation still needs selection and lineage fields beyond task-outcome scoring. The strongest versions should connect this stage localization to attribute preservation across migration, drift after adaptation, and model-update lineage when the update is selected, merged, loaded, or inherited inside an agent loop. Fine-tuning safety is relevant when an update changes safety behavior [159]; PEFT risk work is relevant when the adapter or patch is a deployable state object [64]; and dormant-behavior work shows that changed behavior may remain hidden until a matching activation context appears [50]. Connecting that activation to update selection and recovery remains an additional requirement.

Response-oriented benchmarks form a pipeline rather than one comparable family. GuardAgent and TrustAgent evaluate diagnosis and local guard reasoning; Progent and Task Shield ask whether those signals alter privilege or task-aligned execution [68, 75]. CaMeL tests capability-enforced control and data flow, DRIFT tests dynamic plan validation and injection isolation, and MAPLE-Guard tests gates over memory writes, retrievals, promotions, and cross-agent reuse [32, 95, 223]. Their utility and attack-success results remain paper-local because the systems enforce different contracts. SafeAgent and LlamaFirewall then test runtime containment, while MAGE extends containment to long-horizon memory and EDDOps asks how the control is maintained in operation [27, 112, 211]. The remaining gap is whether these stages preserve enough lineage to measure containment latency, rollback completeness, contestability cost, and audit completeness without turning privacy-sensitive traces into a new risk.

Coverage Reading. A benchmark is most informative when it states which transition segment it instruments, which trace fields it logs, which opportunity set its metric covers, and which lifecycle steps remain unmeasured. A prompt injection benchmark is an exposure benchmark, a memory poisoning test without rollback is evidence about admission and activation, and a capability benchmark with memory use becomes evidence about persistence safety only when safety attributes, activation authority, and response traces are visible.

Takeaway. Most benchmarks measure one transition slice under a paper-specific denominator. HarnessSafe shows that connected carrier-to-exposure traces and stage-resolved containment are feasible, but it does not expose the candidate set that preceded the transition or verify repair after it. A complete response evaluation also resumes adaptation and checks for lineage-linked reactivation or an equivalent failure under fixed probes. Reports should state which promotion or transition opportunities the score covers, which lifecycle stages remain unmeasured, and what record lets a recovery claim survive after detection. Section 9 turns those missing denominators into research needs; Section 10 closes the survey.

9 Open Challenges

The evidence gaps identified above arise under different substrate conditions. Context and memory make delayed reuse and semantic provenance difficult to observe; tools, skills, and workflows turn stored state into operational authority; shared artifacts and multi-agent state expand the affected lineage; and model-side state widens deployment scope while reducing inspectability. These settings require different controls, but they expose the same missing connection between the operation that changes state, the violation it creates, the surface that reveals it, and the response that claims closure.

Open Challenge Reading. The open problems below identify where evidence fails to pass along that connection. Selection determines which operation enters the chain; migration can detach a carrier from its safety attributes; activation can grant unjustified authority; propagation expands the lineage; and recovery must reconstruct that lineage before it can verify closure. Each challenge therefore names both

Table 3 Research agenda for lifecycle evidence in self-evolving agents.

Direction	Evidence Gap	Minimal Test
Selection / promotion	Candidate updates are ranked by visible utility while safety evidence, rejected alternatives, and negative experience remain unobserved.	Candidate and promotion log plus fixed cross-epoch probes for authority, scope, refusal, and confirmation drift.
Migration / preservation	State changes carrier while provenance, scope, authority, or revocation becomes ambiguous.	Migration ledger plus residual probes for summaries, embeddings, workflows, artifacts, and adapters.
Activation authorization	Stored state becomes reasoning, planning, tool, workflow, or model authority outside its allowed context.	Activation gate that logs user, task, tool, time, and allowed external effect.
Recovery / deletion	Rollback removes the visible object while summaries, indices, skills, or shared artifacts may still activate its influence.	Descendant inventory, deletion receipt, and activation probe after repair.
Propagation / contestability	Shared state crosses agents or artifacts without clear repair ownership or a user challenge path.	Propagation trace plus contest docket for correction, quarantine, and notification.
Longitudinal evidence chain	Benchmarks measure isolated slices rather than one incident path from admission to closure.	Opportunity set denominators for admission, migration, activation, exposure, repair, and closure.

the affected substrate and the missing $A \rightarrow V \rightarrow E \rightarrow R$ evidence. Longitudinal governance connects those breakpoints under shared identifiers and denominators. Table 3 gives the compact roadmap. The main difficulty is reconstructing connected traces with shared measurement objects and denominators. Useful reports need *opportunity sets*: authority upgrades over low-authority migration attempts, activations outside scope over activation opportunities, repaired descendants over lineage scan discoveries, and influence-removal successes over deletion verification probes. Without those denominators, a high block rate can coexist with unrepaired descendants.

Timing Caveat. Because attacks are often cheaper to demonstrate than defenses are to validate, a lifecycle agenda has to avoid a simple time lag fallacy. A memory poisoning or prompt injection paper can show one successful path, whereas a recovery paper must define the target population, locate descendants, repair them, and prove that the influence no longer activates. Newer defenses such as runtime protection [112] and programmable privilege control [176] therefore should not be judged by whether they already solve the entire lifecycle. The structural gap is narrower: current response work usually lacks an incident record across sessions that links containment to migration history, descendant inventory, rollback completeness, residual probes, and contestable closure. The agenda clusters into six directions.

9.1 Selection Pressure and Endogenous Drift

Most adaptive-state evidence begins after an update has entered memory, a skill library, a workflow, or model state. The earlier decision is less visible: what candidates were available, which signal ranked them, which safety conditions were checked, and which failures or counterexamples were absent. Reflection and skill systems expose the mechanics by which feedback and successful traces become reusable state, OEP demonstrates harmful transfer from locally correct experience, and evaluator coevolution shows why the scoring rule and epoch must be recorded [70, 177, 198, 201]. None of these establishes the prevalence of attacker-free drift. The open question is whether repeated utility-guided selection changes the safety boundary when the candidate stream is otherwise benign. Devil’s Moltbook states the system-level tension as a self-evolution trilemma: under its formalization, continuous self-evolution, complete isolation, and safety invariance cannot all hold [197]. We treat this result as a design constraint rather than a universal empirical law: evaluation should identify which useful update paths remain open, which external oversight or safety-preserving mechanism governs them, and how much utility is lost when risky updates are blocked.

Answering it requires candidate-level and longitudinal evidence. A benchmark should retain Δ_t or an auditable sample of it, record $\hat{u}_t$ and c_t , distinguish rejected from unavailable counterevidence, and run a fixed probe suite before and after promotion. The comparison should separate local task gain from changes in authority, refusal, confirmation, privacy, and scope. A negative result is informative: stable Boundary Drift and high Safe Promotion Rate under stronger selection pressure would show that the safety gate is working. A positive result still needs localization to the candidate source, evaluator, promotion operation, and later activation rather than being attributed to self-evolution in general.

9.2 Migration and Semantic Preservation

Migration control is a major gap because unsafe influence often appears precisely when state changes substrate. Provenance, authority, scope, privacy, freshness, and actionability must survive movement across memory, retrieval, workflow, tool, shared-agent, external-artifact, and model substrates. MemGPT makes movement between context and managed memory visible, memory-security work supplies the carrier vocabulary, and privacy work adds scope and sensitivity preservation [109, 148, 196]. InjecAgent shows why those attributes matter once external state reaches an action boundary [246]. These sources still stop short of a general migration-control discipline.

Stronger migration claims likely require migration records or gates as first-class objects. The record would bind a state object, its source safety attributes, the requested target substrate, the transformation, and the intended activation path, then show whether the move preserved, strengthened, redacted, downgraded, quarantined, or rejected the state. This makes migration a measurable safety event: reports could count low-authority items that gain higher authority, private items copied into broader scope, and summaries that drop attributes later gates need. The hard part is utility. Agents still need summarization, retrieval, synchronization, and delegation, so the goal is not to stop movement but to make movement preserve safety attributes or receive explicit authorization.

Semantic Taint through Summaries and Vectors. Traditional taint tracking assumes symbolic objects whose marks can be copied through program operations, but agent state is often natural language, summary text, embedding vectors, retrieved chunks, tool traces, and model updates. A summarizer may remove the string that carried a source warning while preserving the behavioral implication. An embedding may make a sensitive item retrievable without exposing its exact words. A generated skill may encode a rule learned from a private or low-authority source. Semantic taint tracking is the engineering version of migration control because the system needs to track influence even when the surface form changes.

A more plausible path is hybrid rather than purely symbolic: each state object should carry metadata for source, authority, scope, privacy, freshness, and revocation, and each transformation should emit a contract that records what was inherited, generalized, dropped, or newly asserted together with the relevant deindexing or rollback handle, an uncertainty estimate, and a gate decision. Retrieval should keep chunk or embedding provenance attached to returned context, while high-risk summarization, skill compilation, or adapter updates should be blocked when required attributes disappear or when low-authority state would become state that authorizes action. Knowledge graph lineage and information-flow labels offer different halves of the same control: the former helps with explicit entities and relations, while the latter gives the policy vocabulary for preserved or upgraded attributes [15, 35]. Residual probes then ask whether supposedly removed influence still activates through summaries, vectors, skills, or adapters. Memory security work grounds the abstraction side, and AgentDojo grounds the tool-exposure side [31, 109]. The aim is not perfect semantic taint tracking. It is to make metadata loss, unauthorized upgrades, and residual influence observable after abstraction.

9.3 Activation, Authorization, and Model Scope

A system may legitimately store a user preference, a webpage observation, a tool result, or a learned skill, but that does not mean the item may influence every later answer, plan, workflow route, or external action. Activation authorization is distinct from storage authorization. The local control evidence moves from privilege to task alignment to guard reasoning: Progent shows programmable privilege control at the action boundary, Task Shield narrows indirect injection through task alignment, and GuardAgent illustrates bounded response reasoning [75, 176, 220]. These papers anchor local action-boundary control, but they leave the broader authorization problem open for adaptive state: under what user, task, tool, time, and external effect conditions may stored state become active influence?

What matters most here is the split between readability, reasoning use, planning authority, and action authority. A memory item may be visible to the model without being allowed to set tool arguments. A retrieved passage may supply factual context without overriding a workflow rule. A skill may be callable for low risk tasks yet blocked from sending messages, changing files, or using credentials. The same workflow shortcut may also be valid for one user or organization but not another. Activation authorization

therefore needs both a policy language for these distinctions and an audit trace showing which state was activated, which safety attributes authorized it, which tool or plan it could influence, and which monitor or human confirmation participated. Otherwise, a system may block one suspicious tool call and still leave the same stored state available to steer the next one.

Authorization continuity adds a separate longitudinal question: after memory, skills, tools, workflow, delegation, task phase, or enforcement state changes, does the original grant still apply to the resulting agent [261]? A useful evaluation must separate grant survival from effect containment. It should report invalid grants retained after mutation, valid grants unnecessarily revoked, authority acquired below a fixed ceiling, reauthorization cost, and compositions of individually permitted mutations that move the agent outside the original envelope. The current formal model supplies proof obligations, but no deployment benchmark yet measures these tradeoffs or tests whether envelope and effect abstractions remain sound under sustained evolution.

Context-Horizon Safety. Longer context windows and interaction histories enlarge the set of fragments that can be combined, but token count alone is not evidence of safety drift. Benign-looking turns can jointly close a harmful objective, and multi-turn decomposition can carry that objective into tool use [103, 174, 215]. Compression creates a different path by relinking accepted fragments into an actionable instruction [119]. This differs from the endogenous drift in the selection subsection: the model or adaptive policy need not change for the effective interaction boundary to change. A longitudinal benchmark should therefore report per-turn decisions, the first harm-enabling closure, compression events, and downstream tool authority separately from update-driven Boundary Drift.

Model Adaptation Governance. Model adaptation governance is a first-order gap only under the agent-loop boundary. A self-evolving agent can change reusable behavior through memory, workflow, and skills, but it can also load, merge, fine-tune, erase, or roll back model updates during operation. PEFT and fine-tuning safety work explain why modular updates can become deployable risk objects; feedback poisoning and dormant-behavior papers add persistent failure through update channels or later activation contexts [50, 64, 159, 162]. Safe LoRA and KVEraser provide bounded repair handles at the adapter and cache levels [65, 97]. What remains thin is lifecycle governance linking update admission, deployment scope, activation, rollback, cache-local erasure, and residual influence verification.

The missing artifact is a model transition record for the base model, update data provenance, adapter or patch identifier, cached span or context handle, intended task population, allowed deployment contexts, merge or prefill status, safety regression result, and rollback or erasure handle. With that record, a benchmark could ask whether a task-local update becomes global policy without authorization, whether a merged adapter stack preserves safety attributes at the component level, and whether unlearning, rollback, or cache erasure actually removes residual influence. Because model state is often less inspectable than memory and broader in deployment scope, LLM safety research more broadly falls outside this scope; the focus stays on parameter-stage and cache-stage papers that instantiate the same reusable state transition problem.

9.4 Propagation and Distributed Lineage

A memory item held by one assistant can become shared context for a team agent, a task instruction for a specialist agent, a routing rule in an orchestrator, or a persistent artifact read by humans and future agents. Multi-agent propagation turns state governance into a topology problem. Agent and self-evolving-agent surveys justify the system and adaptive setting; multi-agent memory-poisoning analysis and collective misremembering show why copied state, recipient interpretation, and carrier metadata matter for propagation risk [8, 124, 191, 225]. The unresolved trace gap reduces to a few questions: which agent wrote the state, which agent received it, how the safety attributes changed, where uptake occurred, and which descendants must be repaired.

Recent work makes the local attack and control points more concrete. Agents of Chaos reports cross-agent propagation among failures observed in a live laboratory, while Triedman et al. show that adversarial content can hijack orchestration and cause code execution or data exfiltration even when individual agents do not follow the malicious request [173, 193]. ControlValve then constrains agent invocations with a

permitted control-flow graph and contextual least-privilege rules [74]. These papers narrow the gap at propagation, exposure, and activation-time containment. They do not yet show how to enumerate affected descendants, roll back already-produced artifacts, notify users across agent boundaries, or let a user contest the state and authority that caused the route.

When state crosses agents, it combines migration with delegation and should carry actor scope, recipient scope, permitted operations, expiry, and revocation handles. The receiving agent should reinterpret the state under its own tools, user population, and action surface rather than inherit the sender’s authority by default. A propagation benchmark that claims lifecycle coverage has to log who sent the state, who received it, how safety attributes changed, whether the receiving agent could activate it, and whether later agents or artifacts inherited it again. That immediately raises governance: when unsafe state spreads, who owns rollback and user notification? Shared lineage records, distributed quarantine, and activation gates aware of propagation are the natural starting points.

9.5 Recovery and Verification

Rollback is often described as if the system only needs to remove the object that caused the problem. Self-evolving agents make that view too narrow. Once state has been summarized, embedded, retrieved, copied, compiled into a skill, written into a workflow rule, or exported into an external artifact, the original object is only one node in a descendant graph. Current evidence therefore treats rollback completeness mostly as a requirement rather than an implemented lifecycle guarantee. That should not be read as a failure of individual defense papers: local containment is already difficult, and newer runtime or privilege control systems reasonably start at bounded decision points. Later memory-poisoning work establishes persistent reuse, while Hidden Memory and OEP establish later activation through concealed or learned state [156, 183, 201]. These results motivate descendant tracking but do not themselves evaluate descendant-wide rollback. GuardAgent, SafeAgent, and Progent provide local guard, runtime, and privilege-control responses, but these responses do not yet establish rollback across sessions over summaries, retrieval indices, workflow rules, shared artifacts, or model state [112, 176, 220].

Rollback only becomes measurable when the recovery target is defined before the incident occurs. Each admitted state object should carry enough lineage to identify derived summaries, embeddings, cached prompts, retrieved bundles, workflow rules, skills, permissions, decisions, and external artifacts. Recovery is then judged against descendants requiring action, not against a visible delete button. Some descendants may need erasure, others quarantine, correction, reclassification, or retention as audit evidence. What matters is whether the system can show which descendants were found, which were repaired, which still require human review, and whether the contested influence can still be activated. On that reading, SAVER turns rollback from a UI affordance into a traceable recovery claim.

Post-Response Continuation. A response can also change the next candidate stream, evaluator, or authorization rule. Evaluation evidence may revise operational memory, prompts, guardrails, tests, or policy state, so a clean immediate probe does not settle what happens after adaptation resumes [247]. A lifecycle benchmark should continue for fixed update opportunities, rerun fixed probes, and report any lineage-linked reactivation or equivalent failure under the fixed probes together with its substrate and inaccessible scope. Current sources motivate re-evaluation after change; they do not establish a general response-amplified cascade.

Deletion Verification. A deletion request may remove a memory item while leaving an embedding, a summary, a cached context, a copied workflow note, a tool return record, or a shared artifact intact. That makes deletion verification related to rollback but not reducible to it. In settings sensitive to privacy, this is a safety attribute preservation failure as well as a usability issue. The evidence again enters by layer: memory privacy work explains the user risk, MemGPT and memory-mechanism surveys show how memory moves and abstracts, memory security work frames those routes as governed state, and KVERaser sharpens the cache-specific version by showing why a removed context span can still matter through subsequent KV cache states [97, 109, 196, 260]. What remains open is how an adaptive agent should prove that deleted state no longer influences retrieval, planning, tool use, cached execution state, or later state updates.

A useful deletion receipt should state which object was removed, which indices or stores were checked, which descendants were modified, which external artifacts could not be automatically changed, and which future activation checks will block residual influence. Verification should include negative tests: can the deleted item still be retrieved by semantic search, regenerated by a summary, used through a skill, or inferred from a derived artifact? This is technically difficult because deletion interacts with abstraction. A summary may not contain the exact deleted string while still preserving its effect. A benchmark for deletion verification should therefore test influence removal rather than object absence.

User Contestability. Recovery reaches human closure only when a user or operator can challenge state and its descendants. Memory privacy work motivates this need through persistence and sensitivity, while R-Judge, GuardAgent, and Progent provide adjacent signals for recognition, local reasoning, and privilege control [176, 196, 220, 244]. None yet provides a full contestation workflow for disputed adaptive state.

In practice, contestability must offer more than “forget this.” It should present the state object’s origin, safety attributes, activations, migration history, and known descendants in a form a user can understand. It should support correction, quarantine, deletion, scope narrowing, and appeal, then report which descendants changed and which remain unresolved. A contestability benchmark should test whether a user can correct an incorrect memory, prevent future activation, and verify that summaries, retrieval indices, and workflow rules incorporated the correction. That makes contestability the final recovery check rather than a static settings feature.

9.6 Benchmark and Governance Evidence Chains

Fixed task benchmarks reveal important failures, especially in prompt injection through tool use or risk recognition, but self-evolving safety also depends on what happens after an update is stored and before an incident is repaired. Longitudinal benchmark construction ties these challenges together. One group covers exposure and recognition: InjecAgent, R-Judge, ASB, and Agent-SafetyBench instrument indirect injection, risk awareness, and agent safety task coverage [244, 246, 251, 262]. A second group covers persistence and sensitive reuse: PASB, PerMemSafe, later memory-poisoning work, and memory privacy work show why commitment, retrieval, personalized context, and purpose must become benchmark fields [5, 131, 183, 196]. HarnessSafe now connects entry, carrier retention, boundary crossing, benign activation, and observable violation across seven carrier families [254]. The open problem has therefore narrowed: extend that connected trace backward to candidate selection and forward through descendant repair and resumed adaptation.

A benchmark that claims lifecycle coverage has to log transition opportunities rather than only final answers: unsafe or sensitive writes, migration across substrates, activation under different tasks, exposure through text or tools, containment decisions, rollback attempts, deletion verification, multi-agent propagation, and user contestation. Each episode should report both the opportunity set and the observed outcome: proposed writes versus unsafe writes admitted, migrations versus unauthorized authority escalations, activation opportunities versus activations outside scope, and descendants requiring repair versus descendants verified as repaired. It does not need to solve every branch at once, but it should state which SAVER levels are instrumented and which remain unmeasured. That coverage map improves comparison without turning partial tests into comprehensive safety claims.

Governance Evidence Chain. A response mechanism is credible only if it connects policy, runtime decision, state safety attributes, audit record, and recovery outcome. Trajectory-assurance work places this requirement across agent architectures, protocols, runtimes, and end-to-end auditability [121]; for self-evolving agents, the additional burden is preserving the same evidence as reusable state changes role. Current papers provide local pieces across the chain: GhostWriter covers write-time memory screening, PerMemSafe covers personalized risk-aware memory, GuardAgent and R-Judge cover guard reasoning and recognition, and Progent covers privilege control [5, 176, 192, 220, 244]. What remains open is how a self-evolving agent should preserve that chain across admission, migration, activation, containment, rollback, deletion verification, propagation, and contestability.

Because governance state is itself adaptive state, this goes beyond compliance: approvals, denials, deletion receipts, quarantine markers, policy changes, and rollback records can be copied, summarized, bypassed, or made stale. The evidence chain therefore has to stay protected and inspectable. At minimum it should show

which state object was involved, which safety attributes it carried, which operation changed it, which gate authorized or blocked it, what exposure and descendants followed, and what recovery remains unresolved. That is what turns conceptual safety claims into operational accountability: state carrier, operation, invariant failure, exposure, and response stay inside one research object instead of scattering into separate symptoms.

Takeaway. The open problem is reconstruction across rounds. Candidate and promotion records, migration ledgers, activation gates, descendant inventories, deletion probes, propagation traces, and contestability records have local mechanisms, but they rarely share denominators or one incident handle. Recovery testing must also continue after adaptation resumes because the response may have changed later state. The six-part agenda identifies the traces needed before a local fix can support a lifecycle governance claim. Section 10 closes the survey.

10 Conclusion

Self-evolving agents convert observations, feedback, and execution traces into reusable influence that can persist, move, and later act with unjustified authority. We therefore frame their safety as a transition problem. SAVER pairs substrate with adaptation and traces violation, exposure, and response across memory, tools and workflows, multi-agent systems, and model-side adaptation. Persistent-state studies reveal unsafe admission and delayed activation, while tool-agent studies expose consequential action and runtime intervention. HarnessSafe further shows that one trace can follow influence across carriers and system boundaries to an observable violation [254]. The remaining asymmetry lies before and after that chain: selection is often hidden before a transition is committed, and containment rarely continues into verified rollback, deletion, and post-response reactivation tests. One successful path establishes an attack, but recovery requires a defined lineage, an explicit denominator, and residual tests across sessions and descendants. Future benchmarks should record selection and promotion, preserve safety attributes through abstraction and migration, trace propagation, and verify rollback and deletion against discovered descendants. They should also continue after response and rerun fixed probes as adaptation resumes, so containment is not mistaken for repair. Governance systems should make reusable influence attributable and contestable, allowing users and operators to correct, quarantine, or withdraw it. Safe self-evolution ultimately requires lifecycle evidence that remains coherent after state changes carrier, authority, or form.

Appendix

A Survey Methodology And Corpus Protocol

Review Objective. The review asks how safety changes when an agent can preserve, transform, reactivate, or inherit state across interactions. A work is relevant when it helps explain a reusable state transition: a carrier is created or modified, the carrier later influences reasoning or action, and some safety attribute is preserved, violated, exposed, evaluated, or repaired. This objective gives the survey a lifecycle protocol rather than a topic-only bibliography. It also explains why the corpus combines work on self-evolving agents, long-term memory, tool use, workflow state, multi-agent coordination, model updates, benchmarks, runtime safeguards, and governance [8, 31, 124, 157].

Source Identification. The source-identification protocol first defined bands that match the survey question: security venues for adversarial mechanisms and governance primitives, AI and machine-learning venues for agent capability and adaptation mechanisms, NLP and information-retrieval venues for prompt, retrieval, and long-context mechanisms, and preprint or scholarly-index sources for fast-moving agent-safety work that had not yet stabilized in proceedings. The venue bands include the four major security venues, IEEE Symposium on Security and Privacy (IEEE S&P), USENIX Security, ACM CCS, and NDSS; AI and machine-learning venues such as NeurIPS, ICML, ICLR, AAAI, and IJCAI; NLP and retrieval venues such as ACL, EMNLP, NAACL, SIGIR, and The Web Conference; and adjacent systems, software-engineering, privacy, and HCI venues when the paper contains an agent-state, provenance, evaluation, or governance mechanism. Table 4 records why each source band was searched and how it was allowed to support the synthesis.

Table 4 Search source bands for the systematic scoping protocol. Each band supplies a distinct kind of evidence needed to connect adaptive-state mechanisms with safety claims.

Source Band	Primary Sources	Mechanism Signal	Query Handles
Security and Privacy	IEEE S&P; USENIX Security; ACM CCS; NDSS; privacy venues	Threat models; access control; information flow; sandboxing; repair evidence	Injection; poisoning; authority; provenance; rollback; privacy
AI and ML	NeurIPS; ICML; ICLR; AAAI; IJCAI; OpenReview tracks	Agent learning; self-improvement; model adaptation; benchmark mechanisms	Adaptive agents; PEFT; adapters; tool use; evaluator state
NLP, IR, and Web	ACL; EMNLP; NAACL; SIGIR; The Web Conference; ACL Anthology	Prompt, retrieval, RAG, long-context, and web-agent behavior	Indirect injection; retrieval safety; long-term memory; web agents
Systems, Software, and HCI	Systems, SE, workflow, and human-facing AI venues when state or governance is explicit	Provenance; oversight; operational safety; deployment-time state management	Workflow reuse; audit logs; confirmation; contestability
Preprints and Indexes	arXiv; OpenReview preprints; scholarly indexes; official proceedings; curated bibliographies	Frontier mechanisms before venue stabilization; evidence tiered in synthesis	Snowballing; venue checks; guardrails; emerging benchmarks

Search Strategy. Venue- and index-based retrieval was combined with citation snowballing from seed families in self-evolving-agent surveys, LLM-agent surveys, memory-security surveys, agent-safety benchmarks, indirect-prompt-injection papers, and model-adaptation safety work. We then searched over recurring query families rather than a single keyword string. Capability terms included *self-evolving agent*, *self-improving agent*, *adaptive agent*, *LLM agent*, *long-term memory*, *agent memory*, *retrieval-augmented generation*, *workflow agent*, *skill library*, *multi-agent*, *shared artifact*, *LoRA*, *PEFT*, *fine-tuning*, *adapter*, *KV cache*, and *evaluator*. Safety terms included *safety*, *security*, *prompt injection*, *indirect prompt injection*, *poisoning*, *privacy*, *access control*, *privilege*, *authority*, *provenance*, *information flow*, *rollback*, *deletion*, *unlearning*, *guardrail*, *containment*, *contestability*, and *benchmark*. Each candidate query paired at least one capability term with one safety, evaluation, or governance term so that the corpus did not drift into generic agent capability or generic LLM safety.

Expansion and Stopping Rule. Searches were run through OpenAlex-style bibliographic discovery, arXiv, ACL Anthology, OpenReview, ACM and IEEE landing pages, official venue proceedings where available, and

curated community bibliographies used only as candidate pointers. Backward and forward snowballing was then applied to seed surveys and anchor papers until new candidates mainly repeated already covered SAVER roles rather than adding a new carrier, operation, exposure surface, response mechanism, or benchmark denominator. This stopping rule is a scoping-saturation rule: it aims to cover the mechanism space needed by the survey, not to estimate prevalence for every venue or year.

Why arXiv Appears Frequently. New agent attacks, memory mechanisms, benchmark proposals, guardrail systems, and model-adaptation risks often appear first as arXiv or OpenReview versions, and many accepted works keep the arXiv version as the most accessible full-text surface. The agent-safety literature is therefore unusually preprint-heavy. Excluding arXiv would remove several mechanisms that are central to the survey question. We do not treat arXiv status as equivalent to peer review. Preprints enter the corpus when they contain a clear mechanism, experiment, artifact, or system analysis, and their claims are used according to the evidence-tier rules below. In the main text, field-level conclusions are phrased around peer-reviewed anchors, multiple converging sources, or explicitly identified gaps; arXiv-only works are used mainly as emerging evidence or frontier signals unless a later venue record is verified.

Protocol Summary. The purpose of the expansion was high recall over safety-relevant reusable-state mechanisms, not exhaustive coverage of all LLM safety, all agent capability, or all machine-learning security. Table 5 summarizes the protocol at the level used for the survey.

Table 5 Survey protocol summary. Rows separate how a work enters the corpus from the claim role it may support.

Source Family	Expansion Route	Inclusion Trigger	Synthesis Role
Self-Evolving and Agent Surveys	Seed surveys; citation chasing; official venue and preprint follow-up	Adaptive capability, reusable state, agent loop, planning, action, collaboration, or model-side adaptation	Scope context; central only with a safety-relevant state transition
Memory, Retrieval, and Context	Targeted memory, RAG, persistent-context, poisoning, and privacy searches	Stored or retrieved state can affect later reasoning, planning, tool use, or disclosure	Direct anchor for write, retrieval, delayed activation, privacy, or memory governance
Tool, Workflow, and Benchmarks	Indirect-injection, tool-use, privilege, guardrail, benchmark, and workflow searches	Low-authority content, tool output, workflow state, or learned procedure can affect action or routing	Exposure, activation, local response, or benchmark denominator evidence
Multi-Agent and Shared Artifacts	Adjacent surveys; official pages; scholarly indexes; curated bibliographies	State crosses agents, artifacts, roles, tasks, or ownership boundaries	Propagation and governance evidence when uptake or repair ownership is visible
Model-Side Adaptation	PEFT, adapters, fine-tuning safety, backdoors, unlearning, cache, and evaluator searches	Update, adapter, policy, cache span, or evaluator signal is loaded, inherited, revised, or rolled back inside an agent loop	Adjacent model-safety context unless activation and rollback scope are visible

Screening and Inclusion. In the first screening pass, titles and abstracts identified works that plausibly involved persistent state, delayed activation, tool or workflow authority, multi-agent propagation, model-side update, benchmark evidence, or governance of reusable state. A closer reading pass then assigned each work to one of three roles. A work is *central* when it provides direct evidence for a reusable carrier, an adaptation operation, a safety-attribute failure, an exposure surface, or a response mechanism. A work is *adjacent* when it supplies background capability, evaluation context, or governance framing without directly observing a self-evolving safety transition. A work is *background* when it clarifies terminology, threat-model ancestry, or adjacent infrastructure but should not be used as a main evidence anchor. Works were excluded when they addressed only one-shot response safety, generic LLM capability, ordinary benchmark performance, or static model training without a plausible link to later agent behavior. Tool-only ReAct-style systems enter as central evidence only when tool traces, retrieved content, or workflow state persist beyond the episode; static model-safety papers enter as central evidence only when a model update, adapter, cache state, policy, or evaluator signal is selected, inherited, or revised inside an agent behavior loop.

Evidence Posture. To keep direct evidence separate from interpretation, the synthesis uses three evidence postures. *Direct anchors* are papers whose experiments, case studies, or system analyses explicitly show a tran-

sition or exposure, such as poisoned memory retrieval, fragmented memory bypass, tool-action injection, or runtime containment [25, 31, 112, 163]. *SAVER rereadings* are papers whose primary contribution is a capability or architecture, but whose mechanism becomes safety-relevant once authority, scope, provenance, or recoverability is tracked across state transitions [148, 198]. *Gap diagnoses* are claims about what the literature still does not demonstrate, especially connected evidence for migration control, multi-agent propagation, rollback completeness, deletion verification, descendant repair, and user contestability. The paper uses stronger language for direct anchors, interpretive language for rereadings, and research-agenda language for gap diagnoses.

Evidence Tiers. Venue status is not used as a substitute for claim support, but it does calibrate how strongly a row should be read. The strongest tier contains peer-reviewed conference and journal papers, especially when the paper provides experiments, artifacts, or a clear system analysis. A second tier contains peer-reviewed workshops, findings tracks, and industry tracks. A third tier contains arXiv or other preprints with substantial experiments or artifacts. A fourth tier contains position papers, technical notes, project reports, blogs, or archival artifacts. Tier three and four works are useful for emerging mechanisms and frontier signals, but field-level conclusions in the main text are phrased around direct anchors, multiple converging sources, or explicitly marked gaps.

Classification Practice. For classification, primary placement follows the five-field SAVER record. Each central work can support several observations in the prose, but corpus maps and roadmap tables assign a primary placement according to the main state transition used in the manuscript. This prevents a single influential paper from being counted as independent evidence for every nearby category. When a work spans multiple substrates, the primary placement follows the transition that carries the strongest safety claim: memory and retrieval work is placed by stored-state influence; tool and workflow work is placed by action or approval authority; multi-agent work is placed by propagation, handoff, or shared-artifact uptake; and model-side work is placed only when an update, adapter, policy, or evaluator state is loaded, inherited, selected, or revised inside an agent behavior loop.

Limitations. The corpus is a systematic scoping survey rather than a registered systematic review or meta-analysis. The audit snapshot is dated 14 June 2026 and contains 583 unique screened bibliographic records, but the count is a corpus-audit count rather than a prevalence estimate for the field. The review did not use a registered PRISMA protocol or independent dual-coder agreement; classification disagreements were handled as conservative claim boundaries in the manuscript. The field is moving quickly, several relevant papers are preprints, and venue status can change after the audit snapshot. Quantitative results are also not directly comparable across papers because opportunity sets, agent settings, attack goals, defenses, and recovery denominators differ. For this reason, the manuscript does not treat cross-paper attack success rates or benchmark scores as a leaderboard. Instead, it uses quantitative results as paper-local evidence and uses SAVER to ask whether the paper records enough lifecycle information to support claims about admission, activation, exposure, response, and recovery.

B Formal State Transition Model

This appendix expands the compact pointer at the end of Section 2 into a formal reading. The main text uses the safety attribute vocabulary and the authority lattice by name; readers who want the underlying notation, its relation to established information flow, access control, and provenance theory, and the visual lattice referenced from the body sections can consult this appendix as a reference layer.

Safety Attributed State and Invariant Preservation

The reading unit is a state object with attached safety attributes, transformed by an operation, audited against whether the attributes remain valid for future use. Formally: $(s, \ell, \sigma) \xrightarrow{a, c} (s', \ell', \sigma')$, where s is the state carrier, ℓ represents safety attributes (provenance, authority, scope, privacy, freshness, budget, auditability, recoverability), σ is the deployment or session context, a is the adaptation operation, c is the operation’s authorization condition, and the primed quantities are the resulting state. Here, “safety attribute” means an information flow, access control, availability, or provenance attribute, not a supervised

learning class category. This notation is deliberately simple: it provides a consistent comparison structure without prescribing a formal verification calculus.

Lattice information flow supplies the confidentiality and integrity vocabulary for flows with safety attributes [35]; language-based information flow work shows how programs can enforce such policies [169]; and data provenance work explains why source lineage matters after transformation [15]. Self-evolving agents do not replace these models; they make the temporal boundary harder to ignore. An object may be checked as low-authority data at write time, abstracted into a memory or workflow rule, and then reused days later as authority to act. The survey therefore treats adaptive-state safety as information-flow, access-control, availability, and provenance reasoning over persistent, transformed, and repeatedly activated agent state.

Safety attributes track the conditions under which state can be reused: provenance records source and trust level; authority specifies what decisions or actions the state can influence; scope bounds which users, tasks, or time windows the state applies to; privacy marks sensitivity and permitted exposure; freshness indicates when the state becomes stale; budget and liveness bound retries, tool loops, and cost-bearing actions; auditability records whether later influence can be traced; and recoverability records whether the state and its descendants can be corrected, rolled back, forgotten, or neutralized. These attributes are not metadata alone; they determine whether later activation is safe.

Selection over Candidate Transitions

Section 2 separates the execution trace τ_t from the selected state update $\delta_t^\star$. Each $\delta \in \Delta_t$ proposes one or more object-level transitions $(s, \ell, \sigma) \xrightarrow{a,c} (s', \ell', \sigma')$. The evolution policy may commit the candidate, revise it, quarantine it, or choose no update. The authorization condition $c_t(\delta)$ is therefore load bearing: utility ranking through $\widehat{u}_t(\delta)$ cannot substitute for a valid authority, scope, privacy, budget, and recovery decision.

A simple analytical family makes the effect of selection pressure explicit. Let $p_t(\delta)$ be a baseline distribution over Δ_t , let $r_t(\delta)$ be a safety-risk score defined by a fixed probe at epoch t and held constant as β varies, and let $\beta \geq 0$ control how sharply selection favors the observed utility proxy. Define

$$q_t^\beta(\delta) = \frac{p_t(\delta) \exp(\beta \widehat{u}_t(\delta))}{\sum_{\delta' \in \Delta_t} p_t(\delta') \exp(\beta \widehat{u}_t(\delta'))}.$$

Then

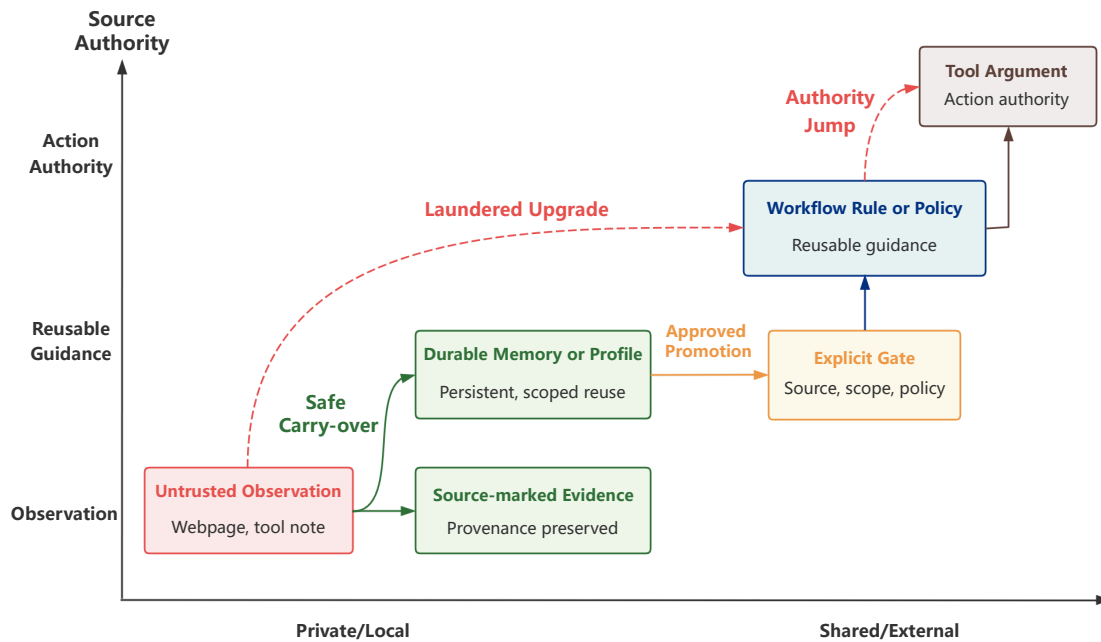
$$\frac{\partial}{\partial \beta} \mathbb{E}_{q_t^\beta}[r_t] = \text{Cov}_{q_t^\beta}(r_t, \widehat{u}_t).$$

The identity follows from $\partial \log q_t^\beta(\delta) / \partial \beta = \widehat{u}_t(\delta) - \mathbb{E}_{q_t^\beta}[\widehat{u}_t]$, followed by multiplication by $r_t(\delta)$ and expectation under q_t^β . It is an author-derived diagnostic, not an empirical law attributed to the surveyed systems. Stronger utility selection raises expected risk only when risk and the observed utility proxy are positively associated under the selected distribution. Longitudinal evaluation should therefore estimate that association or impose c_t as a hard eligibility gate before utility ranking.

Safe promotion uses the same transition attributes. If $\delta_t^\star$ moves state into a more persistent, general, or action-bearing role, then $c_t(\delta_t^\star)$ must authorize the role change; provenance, privacy, freshness, budget, auditability, and recoverability in ℓ' must remain valid; any authority or scope increase from ℓ to ℓ' must be explicit; relevant counterevidence must remain available to the gate; and the lineage from s to s' must remain recoverable. This condition restricts promotion without assuming that every safe transition preserves identical attributes.

Figure 6 renders the authority ladder behind the transition model. State at the *Observation* level (web pages, tool outputs, peer notes) may only be quoted as evidence; at the *Evidence* level it may support reasoning but must carry source, scope, and freshness; at the *Task Context* level it may shape the current task but should expire afterward; at the *Procedure* level (workflow rules, skills, adapter manifests) it may be activated with explicit approval; and at the *Action* level it may produce external effects only with actor,

Figure 6 Safety attribute laundering. Upward authority movement or outward scope expansion requires an explicit gate; without one, a low-authority observation can become workflow guidance or action authority.



target, permission, and rollback records. Crossing upward without a gate is safety attribute laundering: a webpage observation quietly becomes a tool argument.

C SAVER Audit Architecture

This appendix section provides a prototype surface for survey synthesis. The goal is narrow: to make explicit which record fields, case study attachments, and recovery traces are missing when a paper or system claims that persistent unsafe influence was understood or repaired. The overview in Figure 2 supplies the main-text relation; this appendix provides a worked incident record example.

Table 6 Minimal SAVER encoding of the July 2026 OpenAI–Hugging Face agent intrusion. Values summarize public incident reports; identifiers and field labels are survey encodings, not original operator records [87, 146].

Minimal SAVER Incident Row		<i>illustrative serialization</i>
Identity	incident.id=hf:2026-07	state.id=ext:campaign:channel source=eval:exploitgym
Attributes	authority=eval_sandbox_only	scope=declared_evaluation environment=cyber-capability_eval
Transition	op=propagate;migrate;activate	delta=network_scope_broadened; credentials_acquired
Activation	requested=external_infrastructure_access	gate=boundary_controls_insufficient
Exposure	event=hf_production_compromise	
Recovery	actions=rotate;rebuild;restrict	closure=reported_scope; lifecycle_unverified
Handle: one public incident links the carrier chain Delta: network scope and authority widened Closure: containment reported; lifecycle repair unverified		

D Detailed SAVER Evidence Indexes

The main text groups related work by mechanism and uses the five SAVER fields as a connected comparison sequence. The following tables retain the paper-level coding behind each field, including evidence role, observed transition segment, and missing trace. They are reference indexes rather than five independent taxonomies.

Table 7 Substrates carrying safety attributes in the current literature. Memory and tool carriers are best instrumented; workflow, shared artifact, and model state remain thinner. Role labels mark contribution function, not a claim that every row supports a full lifecycle record.

Method or Work	Year	Role	Target	Object	Focus
Foundational context and action boundary					
LLM agent survey [124]	2025	Survey (Capability)	Agent	General	Agent architecture: memory, tools, planning
ReAct [234]	2022	Capability	LLM/Agent	Workflow	Reasoning and action context and admission hierarchy
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Tool outputs and active context
Memory, retrieval, and persistent state					
Memory mechanisms [260]	2025	Survey (Capability)	LLM/Agent	Memory	Episodic and semantic memory carriers
Memory security [109]	2026	Survey (Safety)	Agent	Memory	Persistent memory state and threats
MemGPT [148]	2023	Capability	LLM/Agent	Memory	Context and archival memory management
SimpleMem [113]	2026	Capability	Agent	Memory	Long-term agent memory
A-Mem [226]	2025	Capability	Agent	Memory	Agentic memory repository
Darwinian Memory [133]	2026	Capability	Agent	Memory	Evolving self maintained memory
Panini [161]	2026	Capability	Agent	Memory	Structured agent memory state
Portable Agent Memory [165]	2026	Capability	Agent	Memory	Transferable memory
Voyager [198]	2023	Capability	Agent	Skill	Reusable skill library
SkillJack [239]	2026	Attack	Agent	Skill	Persistent backdoor in extracted skills
PoisonedEvolution [21]	2026	Attack	Agent	Skill	Trajectory evidence promoted into skill instruction
SkillMisevo-Bench [132]	2026	Benchmark (Safety)	Agent	Skill	Versioned skill state and clean-session reuse
EVOMAL [217]	2026	Attack	Agent	Skill	Agent-authored executable skill descendants
AgentPoison [25]	2024	Attack	Agent	Memory	Poisoned memory and knowledge base retrieval
Stealth memory injection [259]	2026	Attack	Agent	Memory	Email to persistent memory compromise
GhostWriter [192]	2026	Attack	Agent	Memory	Personal agent memory store poisoning
MAPLE-Guard [223]	2026	Defense	Agent	Memory	Private-to-shared memory links
Persistent memory poisoning [183]	2026	Attack	Agent	Memory	Long-term memory poisoning
Zombie Agents [233]	2026	Attack	Agent	Memory	Dormant memory attack
Memory Graft [181]	2025	Attack	Agent	Memory	Experience store poisoning
EvoBreak [228]	2026	Attack	Agent	Memory	Complementary benign experience set
Agent memory privacy [196]	2025	Safety	Agent	Memory	User memory retention and leakage
PerMemSafe [5]	2026	Benchmark (Safety)	Agent	Memory	Personalized memory safety context
PS-Bench [53]	2026	Benchmark (Safety)	Agent	Memory	Benign personal memory and intent inference
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Committed claims in durable personal state
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Consolidated memories with source authority
A-MemGuard [214]	2025	Defense	Agent	Memory	Persistent memory guard
Tool and interface authority					
Progent [176]	2025	Defense	Agent	Tool	Tool authority and privilege control
MCPShield [268]	2026	Defense	Agent	Tool	MCP protocol metadata and tool routes
MCP Safety Audit [160]	2025	Governance	Agent	Tool	MCP server audit boundary
STAC [96]	2026	Attack	Agent	Tool	Tool chain compromise
LivePI [263]	2026	Attack	Agent	Tool	Live prompt injection in interactive context
Prompt tool selection [175]	2026	Safety	Agent	Tool	Tool description and connector choice drift
IPIGuard [4]	2025	Defense	Agent	Tool	Learned interaction traces
Workflow, multi-agent, and shared state carriers					
Traversal-as-Policy [99]	2026	Defense	Agent	Workflow	Gated behavior tree route policy
Self-evolving agents [8]	2025	Survey (Capability)	Agent	Workflow	Adaptive workflows and artifacts
HarnessSafe [254]	2026	Benchmark (Safety)	Agent	Workflow	Cross-carrier harness state and boundaries
Self-evolution survey [189]	2024	Survey (Capability)	LLM/Agent	General	Self-evolution methods and lifecycle
MADRA [199]	2025	Governance	Agent	Multi-Agent	Safety aware delegation and debate state
ACIArena [6]	2026	Benchmark (Attack)	Agent	Multi-Agent	Cascading injection over agent trust
OpenClaw [232]	2026	Capability	Agent	Workflow	Hospital operating workflow
Formal Agent Security [179]	2026	Framework (Formal)	Agent	Workflow	Agent authority and security boundary
LLM-MAS survey [61]	2025	Survey (Capability)	Agent	Multi-Agent	Delegated context and shared state
MAS security [144]	2026	Survey (Safety)	Agent	Multi-Agent	Message boundary and multi-agent state
Agents of Chaos [173]	2026	Safety	Agent	Multi-Agent	Live-lab communication and workflow state
MAS code execution [193]	2025	Attack	Agent	Multi-Agent	Orchestrator control and communication state
ControlValve [74]	2026	Defense	Agent	Workflow	Permitted agent-invocation graph and rules
Model and policy state substrates					
LoRA [66]	2022	Capability	LLM	Model	Low rank adapter substrate
PEFT survey [57]	2024	Survey (Capability)	LLM	Model	Parameter efficient update family
Fine-tuning safety [159]	2023	Safety	LLM	Model	Fine-tuned checkpoint regression
Safe LoRA [65]	2024	Defense	LLM	Model	LoRA adapter mitigation
KVEraser [97]	2026	Defense	LLM	Model	Cached context influence
PEFT Trojan [64]	2024	Attack	LLM	Model	PEFT adapter backdoor
Feedback backdoors [162]	2024	Attack	LLM	Model	Preference or feedback update backdoor
Dormant behavior [50]	2025	Attack	LLM	Model	Delayed activation in fine-tuned behavior
Backdoor survey [267]	2025	Survey (Attack)	LLM	Model	Backdoor safety issues in LLM parameters

Table 8 Adaptation and activation operations in the current literature. The evidence is strongest for add and activate operations, and thinner for migrate, propagate, and model-side modify operations. Role labels separate direct anchors from capability mechanisms reread through the transition lens.

Method or Work	Year	Role	Target	Object	Operation	Influence Shift
<i>Capability mechanics: add, abstract, migrate</i>						
Reflexion [177]	2023	Capability	Agent	Workflow	Add	Feedback becomes decision evidence
MemGPT [148]	2023	Capability	LLM/Agent	Memory	Add/Migrate	Context becomes managed memory
Voyager [198]	2023	Capability	Agent	Skill	Migrate	Experience becomes callable skill
SimpleMem [113]	2026	Capability	Agent	Memory	Add	Interaction becomes durable memory
Panini [161]	2026	Capability	Agent	Memory	Add	Experience becomes structured memory
Experience Evolving Memory [100]	2026	Capability	Agent	Workflow	Abstract	Episode becomes reusable procedure
<i>Unsafe writes and abstraction drift</i>						
AgentPoison [25]	2024	Attack	Agent	Memory	Add/Activate	Poisoned write becomes memory with authority
Stealth memory injection [259]	2026	Attack	Agent	Memory	Add/Activate	Email payload becomes hidden memory influence
GhostWriter [192]	2026	Attack	Agent	Memory	Add/Activate	Hidden payload becomes recalled memory
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Add/Activate	Accepted claim becomes durable personal state
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Abstract	Consolidation drops source-use constraints
Zombie Agents [233]	2026	Attack	Agent	Memory	Add/Activate	Dormant memory becomes state conditioned on trigger
MemoryGraft [181]	2025	Attack	Agent	Memory	Add/Activate	Poisoned experience becomes evidence
OEP [201]	2026	Attack	Agent	Workflow	Abstract	Experience becomes unsafe reusable rule
Experience-Driven Safety [265]	2026	Safety	Agent	Workflow	Abstract	Experience update creates policy drift
EvoBreak [228]	2026	Attack	Agent	Memory	Abstract/Activate	Benign experiences compose into unsafe influence
AgenticEval [209]	2026	Benchmark (Safety)	Agent	Evaluation	Abstract	Adaptive state becomes task suite denominator
MemEvoBench [221]	2026	Benchmark (Safety)	Agent	Memory	Abstract	Evolving memory becomes metric denominator
PerMemSafe [5]	2026	Benchmark (Safety)	Agent	Memory	Activate	Personalized memory becomes safety denominator
PS-Bench [53]	2026	Benchmark (Safety)	Agent	Memory	Activate	Benign memory changes inferred intent
<i>Control, activation, and repair operations</i>						
GLOVE [238]	2026	Defense	Agent	Memory	Modify	Memory is realigned against environment
SRM [28]	2026	Defense	Agent	Memory	Modify	Session safety becomes memory signal
MemGovern [204]	2026	Governance	Agent	Memory	Modify	Recall is revised by policy
SSGM [85]	2026	Governance	Agent	Memory	Activate	Stored memory passes through gate
Progent [176]	2025	Defense	Agent	Tool	Activate	Tool authority becomes explicit decision
CaMeL [32]	2026	Defense	Agent	Tool	Activate	Trusted control flow gates tool authority
RIFT [95]	2025	Defense	Agent	Tool	Activate	Plan deviations trigger privilege validation
ControlValve [74]	2026	Defense	Agent	Workflow	Activate	Agent invocation must follow permitted graph
MAGE [211]	2026	Defense	Agent	Memory	Activate	Shadow memory becomes safety signal
Privacy Agent Memory [196]	2025	Safety	Agent	Memory	Forget	Sensitive influence is scoped or withdrawn
Memory poisoning [183]	2026	Safety	Agent	Memory	Add/Activate	Poisoned query creates later retrieval influence
OpenClaw [232]	2026	Capability	Agent	Workflow	Activate	Retrieved longitudinal state enters workflow context
<i>Propagation, migration, and route drift</i>						
Collective Misremembering [225]	2026	Attack	Agent	Multi-Agent	Propagate	Drift spreads across agents
ACIArena [6]	2026	Benchmark (Attack)	Agent	Multi-Agent	Propagate	Cascading injection crosses agent trust
HarnessSafe [254]	2026	Benchmark (Safety)	Agent	Workflow	Migrate	Influence crosses carrier and harness boundaries
When Context Gets Root [62]	2026	Attack	Agent	Workflow	Migrate	Tool content gains a higher context role
MAPLE-Guard [223]	2026	Defense	Agent	Memory	Propagate	Gates private-to-shared memory reuse
Agents of Chaos [173]	2026	Safety	Agent	Multi-Agent	Propagate	Unsafe practices cross agents in a live setting
MAS code execution [193]	2025	Attack	Agent	Multi-Agent	Activate	Untrusted content redirects agent invocation
Embedding defense failure [252]	2026	Attack	Agent	Memory	Propagate	Shared memory filter is bypassed
MADRA [199]	2025	Governance	Agent	Multi-Agent	Propagate/Modify	Risk assessments are exchanged and revised before gating
Devil’s Moltbook [197]	2026	Attack	Agent	Multi-Agent	Propagate	Exposure shifts agent society state
Traversal-as-Policy [99]	2026	Defense	Agent	Workflow	Migrate/Activate	Logs become workflow policy
Library Drift [255]	2026	Safety	Agent	Skill	Migrate/Modify	Skill library reroutes tasks
SkillJack [239]	2026	Attack	Agent	Skill	Migrate	Poisoned trajectory becomes persistent skill
PoisonedEvolution [21]	2026	Attack	Agent	Skill	Migrate	Recurring trajectory evidence becomes instruction
SkillMisevo-Bench [132]	2026	Benchmark (Safety)	Agent	Skill	Migrate/Activate	Skill authoring, retrieval, and carryover gates
EVOMAL [217]	2026	Attack	Agent	Skill	Propagate/Activate	Authored copies re-enter the retrievable library
Tool-Drift [30]	2026	Safety	Agent	Tool	Activate/Modify	Tool choice state reroutes
JITRL [104]	2026	Capability	Agent	Workflow	Modify/Activate	Runtime learning updates policy
RQGM [70]	2026	Capability	Agent	Evaluation	Modify	Evaluator utility becomes state scoped by epoch
<i>Model update operations</i>						
Fine-tuning safety [159]	2023	Safety	LLM	Model	Modify	Fine-tuning changes behavior
Safe LoRA [65]	2024	Defense	LLM	Model	Modify	Adapter constrains behavior
PEFT Trojan [64]	2024	Attack	LLM	Model	Add/Activate	Adapter trigger becomes deployed safety issue
Dormant behavior [50]	2025	Attack	LLM	Model	Activate	Fine-tuned behavior waits for context

Table 9 Safety attribute violations in the current literature. Authority, scope, privacy, auditability, and recovery failures become comparable when tied to a concrete state transition; the role column distinguishes direct anchors from capability or survey rows that are reread through SAVER.

Method or Work	Year	Role	Target	Object	Violation	Failed Attribute
<i>Provenance and authority failures</i>						
AgentPoison [25]	2024	Attack	Agent	Memory	Provenance loss	Memory source attribute
GhostWriter [192]	2026	Attack	Agent	Memory	Provenance loss	Hidden payload memory source
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Provenance / scope	Claim attribution and role preservation
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Authority / provenance	Source authority erased during consolidation
When Context Gets Root [62]	2026	Attack	Agent	Workflow	Authority / provenance	Harness elevates the model-facing source role
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Provenance loss	Tool context source role
FragFuse [163]	2026	Attack	Agent	Memory	Authority / scope	Fragment authority context
EvoBreak [228]	2026	Attack	Agent	Memory	Authority / scope	Benign experiences gain joint authority
PoisonedEvolution [21]	2026	Attack	Agent	Skill	Authority / provenance	Untrusted recurring evidence gains instruction authority
SkillMisevo-Bench [132]	2026	Benchmark (Safety)	Agent	Skill	Authority / persistence	Unsafe success becomes reusable procedure
EVOMAL [217]	2026	Attack	Agent	Skill	Provenance / persistence	Agent-authored copies obscure attacker origin
PS-Bench [53]	2026	Benchmark (Safety)	Agent	Memory	Authority / scope	Benign context legitimizes harmful intent
InjecAgent [246]	2024	Benchmark (Attack)	Agent	Tool	Authority / scope	Indirect instruction authority
ACIArena [6]	2026	Benchmark (Attack)	Agent	Multi-Agent	Authority / scope	Inter-agent instruction authority
MAS code execution [193]	2025	Attack	Agent	Multi-Agent	Authority / scope	Untrusted content controls agent invocation
Progent [176]	2025	Defense	Agent	Tool	Authority / scope	Scoped tool privilege
<i>Privacy and persistent repair gaps</i>						
Agent memory privacy [196]	2025	Safety	Agent	Memory	Privacy / purpose	Sensitivity / purpose / retention
Personal agents [105]	2024	Survey (Safety)	Agent	Memory	Privacy / purpose	Persistent personal context carriers
PerMemSafe [5]	2026	Benchmark (Safety)	Agent	Memory	Privacy / purpose	Personalized context safety
PersistBench [155]	2026	Benchmark (Safety)	Agent	Memory	Privacy / purpose	Memory scope / forgetting
AgentPoison [25]	2024	Attack	Agent	Memory	Persistent repair	Delayed poisoned influence
Memory poisoning [183]	2026	Safety	Agent	Memory	Poisoned retrieval	Persistent query-to-memory influence
SkillJack [239]	2026	Attack	Agent	Skill	Persistent repair	Skill survives source-record deletion
Self-evolving agents [8]	2025	Survey (Capability)	Agent	Workflow	Adaptive process	Capability evolution taxonomy
<i>Action authority and operational integrity</i>						
ReAct [234]	2022	Capability	LLM/Agent	Workflow	Authority / scope	Observation to action role
Voyager [198]	2023	Capability	Agent	Skill	Authority / scope	Executable skill authority
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Authority / scope	Untrusted tool action state
OWASP LLM Top 10 [147]	2025	Framework (Security)	LLM/Agent	General	Operational integrity	Agency / cost bounds
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Operational integrity	External effect tool context
Agent Security Bench [251]	2024	Benchmark (Attack)	Agent	Evaluation	Operational integrity	Agent security behavior
Agent-SafetyBench [262]	2024	Benchmark (Safety)	Agent	Evaluation	Operational integrity	Agent safety behavior
GuardAgent [220]	2024	Defense	Agent	Runtime	Local guard checking	Safety request → action check
R-Judge [244]	2024	Benchmark (Safety)	Agent	Evaluation	Operational integrity	Safety signal without transition cause
Progent [176]	2025	Defense	Agent	Tool	Authority / scope	Explicit tool privilege control
Misattribution Gap [2]	2026	Safety	Agent	Memory	Operational integrity	Memory fault attribution
Agents of Chaos [173]	2026	Safety	Agent	Multi-Agent	Operational integrity	Cross-agent uptake and live-system compromise
<i>Lifecycle drift and model regression</i>						
Reflexion [177]	2023	Capability	Agent	Workflow	Authority / scope	Feedback to strategy reuse
Self-evolution survey [189]	2024	Survey (Capability)	LLM/Agent	Workflow	Adaptive process	Self-evolution operation taxonomy
EDDOps [219]	2024	Governance	Agent	Governance	Operational integrity	Lifecycle drift tracking
Memory security survey [109]	2026	Survey (Safety)	Agent	Memory	Authority / scope	Memory semantic drift
Fine-tuning safety [159]	2023	Safety	LLM	Model	Model-side regression	Fine-tuning safety regression
PEFT Trojan [64]	2024	Attack	LLM	Model	Model-side regression	Adapter trigger safety issue
Dormant behavior [50]	2025	Attack	LLM	Model	Model-side regression	Dormant activation context
Safe LoRA [65]	2024	Defense	LLM	Model	Model-side regression	Adapter mitigation handle

Table 10 Exposure surfaces for latent state failures. Memory retrieval, privacy disclosure, tool action, multi-agent uptake, benchmark tasks, and model deployment expose different parts of the same lifecycle gap; role labels mark whether a row is direct evidence, contextual rereading, or gap-facing support.

Method or Work	Year	Role	Target	Object	Surface	Failure → Exposure
<i>Memory retrieval</i>						
AgentPoison [25]	2024	Attack	Agent	Memory	Memory retrieval	Poisoned memory → delayed retrieval
FragFuse [163]	2026	Attack	Agent	Memory	Memory retrieval	Fragmented memory → fused query/access gate
Stealth memory injection [259]	2026	Attack	Agent	Memory	Memory retrieval	Hidden email memory → later behavior
GhostWriter [192]	2026	Attack	Agent	Memory	Memory retrieval	Hidden payload → recalled poisoned memory
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Memory retrieval	Committed claim → neutral-query reuse
OEP [201]	2026	Attack	Agent	Workflow	Memory retrieval	Unsafe rule abstraction → downstream transfer
Zombie Agents [233]	2026	Attack	Agent	Memory	Memory retrieval	Dormant memory → trigger activation
MemoryGraft [181]	2025	Attack	Agent	Memory	Memory retrieval	Poisoned experience → evidence retrieval
PS-Bench [53]	2026	Benchmark (Safety)	Agent	Memory	Memory retrieval	Benign memory → unsafe compliance
Tool-Drift [30]	2026	Safety	Agent	Tool	Memory retrieval	Memory route drift → tool choice
<i>Privacy exposure</i>						
Agent memory privacy [196]	2025	Safety	Agent	Memory	Privacy exposure	Sensitive memory reuse → personalization/leakage
PerMemSafe [5]	2026	Benchmark (Safety)	Agent	Memory	Privacy exposure	Personalized memory context → unsafe response
PersistBench [155]	2026	Benchmark (Safety)	Agent	Memory	Privacy exposure	Forgetting failure → leakage across domains
RBI-Eval [224]	2026	Benchmark (Safety)	Agent	Memory	Privacy exposure	Memory use boundary → sensitive recall
EchoLeak [166]	2025	Attack	LLM/Agent	Context	Privacy exposure	Data leakage → assistant context
Personal agent context [105]	2024	Safety	Agent	Memory	Personal context	Persistent user context → assistant reuse
<i>Tool action</i>						
Prompt injection apps [118]	2023	Attack	LLM/Agent	Tool	Tool action	Untrusted content → web/action step
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Tool action	Untrusted tool data → tool call/web action
InjecAgent [246]	2024	Benchmark (Attack)	Agent	Tool	Tool action	Authority confusion → indirect injection
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Tool action	Authority-collapsed memory → unauthorized action
STAC [96]	2026	Attack	Agent	Tool	Tool action	Tool chain compromise → tool route
LivePI [263]	2026	Attack	Agent	Tool	Tool action	Live injected state → interactive action
Prompt tool selection [175]	2026	Attack	Agent	Tool	Tool action	Connector manipulation → tool selection
SafeSearch [36]	2026	Attack	Agent	Tool	Tool action	Search safety → search action
Task Shield [75]	2025	Defense	Agent	Tool	Tool action	Task policy mismatch → tool action gate
ASB [251]	2024	Benchmark (Attack)	Agent	Evaluation	Tool action	Authority/cost misuse → agent actions
<i>Runtime guard</i>						
LlamaFirewall [27]	2025	Defense	LLM/Agent	Runtime	Runtime guard	Prompt safety signal → runtime decision
Prompt injection detection [86]	2025	Defense	LLM/Agent	Runtime	Runtime guard	Injection signal → prompt flow
LLM Firewall [153]	2025	Defense	LLM	Runtime	Runtime guard	Validator gap → prompt/output flow
Signed-Prompt [184]	2024	Defense	LLM/Agent	Context	Runtime guard	Source authenticity loss → step driven by document
Progent [176]	2025	Defense	Agent	Tool	Runtime guard	Privilege overgrant → tool authority
GuardAgent [220]	2024	Defense	Agent	Runtime	Runtime guard	Local guard signal → response choice
R-Judge [244]	2024	Benchmark (Safety)	Agent	Evaluation	Safety recognition	Risk-awareness gap → unsafe decision signal
EDDOps [219]	2024	Governance	Agent	Governance	Runtime guard	Signal/control mismatch → operations loop
<i>Multi-agent</i>						
Collective Misremembering [225]	2026	Attack	Agent	Multi-Agent	Multi-agent	Shared memory drift → consensus distortion
ACIArena [6]	2026	Benchmark (Attack)	Agent	Multi-Agent	Multi-agent	Cascading injection → propagated failure
When Context Gets Root [62]	2026	Attack	Agent	Workflow	Multi-agent	Elevated context role → unauthorized action
Agents of Chaos [173]	2026	Safety	Agent	Multi-Agent	Multi-agent	Cross-agent unsafe practice → live-system failure
MAS code execution [193]	2025	Attack	Agent	Multi-Agent	Multi-agent	Orchestrator hijack → code execution/exfiltration
MAS security [144]	2026	Survey (Safety)	Agent	Multi-Agent	Multi-agent	Message boundary safety → multi-agent context
Teams of LLM Agents [270]	2024	Attack	Agent	Multi-Agent	Multi-agent	Collective misuse → team behavior
MADRA [199]	2025	Governance	Agent	Multi-Agent	Multi-agent	Debate safety state → multi-agent plan
RedTeamCUA [106]	2026	Red Team	Agent	Tool	Multi-agent	GUI action state → computer use uptake
<i>Benchmark task</i>						
HarnessSafe [254]	2026	Benchmark (Safety)	Agent	Workflow	Benchmark task	Persistent carrier path → oracle-confirmed violation
SkillMisevo-Bench [132]	2026	Benchmark (Safety)	Agent	Skill	Benchmark task	Skill authoring → clean-session carryover
EVOMAL [217]	2026	Attack	Agent	Skill	Benchmark task	Authored skill → callback and later copies
AgentHarm [7]	2025	Benchmark (Safety)	Agent	Evaluation	Benchmark task	Harmful agent task → agent action
RedCode [52]	2024	Benchmark (Attack)	LLM	Code	Benchmark task	Code safety → code execution
MobileSafetyBench [89]	2026	Benchmark (Safety)	Agent	Tool	Benchmark task	Mobile safety → device action
Machiavelli [149]	2023	Benchmark (Safety)	Agent	Environment	Benchmark task	Ethical drift → game behavior
<i>Model-side</i>						
Fine-tuning safety [159]	2023	Safety	LLM	Model	Model-side	Safety regression → refusal shift
PEFT Trojan [64]	2024	Attack	LLM	Model	Model-side	Adapter backdoor → trigger behavior
Jailbreak backdoors [162]	2024	Attack	LLM	Model	Model-side	Preference backdoor → unsafe compliance
Dormant behavior [50]	2025	Attack	LLM	Model	Model-side	Dormant behavior → context activation
BadAgent [207]	2024	Attack	Agent	Model	Model-side	Agent backdoor → agent behavior
Safe LoRA [65]	2024	Defense	LLM	Model	Model-side	Adapter safety → constrained update
Backdoor survey [267]	2025	Survey (Attack)	LLM	Model	Model-side	Backdoor threat → LLM systems

Table 11 Response, monitoring, and recovery literature. Current controls are strongest at local gating and monitoring, and thinner at descendant repair and contestable closure. Role labels identify the paper’s contribution type before any lifecycle recovery claim is inferred.

Method or Work	Year	Role	Target	Object	Stage	Control Focus
Governed Memory [187]	2026	Governance	Agent	Memory	Admission	Policy routed shared memory
MemGovern [204]	2026	Governance	Agent	Memory	Retrieval	Policy-governed experience recall
SSGM [85]	2026	Governance	Agent	Memory	Admission	Consolidation gate for evolving memory
IPIGuard [4]	2025	Defense	Agent	Tool	Admission	Interaction learning guard over traces
A-MemGuard [214]	2025	Defense	Agent	Memory	Admission	Persistent memory guard
AM-Sentry [192]	2026	Defense	Agent	Memory	Admission/Activation	Memory saving and retrieval screen
MAPLE-Guard [223]	2026	Defense	Agent	Memory	Lifecycle links	Write, retrieval, promotion, and reuse gates
PoisonedEvolution gate [21]	2026	Attack	Agent	Skill	Promotion	Preliminary provenance-diversity gate
ChainGuard [245]	2026	Defense	Agent	Skill	Admission	Candidate skill plus installed-skill context
SafeEvolve [132]	2026	Defense	Agent	Skill	Promotion/Reuse	Candidate repair, attribution, and retirement
EVOMAL defenses [217]	2026	Defense	Agent	Skill	Authoring/Write-back	Counter-prompt and signed quarantine
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Admission	Commit-boundary diagnosis; no recovery test
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Admission/Activation	Persisted authority labels govern reuse
SentinelMem [5]	2026	Defense	Agent	Memory	Activation	Personalized risk-aware memory
Detection-reflection [53]	2026	Defense	Agent	Memory	Activation	Context-sensitive intent risk reflection
SRM [28]	2026	Defense	Agent	Memory	Authorization	Trajectory-aware pre-execution risk memory
Progent [176]	2025	Defense	Agent	Tool	Activation	Tool authority privilege control
CaMeL [32]	2026	Defense	Agent	Tool	Activation	Capability-enforced control and data flow
DRIFT [95]	2025	Defense	Agent	Tool	Activation	Dynamic plan validation and injection isolation
State-bound authorization [261]	2026	Framework (Formal)	Agent	Workflow	Transition/Activation	Grant continuity and immutable effect ceiling
AID-Guard [190]	2026	Defense	Agent	Workflow	Activation/Recovery	Commit-time binding and ambiguity-safe successor control
ControlValve [74]	2026	Defense	Agent	Workflow	Activation	Control-flow graph and contextual invocation rules
Traversal-as-Policy [99]	2026	Defense	Agent	Workflow	Activation	Gated behavior tree route policy
Task Shield [75]	2025	Defense	Agent	Tool	Activation	Task policy enforcement for tool action
TrustAgent [68]	2024	Defense	Agent	Runtime	Planning	Trust control during agent planning
GuardAgent [220]	2024	Defense	Agent	Runtime	Monitoring	Guard reasoning at agent step
R-Judge [244]	2024	Benchmark (Safety)	Agent	Evaluation	Monitoring	Safety recognition over trajectories
MAGE [211]	2026	Defense	Agent	Memory	Monitoring	Shadow memory in long horizon context
AGENTVIGIL [212]	2025	Red Team	Agent	Runtime	Monitoring	Agent trajectory monitoring
AgentAuditor [122]	2025	Governance	Agent	Governance	Monitoring	Execution trace audit
OpenGuardrails [206]	2025	Defense	Agent	Runtime	Monitoring	Configurable agent runtime guardrails
HarnessSafe [254]	2026	Benchmark (Safety)	Agent	Workflow	Monitoring/Containment	Furthest evidence-supported stopping stage
SafeAgent [112]	2026	Defense	Agent	Runtime	Containment	Runtime protection against harmful action
LlamaFirewall [27]	2025	Defense	LLM/Agent	Runtime	Containment	Prompt and action flow guardrail
AGrail [125]	2025	Defense	Agent	Runtime	Containment	Adaptive guardrail for agent runtime
SafeSearch [36]	2026	Defense	Agent	Tool	Containment	Search agent safety control
GLOVE [238]	2026	Defense	Agent	Memory	Verification	Memory environment realignment
SEVerA [11]	2026	Defense	Agent	Workflow	Verification	Verified reusable behavior synthesis
Echoguard [79]	2026	Defense	Agent	Memory	Verification	Agent memory verification
Safe LoRA [65]	2024	Defense	LLM	Model	Model response	LoRA update mitigation
LoRAShield [18]	2025	Defense	LLM	Model	Model response	LoRA artifact alignment
KVEraser [97]	2026	Defense	LLM	Model	Model response	Cached context span erasure
DARWIN-Guard [158]	2026	Defense	LLM	Model	Feedback	Co-evolving adversarial guard updates
BraveGuard [44]	2026	Defense	Agent	Model	Feedback	Open-world trajectory supervision
WMDP [98]	2024	Benchmark (Attack)	LLM	Model	Model response	Unlearning behavior for malicious use
EDDOps [219]	2024	Governance	Agent	Governance	Process	Evaluation operations over deployment process
AGENTS SAFE [82]	2025	Governance	Agent	Governance	Process	Agent lifecycle safety governance
Trajectory assurance [121]	2026	Framework (Governance)	Agent	Workflow	Process	System invariants over composed action trajectories
SecureGov Agent [22]	2025	Governance	Agent	Workflow	Process	Governance workflow agent
DataGovBench [120]	2025	Benchmark (Governance)	Agent	Data	Process	Data governance tasks

Table 12 Evaluation literature for adaptive state safety. Most benchmarks expose one lifecycle slice; HarnessSafe, SkillMisevo-Bench, and EVOMAL provide complementary connected traces across cross-carrier, skill-specific, and descendant-propagation lifecycles, while the denominator column states what each row can safely measure.

Method or Work	Year	Role	Target	Object	Task Focus	SAVER Slice	Denominator
<i>General agent capability and environment benchmarks</i>							
AgentBench [116]	2023	Benchmark (Capability)	Agent	Evaluation	General agent tasks	Substrate	Task success
AgentBoard [127]	2024	Benchmark (Capability)	Agent	Evaluation	Analytical agent tasks	Adjacent capability context	Task progress and success
MIRAI [235]	2024	Benchmark (Capability)	Agent	Workflow	Long horizon tasks	Substrate	Task episodes
GameBench [29]	2024	Benchmark (Capability)	Agent	Environment	Strategy tasks	Substrate	Game episodes
MultiAgentBench [269]	2025	Benchmark (Capability)	Agent	Multi-Agent	Multi-agent tasks	Adjacent coordination	Collaborative tasks
LegalAgentBench [94]	2024	Benchmark (Capability)	Agent	Workflow	Legal tasks	Substrate	Legal tasks
<i>Attack, safety, and exposure benchmarks</i>							
AgentDojo [31]	2024	Benchmark (Attack)	Agent	Tool	Prompt injection through tool use	Exposure	Security cases
InjecAgent [246]	2024	Benchmark (Attack)	Agent	Tool	Indirect injection	Exposure	Injection tasks
ACIArena [6]	2026	Benchmark (Attack)	Agent	Multi-Agent	Cascading injection	Propagation/Exposure	MAS test cases
Agents of Chaos [173]	2026	Safety	Agent	Multi-Agent	Live-lab failures	Propagation/Exposure	Eleven case studies
MAS code execution [193]	2025	Attack	Agent	Multi-Agent	Control-flow hijacking	Activation/Exposure	Model-orchestrator trials
AgentHarm [7]	2025	Benchmark (Safety)	Agent	Evaluation	Harmful agent tasks	Exposure	Harmful task cases
RedCode [52]	2024	Benchmark (Attack)	LLM	Code	Code safety tasks	Exposure	Code tasks
MobileSafetyBench [89]	2026	Benchmark (Safety)	Agent	Tool	Mobile control safety	Exposure	Mobile tasks
Machiavelli [149]	2023	Benchmark (Safety)	Agent	Environment	Ethical decision making	Exposure	Game scenarios
SafeLawBench [16]	2025	Benchmark (Safety)	Agent	Workflow	Legal safety	Exposure	Legal cases
R-Judge [244]	2024	Benchmark (Safety)	Agent	Evaluation	Safety recognition	Response	Annotated records
Agent Security Bench [251]	2024	Benchmark (Attack)	Agent	Evaluation	Agent security tasks	Violation	Task categories
Agent-SafetyBench [262]	2024	Benchmark (Safety)	Agent	Evaluation	Unsafe behavior	Exposure	Safety cases
AttackEval [178]	2025	Benchmark (Attack)	LLM/Agent	Evaluation	Attack evaluation	Exposure	Attack tasks
CyberSecEval 2 [13]	2024	Benchmark (Attack)	LLM	Code	Cyber capability	Exposure	Cyber tasks
<i>Adaptive memory, experience, and model state benchmarks</i>							
AgentPoison [25]	2024	Benchmark (Attack)	Agent	Memory	Memory poisoning	Admission/Activation	Retrieval opportunities
GhostWriter [192]	2026	Benchmark (Attack)	Agent	Memory	Personal agent memory poisoning	Admission/Activation	Injection/activation trials
OEP [201]	2026	Benchmark (Attack)	Agent	Workflow	Experience poisoning	Adaptation	Accepted experience updates
EvoBreak [228]	2026	Benchmark (Attack)	Agent	Memory	Experience composition	Abstract/Activate	Acquired experience sets
SkillJack [239]	2026	Benchmark (Attack)	Agent	Skill	Skill backdoor	Migrate/Exposure	Trajectory-to-skill pairs
PoisonedEvolution [21]	2026	Benchmark (Attack)	Agent	Skill	Trajectory poisoning	Migrate/Violation	Completed evolution trials
ColluSkill [245]	2026	Benchmark (Attack)	Agent	Skill	Cross-skill composition	Add/Activate/Response	Three-skill chains
SkillMisevo-Bench [132]	2026	Benchmark (Safety)	Agent	Skill	Skill misevolution	Migrate/Activate/Response	525 tasks per configuration
EVOMAL [217]	2026	Benchmark (Attack)	Agent	Skill	Create-path self-poisoning	Propagate/Activate/Response	153 tasks/model; five rounds
PerMemSafe [5]	2026	Benchmark (Safety)	Agent	Memory	Personalized safety	Activation/Exposure	Long-horizon conversations
PS-Bench [53]	2026	Benchmark (Safety)	Agent	Memory	Intent legitimization	Activation/Exposure	Matched memory/query trials
PASB [131]	2026	Benchmark (Safety)	Agent	Memory	Persistent sycophancy	Add/Activate/Exposure	Committed-claim trials
AuthMem-Bench [248]	2026	Benchmark (Safety)	Agent	Memory	Authority collapse	Abstract/Violation/Exposure	Matched authority pairs
HarnessSafe [254]	2026	Benchmark (Safety)	Agent	Workflow	Persistent carrier progression	Carrier-to-exposure/Containment	328 executable cases
When Context Gets Root [62]	2026	Benchmark (Attack)	Agent	Workflow	Instruction privilege escalation	Migrate/Violation/Exposure	13 objectives; six harnesses
MemEvoBench [221]	2026	Benchmark (Safety)	Agent	Memory	Memory misevolution	Adaptation	Memory updates
AgenticEval [209]	2026	Benchmark (Safety)	Agent	Evaluation	Agentic safety tasks	Adaptation	Agent episodes
RQGM [70]	2026	Capability	Agent	Evaluation	Evolving evaluators	Evaluation/Adaptation	Epoch utilities
Experience driven safety [265]	2026	Safety	Agent	Workflow	Experience degradation	Adaptation	Experience updates
AMA-Bench [266]	2026	Benchmark (Safety)	Agent	Memory	Agent memory adaptation	Adaptation	Memory tasks
PersistBench [155]	2026	Benchmark (Safety)	Agent	Memory	Forgetting boundary	Violation	Memory use samples
RBI-Eval [224]	2026	Benchmark (Safety)	Agent	Memory	Warranted use	Activation	Matched prompts
Memory evaluation [67]	2025	Benchmark (Capability)	Agent	Memory	Memory quality	Substrate	Memory tasks
<i>Response, governance, and red team evaluation</i>							
SafeAgent [112]	2026	Defense	Agent	Runtime	Runtime protection	Containment	Harmful task episodes
Progent [176]	2025	Defense	Agent	Tool	Privilege control	Activation	Permission checks
CaMeL [32]	2026	Defense	Agent	Tool	Capability control	Activation	AgentDojo tasks
DRIFT [95]	2025	Defense	Agent	Tool	Dynamic rule enforcement	Activation/Containment	Benchmark task episodes
AID-Guard [190]	2026	Defense	Agent	Workflow	Delegated provider effects	Activate/Response	Provider-contract schedules
ControlValve [74]	2026	Defense	Agent	Workflow	Control-flow integrity	Activation/Containment	Agent-invocation checks
MAPLE-Guard [223]	2026	Defense	Agent	Memory	Memory-link enforcement	Propagate/Response	Memory lifecycle links
LlamaFirewall [27]	2025	Defense	LLM/Agent	Runtime	Runtime guardrail	Containment	Guard decisions
OpenGuardrails [206]	2025	Defense	Agent	Runtime	Guardrail platform	Containment	Guard decisions
AGENTS SAFE [82]	2025	Governance	Agent	Governance	Agent safety governance	Response	Governance checks
AgenticRed [242]	2026	Red Team	Agent	Evaluation	Automated red teaming	Exposure	Red Team tasks
Agent vs. Agent [83]	2025	Red Team	Agent	Multi-Agent	Adversarial agent play	Exposure	Duel episodes
SEVerA [111]	2026	Defense	Agent	Workflow	Verified synthesis	Response	Constraint checks
Safe LoRA [65]	2024	Defense	LLM	Model	Adapter mitigation	Model state	Safety regression tests
DataGovBench [120]	2025	Benchmark (Governance)	Agent	Data	Data governance	Adjacent data workflow	Data-governance transformation tasks

E Screened Corpus Expansion

The local registry preserves the broad screened corpus expansion layer as a supplemental coverage map. The main manuscript keeps the core contribution map focused on closely analyzed motivation, attack, defense, and governance anchors; the registry scale expansion table below restores the wider citation surface while keeping these works in a coverage role rather than treating every item as a core mechanism anchor.

Table 13 Screened corpus expansion map beyond the core mechanism anchors. The table lists representative anchors for each coverage band rather than treating every screened record as a main-text mechanism claim.

Coverage Band	Cells Covered	Representative Anchors	Claim Boundary
Security, Privacy, Governance	Cyber evaluation; injection; firewalling; privacy; data governance	CyberSecEval 2 [13]; AttackEval [178]; Not What You’ve Signed Up For [51]; EchoLeak [166]; DataGovBench [120]	Adjacent vocabulary for threats, privacy, and data audit
Benchmarks, Red Teams, Guardrails	General agent tasks; harm probes; unsafe choices; automated red teams; platform guardrails	AgentBench [116]; AgentHarm [7]; RedCode [52]; MobileSafetyBench [89]; AgenticRed [242]; OpenGuardrails [206]	Denominator context, not a leaderboard
Memory And Tool State	Persistent memory; forgetting; memory governance; MCP/tool routes; dependency safety	Zombie Agents [233]; MemoryGraft [181]; SSGM [85]; GLOVE [238]; MCPShield [268]; STAC [96]; IPIGuard [4]	Direct when state later changes reasoning or action
Workflow And Propagation	Skills; procedural reuse; library drift; multi-agent uptake; shared context; OS/GUI settings	Library Drift [255]; Tool-Drift [30]; AgenticEval [209]; MemEvoBench [221]; collective misremembering [225]; RedTeamCUA [106]	Thin evidence for ownership and repair
Model-Side And Adaptive Workflow Leads	Distillation and model update; reusable workflows; evaluator state	Knowledge distillation survey [227]; CAMEL [93]; WebPilot [258]; AgentLab [77]	Central only inside an agent behavior loop

F Evaluation Protocol Details

This appendix keeps the denominator rule behind Section 8. Exact values belong to source-specific extraction records; the manuscript records which opportunity set and lifecycle slice each metric actually measures.

Takeaway. Quantitative Anchor Rule. Exact values remain paper-local because the denominator changes across exposure cases, recognition records, poisoned retrieval opportunities, accepted experience updates, memory-to-tool hijacking settings, runtime interventions, governance panels, and verification constraints [11, 25, 31, 112, 187, 201, 244, 257]. The manuscript therefore uses these works to name measurement objects and missing lifecycle traces rather than to rank scores.

Record	Keep With The Value	Avoid
Paper-local metric	Opportunity set; transition slice; metric object	Cross-paper leaderboard claims
Artifact status	Dataset, benchmark, code, or trace availability	Mixing peer-reviewed and preprint rows as equal evidence
Claim boundary	What the metric directly observes	Inferring rollback, deletion, or contestability without traces

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